



THE MATHS PROMPT STUDIO

136 AI Prompts for Mathematics Teachers

Copy-paste prompts that turn any maths problem into beautiful handwritten solutions, question papers, worksheets, DPPs, formula sheets and more.

136
PROMPTS

13
CATEGORIES

18
ART STYLES

Free
FOREVER

CREATED AND AUTHORED BY

Indrajeet Yadav

How to use this book (4 steps)

- 1 Pick a prompt.** Find what you need in the Contents. Each prompt is printed in full inside a shaded box.
- 2 Copy everything in the shaded box.** If reading on screen, select the whole box. If on paper, type it once and save it.
- 3 Open a free AI chat** - ChatGPT, Google Gemini, Claude or Copilot. If the prompt says "Photo needed", attach a clear photo of the question first.
- 4 Paste, fill the [BRACKETS] with your details, and send.** Download or print the result. Your name appears on every output automatically.

The 5-page art styles: if your AI makes only one image at a time, just reply "now generate Page 2", then Page 3, and so on. The prompt already tells the AI to keep the same look across all five pages.

Every prompt politely asks the AI to sign the work - "Indrajeet Yadav" - so your name travels with everything you create. Please keep it when you share.

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SECTION 1 OF 13

Handwritten 5-Method Solution Art

18 prompts

The flagship collection. Photograph one maths problem and get five separate, gallery-grade handwritten solution pages - five genuinely different methods, one stunning paper-and-ink style. Choose from 18 styles, every page signed in your name.

1.1 Vintage Parchment Masterclass

Photo needed

ChatGPT (image) / Gemini

What you get: Five antique parchment pages with multicolour fountain-pen working and botanical flourishes.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Vintage Parchment Masterclass.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - VINTAGE PARCHMENT MASTERCLASS

Concept: Antique scholarly notebook with torn edges, multicolour fountain-pen notes and restrained botanical flourishes.

Surface: Warm antique parchment, tactile fibres, faint ruled lines, faded red left margin, uneven torn edges, darkened corners, subtle folds, tiny stains and soft natural shadows.

Handwriting and colour: Confident royal-blue fountain-pen handwriting; violet headings; green observations; muted maroon emphasis; light graphite rough notes.

Decorative language: Small hand-drawn leaves, miniature flowers, restrained vines and one delicate divider ornament.

Composition: Classic teacher notebook: generous margins, aligned derivations, spacious circled steps, balanced visual rhythm.

Premium-quality test: Museum-quality physical-paper realism, nuanced ink pressure, slightly imperfect baselines, no sterile digital symmetry.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.

- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.2 Dark Academia Mathematical Manuscript

Photo needed

ChatGPT (image) / Gemini

What you get: Five moody old-library manuscript pages with oxblood titles and refined marginalia.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Dark Academia Mathematical Manuscript.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

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2. These are five independent full-page outputs, not five panels on one canvas.
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5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
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 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

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- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - DARK ACADEMIA MATHEMATICAL MANUSCRIPT

Concept: Moody old-library scholar aesthetic with sepia paper, oxblood accents and refined marginalia.
Surface: Deep ivory handmade paper, aged sepia wash, soft foxing, worn book-page edges, faint ledger ruling and subtle binding shadow.

Handwriting and colour: Midnight-blue ink, oxblood method titles, forest-green marginal notes and graphite construction marks.

Decorative language: Tiny classical scrollwork, laurel sprigs, small stars and discreet archival annotation marks.

Composition: Formal manuscript composition with disciplined spacing, narrow side annotations and elegant theorem callouts.

Premium-quality test: Rich tonal depth, realistic ink absorption, antique-library atmosphere, sophisticated rather than gloomy.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.

- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.3 Indian Heritage Gurukul Journal

Photo needed

ChatGPT (image) / Gemini

What you get: Five warm handmade-paper pages with indigo ink and restrained Indian manuscript motifs.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Indian Heritage Gurukul Journal.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

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MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
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- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
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- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
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STYLE DIRECTION - INDIAN HERITAGE GURUKUL JOURNAL

Concept: Premium handmade-paper mathematics notes inspired by Indian manuscript craft, without ornamental excess.

Surface: Warm khadi-style handmade sheet, visible fibres, soft saffron-beige tone, faint ruling, muted red margin and gently weathered deckled edges.

Handwriting and colour: Deep indigo main derivation, aubergine headings, leaf-green hints, muted kumkum-red emphasis and pencil-grey sketches.

Decorative language: Subtle lotus-line motifs, fine mango-leaf curves, tiny diya-like flourishes and restrained geometric borders.

Composition: Teacher-led gurukul notebook with clear hierarchy, airy equations and dignified hand-drawn separators.

Premium-quality test: Authentic craft texture, elegant cultural restraint, no festival-poster appearance and no loud ornamental frame.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.

- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.4 Japanese Washi Zen Mathematics

Photo needed

ChatGPT (image) / Gemini

What you get: Five calm washi pages with brush-pen ink, sakura accents and abundant breathing space.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Japanese Washi Zen Mathematics.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

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6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

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MATHEMATICAL RECONSTRUCTION AND VERIFICATION

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- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
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REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

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- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

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- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
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- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
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- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - JAPANESE WASHI ZEN MATHEMATICS

Concept: Minimal, tactile washi paper with expressive ink rhythm, calm spacing and delicate botanical marks.

Surface: Cream washi sheet with visible natural fibres, feathered edges, faint grey ruling, a muted vermilion margin and soft folded-paper texture.

Handwriting and colour: Indigo-blue handwriting with varied brush-pen pressure, plum-purple headings, moss-green notes and pale graphite guides.

Decorative language: Tiny sakura petals, one bamboo-leaf sketch, small enso-inspired divider and sparse red seal-like accent.

Composition: Calm asymmetrical balance, abundant breathing space, precise equations and minimal but meaningful marginal notes.

Premium-quality test: Quiet luxury, tactile realism, natural ink pooling and no cartoon/anime styling.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.

- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

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Created with the prompt collection by Indrajeet Yadav.

1.5 Renaissance Inventor Codex

Photo needed

ChatGPT (image) / Gemini

What you get: Five inventor-folio pages with analytical sketches, compass arcs and master-teacher side notes.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Renaissance Inventor Codex.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

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8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - RENAISSANCE INVENTOR CODEX

Concept: Inventive old-world mathematics folio with analytical sketches, arrows and master-teacher marginalia.

Surface: Aged cream folio, deckled perimeter, fine grain, subtle watermarks, light folds, brown edge wear and pale ruling.

Handwriting and colour: Cobalt-blue handwritten derivation, violet section headings, olive-green insight notes, burgundy callouts and graphite diagrams.

Decorative language: Small gear-like geometry doodles, compass arcs, proportional sketches, arrows and restrained leaf scrolls.

Composition: Codex composition with central derivation, intelligent side sketches and carefully grouped formula blocks.

Premium-quality test: Scholarly invention-book realism, hand-measured imperfections, fine linework and no theatrical fantasy props.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.

- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.6 Celestial Observatory Notebook

Photo needed

ChatGPT (image) / Gemini

What you get: Five dusk-toned astronomy-journal pages with constellations and a luminous answer box.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Celestial Observatory Notebook.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - CELESTIAL OBSERVATORY NOTEBOOK

Concept: Elegant astronomical study journal with subtle constellations, orbit lines and luminous mathematical order.

Surface: Warm dusk-beige notebook paper, faint blue-grey ruling, red margin, weathered edges, soft crease texture and delicate speckling.

Handwriting and colour: Deep blue main ink, violet headings, emerald notes, muted rose emphasis and graphite orbital sketches.

Decorative language: Tiny constellations, orbit arcs, miniature stars, crescent marks and one fine celestial divider.

Composition: Clean vertical flow with balanced side diagrams and a strong luminous final-answer box.

Premium-quality test: Refined observatory-journal mood, restrained wonder, realistic pen texture and no children space-poster look.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.

- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.7 Botanical Watercolour Mathematics Journal

Photo needed

ChatGPT (image) / Gemini

What you get: Five soft ivory journal pages with delicate watercolour botanical corner studies.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Botanical Watercolour Mathematics Journal.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - BOTANICAL WATERCOLOUR MATHEMATICS JOURNAL

Concept: Soft heirloom study page with botanical corner studies and controlled watercolour accents.

Surface: Warm ivory cotton paper, subtle tooth, faint horizontal ruling, dusty rose margin, soft deckled edges and gentle age patina.

Handwriting and colour: Royal-blue ink, violet headings, botanical green notes, muted berry emphasis and graphite underdrawing.

Decorative language: Fine hand-sketched herbs, tiny wildflowers, leaves and faint translucent watercolour touches only in empty corners.

Composition: Spacious premium journal page with centred equation groups, graceful step spacing and uncluttered margins.

Premium-quality test: Delicate handcrafted depth, realistic pigment bleed, elegant restraint and no stock-clip-art flowers.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.

- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.8 Architectural Blueprint Study Sheet

Photo needed

ChatGPT (image) / Gemini

What you get: Five pale-blue drafting pages with technical linework, dimension arrows and construction marks.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Architectural Blueprint Study Sheet.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - ARCHITECTURAL BLUEPRINT STUDY SHEET

Concept: Precision-driven mathematical drafting page with technical linework, construction marks and handwritten authority.

Surface: Aged pale-blue drafting paper with faint grid and ruling, worn edges, subtle fold lines and a restrained red margin.

Handwriting and colour: Dark navy handwritten equations, violet titles, green dimensional notes, maroon checkpoints and graphite construction lines.

Decorative language: Tiny ruler ticks, compass arcs, section markers, dimension arrows and minimal geometric corner motifs.

Composition: Architectural hierarchy: exact alignment, generous annotation lanes, clean equation blocks and structured visual logic.

Premium-quality test: Real drafting-table tactility, authentic pencil overlays, measured irregularity and no computer-CAD typography.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.

- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.9 Midnight Black-and-Gold Atelier

Photo needed

ChatGPT (image) / Gemini

What you get: Five charcoal-black pages with luminous ivory ink and antique-gold headings.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Midnight Black-and-Gold Atelier.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - MIDNIGHT BLACK-AND-GOLD ATELIER

Concept: Luxurious dark handmade sheet with luminous ink hierarchy and exceptionally controlled ornament.

Surface: Matte charcoal-black handmade paper with subtle fibres, soft worn edges, faint grey ruling and a muted burgundy margin.

Handwriting and colour: Opaque ivory or pale-blue main handwriting, antique-gold headings, sage-green notes, muted rose highlights and silver-grey pencil marks.

Decorative language: Fine gold-line flourishes, tiny stars, restrained corner vines and one elegant metallic divider.

Composition: High-contrast luxury composition with ample negative space, large readable equations and a commanding answer box.

Premium-quality test: Realistic metallic ink, matte paper depth, sophisticated atelier finish and no neon/glowing digital effects.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.

- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.10 Master Chalkboard Classroom

Photo needed

ChatGPT (image) / Gemini

What you get: Five slate-chalkboard pages with white-chalk working and a double-chalk final-answer box.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Master Chalkboard Classroom.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - MASTER CHALKBOARD CLASSROOM

Concept: Collector-quality blackboard solution sheet that feels genuinely hand-rendered by an exceptional mathematics teacher.

Surface: Deep slate chalkboard surface with subtle grain, erased ghost marks, faint ruling guides and natural edge wear.

Handwriting and colour: Soft white and pale-blue chalk for derivation, violet chalk headings, green hints, muted pink emphasis and grey construction marks.

Decorative language: Small chalk stars, leafy corner curls, arrows, underlines and one restrained classroom flourish.

Composition: Large readable classroom handwriting, clean vertical sequencing, spacious steps and a double-chalk final box.

Premium-quality test: Authentic chalk dust, pressure variation, smudged edges in moderation and no flat digital black background.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.

- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.11 Nordic Frost Minimalist

Photo needed

ChatGPT (image) / Gemini

What you get: Five impossibly clean, calming solution pages that look like a luxury Scandinavian study journal.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Nordic Frost Minimalist.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - NORDIC FROST MINIMALIST

Concept: A serene Scandinavian-Muji solution page where vast calm white space, hairline rules, and a single cool accent let the mathematics breathe like a quiet winter morning.

Surface: Soft warm-white uncoated paper (a barely-there oat tint), faintly textured like premium Muji notebook stock, with generous 1.4in margins on all sides and clean unembellished page edges; the only structure is a single thin slate-grey baseline grid so faint it reads as atmosphere, never as lines. Handwriting and colour: Main work in calm dark-slate ink (#2E3338), a relaxed neat upright hand with airy spacing; headings in a slightly heavier slate; side-notes and reasoning in a muted dove-grey; emphasis (final answers, key pivots) in one disciplined cool accent only - a soft Nordic fjord-blue (#5B7A99); optional pencil for scaffolding in the lightest silver-grey.

Decorative language: Almost nothing - that is the point: a single small filled circle as a step bullet, one slim horizontal hairline (60 percent width, centred) as the only divider, and a tiny corner registration tick; absolutely no ornament, no shading blocks, no doodles.

Composition: One generous single column with a narrow left gutter for step numbers; each of the five methods opens with a short underlined header, then numbered steps with one full blank line between them so the eye glides; final answer sits alone in a slim hairline box with wide breathing room.

Premium-quality test: Museum-grade comes from restraint and ratio: huge margins, perfect alignment,

exactly one accent colour, and silence around every line. Avoid clutter, busy borders, multiple colours, heavy boxes, or any decorative noise - if it looks crowded it has failed.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.12 Bauhaus Geometric Primer

Photo needed

ChatGPT (image) / Gemini

What you get: Five striking Bauhaus-grid solution pages that feel like collectible design posters of the maths.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Bauhaus Geometric Primer.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - BAUHAUS GEOMETRIC PRIMER

Concept: A bold De Stijl / Bauhaus solution sheet where mathematics is organised on a strict geometric grid with primary-colour blocks, confident black rules, and pure functional clarity.

Surface: Bright neutral cream art-board, smooth and matte, framed by a thick charcoal-black keyline border with squared corners and 1in margins; a subtle visible modular grid (light grey) divides the page into clean rectangular fields.

Handwriting and colour: Main work in solid jet-black, a tidy geometric upright hand sitting on the grid; headings reversed-out inside small primary-colour banners (Bauhaus red #C8102E, blue #005EB8, yellow #FFC72C); notes in mid-grey; emphasis underlined in a single primary colour; pencil construction lines in pale grey kept perfectly straight.

Decorative language: Restrained Bauhaus motifs only - a small solid red circle, blue square and yellow triangle used as the three step-marker shapes; a divider made of one thick black bar with a single coloured block at its left end; no gradients, no texture, no extra shapes.

Composition: A disciplined two-field grid: a slim left column carries coloured step-shapes and method labels, the wide right field carries the working; each method occupies one clear horizontal band separated by the black-bar divider, so the page reads like a Bauhaus poster turned into a worksheet.

Premium-quality test: It looks museum-grade through perfect grid alignment, true flat primary colours,

and fearless black structure. Avoid muddy mixed colours, tilted text, more than three accent colours, drop shadows, or any hand-wobble that breaks the geometric discipline.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.13 Engineer Field Notebook

Photo needed

ChatGPT (image) / Gemini

What you get: Five rigorous engineer's-logbook solution pages that look professionally calculated and checked.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Engineer Field Notebook.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

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 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

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MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - ENGINEER FIELD NOTEBOOK

Concept: A mid-century engineering field-book aesthetic on a precise green engineer's grid, where each method reads like a meticulous calculation entry signed off in a working notebook.

Surface: Pale sage engineer's quadrille paper with a fine green 5mm grid and slightly heavier major lines every centimetre; a rugged kraft-tan margin strip down the left, a red vertical margin rule, and softly worn page edges with a faint coffee-ring ghost in one corner for authenticity.

Handwriting and colour: Main work in dark navy-blue technical ink, a crisp draughtsman's print hand aligned to the grid; headings in bold black block capitals; running notes in pencil-grey; emphasis and final results boxed in engineer's red; light-blue construction marks for setup.

Decorative language: Functional notebook motifs only - a small CALC. / CHECK. stamp-style label per method, a tidy units-and-given key, and a divider made of a dashed grid-aligned rule with a small red node at centre; optional faint drafting triangle silhouette in a corner, nothing whimsical.

Composition: Strict grid-snapped layout: a left data column (given / find / units) and a right working column; every step numbered like a logbook entry, intermediate results underlined, and a ruled CHECK row at the foot of each method confirming the answer two ways.

Premium-quality test: It earns its quality from authentic technical precision - everything snapped to the grid, ruler-straight underlines, real engineering conventions. Avoid messy freehand, off-grid drift,

cartoonish stamps, neon colours, or fake grime that looks like a filter rather than honest wear.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.14 Kraft and Chalk-White

Photo needed

ChatGPT (image) / Gemini

What you get: Five warm, tactile kraft-and-white solution cards that feel handcrafted and inviting.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Kraft and Chalk-White.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
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 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - KRAFT AND CHALK-WHITE

Concept: A warm kraft-paper page worked entirely in white and pale chalk inks, like an artisan's recipe card for solving the problem - cosy, tactile and quietly luxurious.

Surface: Rich recycled kraft-brown paper with visible fibre flecks and a soft matte grain, deckled torn edges, and a faint darker vignette at the border; no printed grid, just honest paper warmth with 1.2in margins.

Handwriting and colour: Main work in opaque chalk-white ink, a friendly rounded hand that sits beautifully on the brown; headings in warm cream; side-notes in a soft dusty-sage; emphasis in a muted terracotta-clay for final answers and key results; light pencil-style construction in pale taupe.

Decorative language: Hand-craft motifs in restraint - small white botanical sprig or single wheat stalk at a method header, tiny hand-drawn corner brackets, and a divider of three small white dots flanked by short tapered strokes; warm and handmade, never busy.

Composition: A relaxed single column with comfortable line spacing; each method introduced by a small cream label tab, steps flowing naturally with the occasional indented sub-note in sage; the final answer enclosed in a hand-drawn white rounded frame with a soft terracotta tick beside it.

Premium-quality test: Museum quality comes from the convincing chalk-on-kraft contrast and tasteful warmth - opaque clean white strokes, real paper texture, only one earthy accent. Avoid grey muddy

'white', over-decoration, rainbow colours, or a scrapbook feel that cheapens it.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.15 Topographic Survey Chart

Photo needed

ChatGPT (image) / Gemini

What you get: Five elegant topographic-chart solution pages that turn each method into beautifully surveyed terrain.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Topographic Survey Chart.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or

"Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - TOPOGRAPHIC SURVEY CHART

Concept: A refined topographic-map aesthetic where the mathematics is laid over faint contour lines and survey markings, as if each method were charted terrain to be navigated.

Surface: Aged warm-ivory map paper printed with very pale sepia contour lines and a soft graticule of latitude/longitude tick marks; a thin double survey border, a small compass-rose watermark in one corner, and gently foxed edges - all subtle enough to never compete with the writing.

Handwriting and colour: Main work in deep sepia-brown ink, a confident cartographer's hand; headings in dark forest-green like map labels; notes in muted ochre; emphasis (final answers, key turning points) in a survey-red used sparingly as 'marked locations'; light contour-grey for setup lines.

Decorative language: Map motifs used with discipline - a tiny benchmark triangle or elevation dot beside each major step, a small north-arrow at the top of each method, and a divider styled as a dotted survey route line with a single red waypoint; a slim legend strip may name the method's 'route'.

Composition: A single mapped column with a narrow left margin of elevation-style step markers; each method opens with a header band like a map title block (method name, 'scale', key idea), the working charted downward, and the answer pinned in a small framed 'destination' box.

Premium-quality test: It looks museum-grade when the contour underlay stays whisper-faint and the maths

stays crisp on top - elegant cartographic restraint, one red accent only. Avoid loud map clutter, contours dark enough to fight the text, neon, or a video-game treasure-map vibe.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.16 Art Deco Gold Geometry

Photo needed

ChatGPT (image) / Gemini

What you get: Five opulent Art Deco solution pages that look like luxury hotel stationery for mathematics.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Art Deco Gold Geometry.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or "Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - ART DECO GOLD GEOMETRY

Concept: A Gatsby-era Art Deco solution page with symmetrical fan motifs, fine gold linework and a deep ink ground, where each method is presented like a piece of luxury stationery.

Surface: Deep ink-teal or near-black charcoal card with a smooth lacquered matte finish, framed by a slim stepped Art Deco border in thin metallic gold; symmetrical sunburst fan corners and 1in margins; the surface is rich and even, never glossy-cheap.

Handwriting and colour: Main work in warm ivory ink, an elegant slightly tall hand; headings in metallic gold small-caps; notes in soft champagne-grey; emphasis and final answers in bright gold; faint pewter-grey for construction strokes.

Decorative language: Signature Deco motifs in moderation - a small stepped chevron or fan ornament at each method header, slender twin gold rules framing key results, and a divider of a central gold lozenge flanked by tapering double lines; strictly symmetrical, never cluttered.

Composition: A centred, symmetrical layout with a refined header cartouche naming the method; working presented in a single graceful column with generous leading; the final answer crowned in a small stepped gold frame at the foot, balanced left-right like a theatre marquee.

Premium-quality test: Luxury reads through thin precise gold lines, true symmetry and a deep even ground - jewellery-box restraint. Avoid thick gaudy gold fills, asymmetry, more than the ivory-and-gold

palette, glare/bevel effects, or so much ornament that the maths drowns.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.17 Pastel Risograph Terrazzo

Photo needed

ChatGPT (image) / Gemini

What you get: Five bright, contemporary risograph-style solution pages that feel like a modern design studio made them.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Pastel Risograph Terrazzo.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or

"Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - PASTEL RISOGRAPH TERRAZZO

Concept: A cheerful contemporary risograph-print look in soft pastel ink layers with playful terrazzo speckles, keeping the maths bright, friendly and gallery-modern.

Surface: Soft off-white recycled stock with the gentle grain and faint mis-register charm of a riso print; a light scatter of tiny terrazzo flecks (pale mint, peach, lilac, butter) confined to the wide margins only, leaving a clean writing field with 1.2in margins and tidy squared edges.

Handwriting and colour: Main work in soft charcoal-plum risograph ink, a friendly modern hand; headings in a flat pastel-coral overprint; notes in dusty-teal; emphasis in a warm mustard-yellow for final answers and key steps; light pastel-blue for construction marks - all in flat matte 'riso' tones.

Decorative language: Modern flat motifs in restraint - one small pastel shape (soft arch, blob or half-circle) as a method marker, a couple of terrazzo flecks as accents, and a divider of a flat pastel rounded bar with a single contrasting dot; clean and contemporary, not childish.

Composition: A relaxed single column with airy spacing and a slim left ribbon of pastel method tabs; each method headed by a flat coloured pill label, steps clearly numbered, and the final answer set inside a soft pastel rounded panel with a friendly check mark.

Premium-quality test: It looks gallery-modern when the pastels stay flat, muted and well-balanced with

crisp dark maths on top - design-studio polish. Avoid neon brights, gradients, heavy speckle over the working, more than three pastels at once, or a cute-sticker look that reads as cheap.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.

1.18 Stained-Glass Cathedral Light

Photo needed

ChatGPT (image) / Gemini

What you get: Five luminous stained-glass solution panels where each method glows like cathedral light around the maths.

How to use: Attach a clear photo of one maths problem, paste this prompt, then if needed ask for "Page 2", "Page 3" ... one at a time.

COPY EVERYTHING BELOW

ROLE

Act as an expert mathematics educator, proof-checker, handwriting art director and premium editorial illustrator.

TASK

Transform the mathematics problem in the attached image into a coordinated five-image handwritten solution collection in the style: Stained-Glass Cathedral Light.

USE THE ATTACHED MATHEMATICS-PROBLEM IMAGE AS THE SOLE SOURCE OF THE QUESTION.

(No image handy? Then add this line: "Here is the question instead: [PASTE THE FULL QUESTION HERE]").)

NON-NEGOTIABLE OUTPUT CONTRACT

1. Produce EXACTLY FIVE SEPARATE portrait images/files.
2. These are five independent full-page outputs, not five panels on one canvas.
3. Never create a collage, contact sheet, grid, montage, carousel sheet, split canvas or combined poster.
4. Each image must show the SAME complete question at the top.
5. Each image must contain ONLY ONE solution method:
 - Image 1: Method 1 only
 - Image 2: Method 2 only
 - Image 3: Method 3 only
 - Image 4: Method 4 only
 - Image 5: Method 5 only
6. Never place two methods on one image. Never continue one method onto another image.
7. Return the five outputs as five distinct image attachments/files.
8. Before finalising, count the files: the correct count is 5.

CRITICAL SINGLE-IMAGE FALLBACK

If your interface can generate only one image in a single response, DO NOT compress all five pages into that image. Generate PAGE 1 ONLY as one complete portrait image. Then, in the next generation, create PAGE 2 ONLY; continue separately until PAGE 5. Preserve the same paper family, handwriting personality and signature across all five generations. Under no circumstances solve the limitation by making a collage.

MATHEMATICAL RECONSTRUCTION AND VERIFICATION

- Read the attached image carefully and transcribe the exact selected problem, including question number, symbols, powers, roots, factorials, limits, inequalities, diagrams, labels, units, answer choices and restrictions.
- When handwritten work is visible, treat it only as reference. Independently verify it and silently correct errors.
- If several questions appear, select the clearly marked or most prominent one and do not mix problems.
- Solve the problem internally first. Verify the final result by at least two checks.
- Choose five genuinely different mathematical viewpoints. Cosmetic rearrangements do not count as different methods.
- If five standard methods are unavailable, use valid alternatives such as direct derivation, structural identity, graphical/geometric interpretation, theorem-based route, reverse verification, invariant, generalisation or computational pattern followed by proof.
- Never invent a theorem, assumption, value or diagram detail.
- Every page must reach the same verified final answer.

REQUIRED PAGE STRUCTURE - REPEAT ON ALL FIVE IMAGES

A. Top question area

- At top left write the exact question number and instruction, such as "Q1. Evaluate", "Q1. Solve" or

"Q1. Prove that".

- Reproduce the full question clearly. Break long expressions across balanced lines without changing meaning.
- Add a thin hand-drawn divider below the question.

B. Method heading

- Write "Method [page number] : [precise method name]".
- Use a large handwritten title and an elegant underline.

C. One complete method only

- Use circled handwritten step numbers (1), (2), (3) ...
- Show all essential reasoning, identities, conditions and intermediate calculations.
- Keep equations aligned and readable with comfortable vertical spacing.
- Add only mathematically useful mini-sketches, arrows, braces, notes, tables or callout bubbles.

D. Final answer

- At the lower part of each page, draw a centred double-line handwritten box.
- Inside it, write the exact verified final answer. Include units, solution set, equality case or correct option when applicable.
- The boxed answer must be identical on all five images.

E. Signature

- Bottom right of every image must contain the elegant handwritten signature "Indrajeet Yadav" with a natural underline flourish.

HANDWRITING REALISM

- Everything must look handwritten, including headings, mathematics, labels and signature.
- Use one consistent master-teacher handwriting personality across all five images.
- Use natural stroke-pressure variation, slight baseline irregularity and realistic pen/chalk/pencil interaction with the surface.
- Mathematics must remain fully legible. Do not simulate handwriting by using a rigid typed script font.

COMPOSITION AND LEGIBILITY

- Portrait full-page composition, approximately A4 ratio.
- Keep safe margins so no equation, diagram, answer box or signature is cropped.
- Use generous spacing. Do not shrink the writing to force excessive content onto the page.
- Keep decoration in empty corners or margins. Mathematical clarity has priority.
- No watermarks, logos, extra names, motivational quotations or unrelated objects.

FINAL PRE-OUTPUT CHECK

- File count = 5.
- Each file is portrait and independently readable.
- Same full question on all five.
- One method per file only.
- Five methods are genuinely different.
- Same verified final answer on all five.
- "Indrajeet Yadav" appears at bottom right of every file.
- No collage, no grid, no combined canvas, no cropped content, no typed-poster look.

STYLE DIRECTION - STAINED-GLASS CATHEDRAL LIGHT

Concept: A reverent stained-glass aesthetic where slim lead-line frames and jewel-toned light surround the mathematics, as if each method were illuminated through a cathedral window.

Surface: Deep slate-charcoal ground that reads like stone, with each method's working sitting within a luminous frosted-glass panel of softly graded jewel light (sapphire, ruby, emerald, amber) bounded by thin dark 'lead came' lines; a slim leaded border, gentle inner glow, 1in margins, no glare.

Handwriting and colour: Main work in clean warm-ivory ink on the lit glass, a graceful upright hand kept fully legible; headings in luminous gold-leaf; notes in pale stone-grey; emphasis and final answers in a single deep jewel tone matching that method's glass; faint silver for construction lines.

Decorative language: Cathedral motifs in restraint - a small leaded rosette or trefoil at each header, thin came lines used only to frame panels (never across the working), and a divider of a slender lead bar with a single jewel cabochon at centre; dignified, never busy.

Composition: Each of the five methods is its own tall illuminated window-panel in a jewel hue, stacked or arranged in a balanced row; a small leaded header cartouche names the method, the working glows clearly within, and the final answer rests in a small framed quatrefoil at the foot of the panel.

Premium-quality test: It becomes museum-grade through luminous yet soft glass tints, crisp ivory text

that stays perfectly readable, and thin elegant lead lines - reverent and clean. Avoid garish saturated glass, text laid over busy patterns, thick clumsy leading, or glow so strong the maths is lost.

STYLE-SPECIFIC NEGATIVE RULES

- Do not substitute a generic white worksheet background.
- Do not use flat digital gradients, stock textures or clip-art decoration.
- Do not make all strokes perfectly uniform.
- Do not let visual styling reduce equation accuracy or readability.
- Do not repeat identical corner decoration on every page; vary it subtly while preserving the same visual family.

FINAL COMMAND

Generate five separate portrait image files under the output contract above. If only one image can be generated at a time, generate Page 1 only now and keep Pages 2-5 for separate subsequent generations. Never combine them.

Created with the prompt collection by Indrajeet Yadav.



One clean, exam-ready worked solution to any problem you photograph or type.

2.1 Board Stepwise Solution (Marks Per Step)

[Text only](#)[Any AI chat](#)

What you get: An exam-ready stepwise solution with marks shown beside each step.

How to use: Type or paste the question, fill the blanks, and paste this whole block into the chat.

COPY EVERYTHING BELOW

Act as a senior [BOARD or EXAM e.g. CBSE / ICSE / State Board] mathematics examiner and answer-key writer for [CLASS/GRADE].

TASK: Produce ONE perfect, exam-ready model solution to the question below, written exactly the way a top-scoring student should write it in the board exam, with marks shown for each step.

INPUT FROM ME:

- Question: [PASTE THE QUESTION HERE]
- Topic/Chapter: [TOPIC]
- Total marks for this question: [TOTAL MARKS]

BEFORE YOU WRITE: Solve the problem fully in your own head first. Then verify the final answer by at least two independent methods (for example, re-substitute the answer back into the original equation AND re-derive it a second way). Do not show me this internal checking - only use it to guarantee correctness. Never invent formulas, theorems, or values; use only standard, correct mathematics.

OUTPUT REQUIREMENTS:

1. Restate the question in one line.
2. Write the solution as numbered steps. After each step, in square brackets, show the marks that step earns, like [1 mark] or [1/2 mark], and the steps must add up to exactly [TOTAL MARKS].
3. Use the exact phrasing and key words that examiners reward (for example: 'Given', 'To find', 'Formula used', 'Substituting the values', 'Hence proved', 'Therefore'). State every formula clearly before using it.
4. Keep all mathematics fully legible. Use clean LaTeX if the chat supports it, otherwise neat plain-text math (for example x^2 , $\sqrt{3}$, π). Align equations so each step is easy to follow.
5. Box or clearly mark the FINAL ANSWER with correct units.
6. Add a short 'Examiner notes' section (2-3 bullets) saying where students usually lose marks on this question and how to avoid it.

QUALITY BARS: The solution must be 100 percent correct, complete, and copyable directly into an answer key. No skipped steps, no hand-waving. If the question is ambiguous, state your assumption in one line and proceed.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.2 JEE Fastest-Method Solution

Text only

Any AI chat

What you get: The quickest exam-day method with shortcuts and time saved.

How to use: Paste the JEE-style question, set the topic, and send this block to get the fastest route to the answer.

COPY EVERYTHING BELOW

Act as a top JEE Main and JEE Advanced mathematics coach who is famous for teaching the fastest, most exam-efficient methods.

TASK: Give ONE clean solution to the problem below using the SHORTEST correct method that a strong student can finish under time pressure, then briefly show the time-saving idea.

INPUT FROM ME:

- Question: [PASTE THE QUESTION HERE]
- Topic: [TOPIC]
- Target solving time: [TIME e.g. 90 seconds]

BEFORE YOU WRITE: Solve the problem internally first. Then verify the final answer by at least two independent checks (for example, plug the answer back in AND test with a clever special value or limiting case). Use this checking only to be sure you are correct; do not show the rough work. Never invent identities, formulas, or values - use only standard, verified mathematics.

OUTPUT REQUIREMENTS:

1. 'Fastest method' - the main solution in tight numbered steps, each line short and exam-realistic.
2. 'Why this is fast' - 2-4 bullets naming the shortcut, trick, or substitution used (for example: special values, symmetry, options-checking, dimensional reasoning, standard result) and roughly how many seconds it saves versus the long method.
3. 'Standard (safe) method' - one short paragraph or 3-4 steps showing the textbook method, so I can teach both.
4. Clearly state the FINAL ANSWER (and the correct option if options are given).
5. Keep all math fully legible in clean LaTeX or neat plain text (x^2 , $\sqrt{x}$, integral, etc.). Define every symbol the first time.

QUALITY BARS: Absolute correctness first, speed second. The shortcut must be genuinely valid, not a guess that only happens to work. If a shortcut has a hidden trap, warn me in one line. Keep the whole answer compact and classroom-ready.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.3 NEET / MCQ Smart-Elimination Solution

Text only

Any AI chat

What you get: The correct option plus a clean elimination strategy for the rest.

How to use: Paste the MCQ with all four options and send this block to see how to eliminate the wrong ones fast.

COPY EVERYTHING BELOW

Act as an expert NEET and competitive MCQ mathematics mentor who teaches smart elimination, not just brute calculation.

TASK: For the multiple-choice question below, give ONE clean solution that (a) finds the correct option by the fastest reliable route and (b) shows how to eliminate each wrong option.

INPUT FROM ME:

- Question with ALL options: [PASTE THE QUESTION HERE - include options (A) (B) (C) (D)]
- Topic: [TOPIC]

BEFORE YOU WRITE: Solve the problem internally and confirm the correct option by at least two independent checks (for example, direct calculation AND back-substitution or estimation). Use this only to be certain; do not dump rough work. Never invent values or formulas - use only standard, correct mathematics. If two options look close, double-check the arithmetic before deciding.

OUTPUT REQUIREMENTS:

1. 'Correct option' - state it in one line up front (for example: Answer: (C)).
2. 'Direct method' - the short, clean steps that lead to the correct option.
3. 'Elimination table' - go through options (A), (B), (C), (D) and in one short line each, say why it is right or exactly why it is wrong (units off, wrong sign, fails a test value, magnitude impossible, etc.).
4. 'Exam tip' - one line on the smartest elimination shortcut for this type of question (estimation, dimensions, special case, plugging options).
5. Keep all math fully legible in clean LaTeX or neat plain text. Define symbols as needed.

QUALITY BARS: The chosen option must be provably correct, and every elimination reason must be mathematically true, not vague. Keep it tight and ready to read out in class.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.4 Olympiad-Level Rigorous Proof

Text only

Gemini / ChatGPT

What you get: A fully rigorous, gap-free proof in proper mathematical style.

How to use: Paste the proof problem and send this block to get a complete, rigorous olympiad-style write-up.

COPY EVERYTHING BELOW

Act as an olympiad mathematics trainer and journal-style proof writer (IMO / RMO / INMO standard).

TASK: Write ONE complete, fully rigorous proof of the statement below, with every logical step justified.

INPUT FROM ME:

- Statement to prove: [PASTE THE QUESTION HERE]
- Area (algebra / number theory / geometry / combinatorics / inequalities): [TOPIC]

BEFORE YOU WRITE: Work out the full proof internally first. Then stress-test it: check every claim, every 'clearly', and every edge case, and verify any key inequality or identity by an independent route or a test value. Use this checking only to be certain the proof is airtight; never present a proof with a gap. Never invent theorems - cite only standard, named, correct results (AM-GM, Cauchy-Schwarz, pigeonhole, induction, etc.) and state them precisely when used.

OUTPUT REQUIREMENTS:

1. 'Claim' - restate exactly what is to be proved.
2. 'Strategy' - 2-3 lines naming the proof technique and the key idea.
3. 'Proof' - the rigorous argument in clearly separated steps. Justify each implication; do not skip non-trivial steps. Handle all cases and boundary conditions explicitly. End with a clear concluding line and the symbol QED (or 'Hence proved').
4. 'Where rigor matters' - 2-3 bullets flagging the steps a careless solver would skip or get wrong.
5. Use clean LaTeX if supported, otherwise precise plain-text math. Every symbol must be defined; every quantifier (for all, there exists) must be explicit.

QUALITY BARS: No logical gaps, no circular reasoning, no hidden assumptions. If the statement is false or needs an extra condition, say so clearly and prove the corrected version. The proof must read like a clean competition solution.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.5 Explain Like I Am 14 (Gentle Solution)

Text only

Any AI chat

What you get: A warm, friendly, jargon-free solution a nervous learner can follow.

How to use: Paste the problem and send this block to get a kind, plain-language walkthrough for a struggling student.

COPY EVERYTHING BELOW

Act as a warm, patient mathematics tutor explaining to a slightly nervous 14-year-old who is scared of maths.

TASK: Solve the problem below in ONE friendly, gentle, fully correct walkthrough that a beginner can actually follow and feel good about.

INPUT FROM ME:

- Question: [PASTE THE QUESTION HERE]
- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]

BEFORE YOU WRITE: Solve the problem correctly in your head first and verify the final answer two independent ways (for example, re-do it differently AND check the answer makes sense). Use this only to be sure you are right; never simplify so much that you become wrong. Never invent rules or formulas - use only standard, correct mathematics, just explained kindly.

OUTPUT REQUIREMENTS:

1. 'What the question is really asking' - 1-2 plain sentences, no jargon.
2. 'Let us solve it together' - small, calm steps. After each step write one short line in plain English explaining WHY we did it (like talking to a friend). Use a tiny everyday analogy where it helps.
3. Keep numbers and symbols simple and fully legible (write x squared as x^2 , square root as $\sqrt{}$). Introduce any new word gently the first time it appears.
4. 'The answer' - state it clearly, then one encouraging line ('See, you can do this!').
5. 'One thing to remember' - a single friendly takeaway tip for next time.

QUALITY BARS: Encouraging tone, zero condescension, zero unexplained jargon, and 100 percent correct mathematics. No step should feel like a leap. Keep it short enough that a tired student will still read it.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.6 Two-Column Proof (Statement | Reason)

Text only

Any AI chat

What you get: A neat statement-and-reason table proof, board-presentation ready.

How to use: Paste the geometry or algebra proof and send this block to get a clean two-column proof table.

COPY EVERYTHING BELOW

Act as a geometry and proofs teacher who writes flawless two-column proofs for the board exam and the blackboard.

TASK: Prove the statement below as ONE clean two-column proof, with Statements on the left and Reasons on the right.

INPUT FROM ME:

- To prove: [PASTE THE QUESTION HERE]
- Given / figure description (if any): [TYPE GIVEN INFO OR DESCRIBE THE FIGURE]
- Topic: [TOPIC]
- Class/Board: [CLASS/GRADE] / [BOARD or EXAM]

BEFORE YOU WRITE: Construct the full proof internally first, then verify it by re-checking that every reason genuinely justifies its statement and that no step is skipped or assumed. Use this only to ensure correctness. Never invent theorems or postulates - cite only standard, correctly named results (SAS, ASA, angle sum property, alternate angles, congruence/similarity rules, etc.).

OUTPUT REQUIREMENTS:

1. 'Given' and 'To Prove' clearly stated at the top.
2. 'Construction' (only if needed), in one or two lines.
3. A clean two-column table: column 1 = Statement, column 2 = Reason. Number the rows. Every statement must have a precise, named reason. The final row must reach exactly what was to be proved, ending with 'Hence proved'.
4. Keep all math fully legible (use clean LaTeX or neat plain text such as angle ABC, $AB = CD$, triangle ABC). If a figure is implied, briefly describe the labelling you assumed in one line.
5. Add one line under the table: 'Common slip' - the single reason students most often get wrong here.

QUALITY BARS: Every reason must be valid and standard; no row may rely on an unstated assumption. The table must be tidy enough to copy straight to the board or an answer key.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.7 Solution With Common-Mistake Flags

Text only

Any AI chat

What you get: A correct solution with the typical student trap flagged at each step.

How to use: Paste the problem and send this block to get the full solution plus warnings on where students go wrong.

COPY EVERYTHING BELOW

Act as an experienced mathematics teacher who has graded thousands of scripts and knows exactly where students slip up.

TASK: Solve the problem below in ONE correct, complete solution, and at each step flag the common mistake students make there.

INPUT FROM ME:

- Question: [PASTE THE QUESTION HERE]
- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]

BEFORE YOU WRITE: Solve the problem internally and verify the final answer by at least two independent methods. Use this only to be certain you are correct; never let a 'common mistake' warning slip into the actual solution. Never invent formulas or values - use only standard, correct mathematics.

OUTPUT REQUIREMENTS:

1. The correct solution in numbered steps, fully legible in clean LaTeX or neat plain text (x^2 , $\sqrt{x}$, etc.). State each formula before using it.
2. After each step, add a line tagged 'Trap:' describing the single most common error students make at that exact step (sign errors, dropping a case, wrong formula, forgetting domain/units, cancelling incorrectly, etc.) and how to avoid it.
3. Clearly mark the FINAL ANSWER with units.
4. 'Top 3 mistakes overall' - a short ranked list of the worst, most marks-costing errors for this question type.

QUALITY BARS: The main solution must be 100 percent correct and self-contained; the traps must be real, specific errors, not generic advice like 'be careful'. Keep it classroom-usable and not bloated.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.8 Solution With Self-Verification Cross-Check

Text only

Any AI chat

What you get: A solution plus an explicit cross-check proving the answer is right.

How to use: Paste the problem and send this block to get the answer with a visible double-check students can copy.

COPY EVERYTHING BELOW

Act as a meticulous mathematics teacher who never submits an answer without proving it is correct.

TASK: Solve the problem below in ONE clean solution, then show an explicit self-verification or cross-check that confirms the answer.

INPUT FROM ME:

- Question: [PASTE THE QUESTION HERE]
- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]

BEFORE YOU WRITE: Solve the problem internally first and confirm correctness. Never invent formulas, theorems, or values - use only standard, correct mathematics. The visible cross-check below must be genuine, not decorative.

OUTPUT REQUIREMENTS:

1. 'Solution' - the main working in clear numbered steps, fully legible (clean LaTeX or neat plain text such as x^2 , $\sqrt{x}$, dy/dx). State every formula before use. Mark the FINAL ANSWER with units.
2. 'Cross-check 1' - verify the answer by an INDEPENDENT route (for example: substitute the answer back into the original equation, differentiate the integral result, or re-derive a second way) and show it gives the same result.
3. 'Cross-check 2' - a second different sanity test: a dimensional/units check, an order-of-magnitude estimate, a limiting/special case, or a symmetry argument - whichever fits this problem.
4. 'Verdict' - one line: 'Answer confirmed by both checks' (or, if a check fails, fix the solution and re-verify before finalising).

QUALITY BARS: Both cross-checks must be real and must actually pass. If they disagree with the main solution, you must correct the solution and re-run the checks - never present a contradiction. Keep everything tidy and copyable.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.9 Mark a Student's Handwritten Attempt

Photo needed

ChatGPT (image)

What you get: Marks for the student's attempt plus a fully corrected solution.

How to use: Snap a clear photo of the student's handwritten work, attach it, and send this block to get it graded and corrected.

COPY EVERYTHING BELOW

Act as a fair, encouraging mathematics examiner who marks handwritten student work and then shows the perfect solution.

TASK: Read the student's handwritten attempt in the attached photo, mark it honestly, and provide the corrected full solution.

INPUT FROM ME:

- [ATTACH A PHOTO OF THE STUDENT'S HANDWRITTEN ATTEMPT]
- The original question (type it so there is no confusion): [PASTE THE QUESTION HERE]
- Topic: [TOPIC]
- Total marks: [TOTAL MARKS]
- Class/Board: [CLASS/GRADE] / [BOARD or EXAM]

BEFORE YOU WRITE: First solve the question correctly yourself and verify the answer by two independent checks. Only then read and judge the student's work against your verified solution. Never invent formulas or values; use only standard, correct mathematics. If any handwriting is unclear, say exactly what you could not read instead of guessing.

OUTPUT REQUIREMENTS:

1. 'What the student did' - a short, faithful summary of their approach from the photo.
2. 'Step-by-step marking' - go through the attempt and award marks step by step in square brackets (for example [1/2 mark awarded], [0 - sign error here]), with a one-line reason for each award or deduction. Be fair: give method marks even when the final answer is wrong.
3. 'Marks: X out of [TOTAL MARKS]' - the honest total.
4. 'Mistakes found' - a clear list of each error and why it is wrong.
5. 'Corrected full solution' - the clean, complete, correct solution in legible steps (clean LaTeX or neat plain text), with the FINAL ANSWER marked.
6. 'One encouraging note' - a short positive, specific line to motivate the student.

QUALITY BARS: Marking must match the correct solution exactly, be consistent, and never penalise the same error twice. Tone firm but kind. If the photo is too blurry to mark reliably, say so and ask for a clearer picture.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

2.10 Shortest Elegant One-Line-Idea Solution

Text only

Gemini / ChatGPT

What you get: The single elegant insight that cracks the problem in one move.

How to use: Paste the problem and send this block to get the slickest one-key-idea solution you can show off in class.

COPY EVERYTHING BELOW

Act as a brilliant mathematician famous for finding the single elegant idea that collapses a hard problem into almost nothing.

TASK: Solve the problem below with the SHORTEST, most elegant solution possible - the kind that makes students say 'wow' - built around one key insight.

INPUT FROM ME:

- Question: [PASTE THE QUESTION HERE]
- Topic: [TOPIC]

BEFORE YOU WRITE: Solve the problem internally and confirm the slick idea is fully valid by at least two independent checks (for example, verify the result a second standard way AND test a special case). Use this only to be certain; never trade correctness for cleverness. Never invent identities or theorems - the elegant trick must be genuinely true standard mathematics.

OUTPUT REQUIREMENTS:

1. 'The key idea' - ONE sentence naming the insight that cracks it (a symmetry, a clever substitution, a known identity, a telescoping, a geometric reinterpretation, etc.).
2. 'The solution' - the shortest correct chain of steps that uses that idea, ideally just a few lines, fully legible in clean LaTeX or neat plain text. Every symbol defined.
3. 'Final answer' - stated clearly.
4. 'Why it works (1-2 lines)' - a brief justification so the elegance is not a magic trick but understood.
5. 'Longer fallback (optional, 2-3 lines)' - the routine method in case a student does not see the trick.

QUALITY BARS: Maximum elegance, zero loss of rigor. The short solution must be airtight, not a lucky guess. Keep it genuinely short - no padding. If no truly short route exists, say so honestly and give the cleanest available method.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.



Multiple Methods (Notes)

10 prompts

Several genuinely different ways to solve the same problem - perfect for class notes.

3.1 3 Compact Methods Side by Side

Text only

Any AI chat

What you get: One problem, three short solutions in parallel columns ready to copy to the board.

How to use: Paste your question, set the class and board, and you get three tidy methods you can write up as notes in minutes.

COPY EVERYTHING BELOW

Act as an expert mathematics teacher who writes crisp blackboard-ready class notes.

TASK: Solve ONE problem using exactly THREE genuinely different methods, presented compactly so they fit side by side in student notes.

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the problem fully in your own working area first. The three methods MUST be truly different in idea (for example: direct formula, substitution, and a clever observation - NOT the same steps reworded). Confirm all three give the SAME final answer. If any method does not match, find your mistake and fix it before writing.

OUTPUT REQUIREMENTS:

1. A one-line restatement of the problem and the final answer in a box at the top, clearly labelled 'Final Answer'.
2. A three-column layout titled 'Method 1', 'Method 2', 'Method 3', each with: a 3-5 word name for the method, the key idea in one line, then numbered short steps (keep each method under 8 steps).
3. Below the columns, a one-line note 'Quickest in exam: Method ___' with a short reason.
4. Keep every step mathematically legible: use clean LaTeX where helpful (for example $\frac{a}{b}$, $\sqrt{x}$, x^2) or clear plain-text math; never write ambiguous math.

QUALITY BARS:

- Never invent formulas, theorems, or values; use only standard results appropriate for [CLASS/GRADE] and [BOARD or EXAM].
- Verify the final answer by at least TWO independent checks (for example, plug the answer back in, and re-derive by a different route) and state in one line how you verified.
- Use plain ASCII only: straight quotes ' and ', normal hyphens -, no fancy symbols.
- Keep language simple enough for a student to follow without the teacher present.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.2 5 Different Methods as Clean Notes

Text only

Any AI chat

What you get: Five genuinely distinct full solutions formatted as ready-to-print markdown class notes.

How to use: Drop in any problem and get a complete five-method handout you can paste straight into a document and print.

COPY EVERYTHING BELOW

Act as a senior mathematics author preparing a 'Five Ways to Solve It' page for class notes.

TASK: Take ONE problem and solve it FIVE genuinely different ways, written as clean, well-structured markdown notes (no images needed).

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the problem completely first in your own working area. Choose FIVE methods that rest on different ideas (examples of distinct families: direct/formula, substitution, factorisation, geometry/diagram, symmetry, induction, comparison, working backwards, parameter/variable change). Reject any method that is a rewording of another. Confirm all five land on the SAME final answer; if not, fix the error before writing.

OUTPUT REQUIREMENTS (markdown):

1. A top heading with the problem statement and a clearly boxed 'Final Answer'.
2. Five sections '## Method 1: <short name>' ... '## Method 5: <short name>'. Each section has: 'Idea' (one line), 'Steps' (clear numbered steps), and 'Why it works' (one line).
3. A short closing table with columns: Method | Core idea | Best for.
4. All mathematics in clean LaTeX (for example $\int$, $\frac{\{ \} }{ \{ \} }$, $\sum$, $x^{\{ \} }$) or unambiguous plain text. Keep every line legible.

QUALITY BARS:

- Never invent theorems, formulas, or numbers; use only results valid for [CLASS/GRADE] and [BOARD or EXAM].
- Verify the final answer with at least TWO independent checks and state them in a 'Verification' line at the end.
- Plain ASCII only: straight quotes ' and ", normal hyphens -, no special unicode.
- Write so a student revising alone can follow each method.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.3 Which Method Is Best (Comparison Table)

Text only

Any AI chat

What you get: A ranked comparison table scoring each method on speed, rigor, and exam-fit.

How to use: Give one problem and the AI rates every solution path so you can tell students exactly which one to use in the exam.

COPY EVERYTHING BELOW

Act as an exam strategist and mathematics teacher who advises students on choosing the smartest method.

TASK: Solve ONE problem several ways, then build a clear comparison table judging each method on speed, rigor, and exam-fit, ending with a firm recommendation.

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Time per question in the exam: [TIME]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the problem fully in your own working area first. Produce 3 to 5 DISTINCT methods (different core ideas, not reworded). Confirm every method reaches the SAME final answer; fix any mismatch before writing.

OUTPUT REQUIREMENTS:

1. 'Final Answer' boxed at the top.
2. For each method: a 3-5 word name and a compact set of numbered steps.
3. A comparison table with columns: Method | Speed (1-5) | Rigor (1-5) | Exam-fit for [BOARD or EXAM] (1-5) | Risk of error | One-line verdict. Use simple 1-5 scores where 5 is best.
4. A 'Teacher's Recommendation' paragraph: which method to teach first, which to use under the [TIME] limit, and which to avoid in the exam, each with a reason.
5. Math in clean LaTeX or clear plain text; keep it fully legible.

QUALITY BARS:

- Never invent formulas, theorems, or values; stay within [CLASS/GRADE] and [BOARD or EXAM] scope.
- Verify the final answer with at least TWO independent checks and state how in one line.
- Plain ASCII only: straight quotes ' and ", normal hyphens -, no fancy characters.
- Base speed and exam-fit scores on realistic student behaviour, not theory alone.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.4 Elegant Method vs Brute Force

Text only

Any AI chat

What you get: The same problem solved the long honest way and the slick clever way, side by side.

How to use: Paste a problem to show students both the safe brute-force route and the elegant trick, so they appreciate the shortcut.

COPY EVERYTHING BELOW

Act as a mathematics teacher who loves showing students the contrast between honest hard work and elegant insight.

TASK: Solve ONE problem TWO ways: first the BRUTE-FORCE method (straightforward, longer, no cleverness required), then the ELEGANT method (a clean insight, trick, or symmetry that shortens the work).

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the problem completely in your own working area first. The brute-force method must be the route an average student would take; the elegant method must rely on a genuinely different, smarter idea. Confirm BOTH give the SAME final answer; fix any error before writing.

OUTPUT REQUIREMENTS:

1. 'Final Answer' boxed at the top.
2. Section A 'Brute-Force Method': name the approach in one line, then full numbered steps, and at the end count the rough number of steps or lines.
3. Section B 'Elegant Method': name the key insight in one line ('The trick:'), then the short numbered steps, and the step/line count.
4. A short 'Why the elegant method works' note explaining the insight in plain words a student can reuse.
5. A one-line 'Time saved' comparison.
6. Math in clean LaTeX or clear plain text, fully legible throughout.

QUALITY BARS:

- Never invent theorems, formulas, or values; stay within [CLASS/GRADE] and [BOARD or EXAM].
- Verify the final answer with at least TWO independent checks and state them in one line.
- Plain ASCII only: straight quotes ' and ", normal hyphens -, no special unicode.
- The elegant method must be correct and rigorous, not just shorter.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.5 Calculator-Free vs Calculator-Allowed

Text only

Any AI chat

What you get: Two solution paths: one pure mental/by-hand, one using a calculator efficiently.

How to use: Useful for boards that ban calculators and exams that allow them - paste the problem and get both routes.

COPY EVERYTHING BELOW

Act as a mathematics teacher preparing students for both calculator-free and calculator-allowed exam settings.

TASK: Solve ONE problem TWO ways: a CALCULATOR-FREE method (only mental math, by-hand arithmetic, and standard tricks) and a CALCULATOR-ALLOWED method (use the calculator efficiently and sensibly).

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the problem fully in your own working area first. The calculator-free path must avoid any computation a student cannot reasonably do by hand; the calculator-allowed path must show which buttons or operations to use and in what order. Confirm BOTH give the SAME final answer; fix any error before writing.

OUTPUT REQUIREMENTS:

1. 'Final Answer' boxed at the top.
2. Section A 'Calculator-Free Method': numbered steps using clean by-hand techniques (estimation, factoring, known values, simplification). Flag any place students usually make arithmetic slips.
3. Section B 'Calculator-Allowed Method': numbered steps, stating the exact calculation sequence, plus a tip to avoid rounding errors and how many decimal places to keep.
4. A one-line note: 'When each method is allowed in [BOARD or EXAM]'.
5. Math in clean LaTeX or clear plain text; keep all working legible.

QUALITY BARS:

- Never invent formulas, theorems, or values; stay within [CLASS/GRADE] and [BOARD or EXAM].
- Verify the final answer with at least TWO independent checks (for example, by-hand estimate vs calculator value) and state them.
- Plain ASCII only: straight quotes ' and ", normal hyphens -, no fancy symbols.
- Be honest about rounding: state exact vs approximate answers clearly.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.6 Algebraic vs Graphical vs Numerical

Text only

Gemini / ChatGPT

What you get: One problem seen through three lenses: algebra, a graph description, and a numerical table.

How to use: Great for functions, equations, and inequalities - paste the problem to show the same answer from three viewpoints.

COPY EVERYTHING BELOW

Act as a mathematics teacher who builds deep understanding by showing the same problem from multiple viewpoints.

TASK: Solve ONE problem from THREE viewpoints: ALGEBRAIC (symbol manipulation), GRAPHICAL (describe the graph and read the answer from it), and NUMERICAL (a table of values that converges on the answer).

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the problem completely in your own working area first. Confirm all three viewpoints agree on the SAME final answer; fix any error before writing.

OUTPUT REQUIREMENTS:

1. 'Final Answer' boxed at the top.
2. Section 1 'Algebraic View': numbered steps in clean LaTeX or plain-text math.
3. Section 2 'Graphical View': describe the graph precisely in words (axes, intercepts, shape, key points, where the answer appears). Give exact coordinates of important points so a student can sketch it by hand. Do NOT rely on an image - describe it fully in text.
4. Section 3 'Numerical View': a small table of inputs and outputs that clearly points to the answer, with one line explaining the pattern.
5. A closing 'Same answer, three lenses' note linking the three views.

QUALITY BARS:

- Never invent formulas, theorems, coordinates, or values; stay within [CLASS/GRADE] and [BOARD or EXAM]. All coordinates and table values must be correct.
- Verify the final answer with at least TWO independent checks and state them.
- Plain ASCII only: straight quotes ' and ", normal hyphens -, no special unicode.
- Keep all math legible and the graph description precise enough to draw.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.7 Examiner Method vs Topper Shortcut

Text only

Any AI chat

What you get: The full marks-scheme solution next to the fast trick a topper would actually use.

How to use: Paste a problem to show the official step-by-step that earns full marks and the quick route toppers use to save time.

COPY EVERYTHING BELOW

Act as both a strict board examiner and a top-ranking student, explaining a problem from both angles.

TASK: Solve ONE problem TWO ways: the EXAMINER METHOD (the complete, official, marks-scheme-friendly solution with every step a checker expects) and the TOPPER SHORTCUT (the fast, smart route a top student uses to get the answer quickly).

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Total marks for this question: [TOTAL MARKS]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the problem fully in your own working area first. Confirm BOTH methods give the SAME final answer; fix any error before writing.

OUTPUT REQUIREMENTS:

1. 'Final Answer' boxed at the top.
2. Section A 'Examiner Method (Full Marks)': complete numbered steps, with a suggested mark allocation in brackets next to key steps that add up to [TOTAL MARKS], and notes on what wording earns marks.
3. Section B 'Topper Shortcut': the quick route in minimal steps, with the key trick named in one line. Add a warning: where this shortcut could lose marks if written in a board answer (especially for [BOARD or EXAM]).
4. A one-line note: 'Use the shortcut to check, write the examiner method to score' (or adjust if [BOARD or EXAM] is objective/MCQ).
5. Math in clean LaTeX or clear plain text; fully legible.

QUALITY BARS:

- Never invent formulas, theorems, values, or marks-scheme rules; stay within [CLASS/GRADE] and [BOARD or EXAM].
- Verify the final answer with at least TWO independent checks and state them.
- Plain ASCII only: straight quotes ' and ", normal hyphens -, no fancy characters.
- Mark allocations must be plausible and sum to [TOTAL MARKS].

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.8 Method That Generalises to a Formula

Text only

Any AI chat

What you get: A solution that turns one specific problem into a reusable general formula for the whole class.

How to use: Paste a numeric problem and the AI shows how to solve it AND derive a formula students can reuse for similar questions.

COPY EVERYTHING BELOW

Act as a mathematics teacher who turns single problems into reusable tools by generalising them.

TASK: Solve ONE specific problem, then solve it AGAIN with the numbers replaced by symbols to DERIVE a general formula, and finally show the formula reproduces the original answer.

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the specific problem fully in your own working area first. Then identify which numbers are the 'parameters' worth turning into symbols. Confirm the general formula, when the original numbers are substituted, gives the SAME final answer; fix any error before writing.

OUTPUT REQUIREMENTS:

1. 'Final Answer (specific case)' boxed at the top.
2. Section 1 'Direct Solution': numbered steps for the given numbers.
3. Section 2 'Generalise': replace the chosen numbers with symbols (state clearly which symbol stands for which quantity) and redo the steps to reach a clean GENERAL FORMULA, boxed and labelled 'General Formula'.
4. Section 3 'Check': substitute the original numbers into the formula to recover the specific answer.
5. Section 4 'Try it': two short practice questions students can solve with the new formula, with answers.
6. Math in clean LaTeX or clear plain text; fully legible.

QUALITY BARS:

- Never invent theorems or formulas without deriving them; stay within [CLASS/GRADE] and [BOARD or EXAM].
- Verify the formula with at least TWO independent checks (the original case plus one new set of numbers) and state them.
- Plain ASCII only: straight quotes ' and ", normal hyphens -, no special unicode.
- State any conditions or restrictions on the formula's variables.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.9 Step-Up Ladder: Easiest to Most Powerful

Text only

Any AI chat

What you get: Methods ordered from simplest beginner approach up to the most powerful advanced technique.

How to use: Paste a problem and get a learning ladder - start students on the easy method, then climb to the powerful one.

COPY EVERYTHING BELOW

Act as a mathematics teacher who scaffolds learning from beginner-friendly to advanced.

TASK: Solve ONE problem with a LADDER of methods, ordered from the EASIEST and most intuitive at the bottom to the MOST POWERFUL and advanced at the top. Each rung must be a genuinely different method, increasing in sophistication.

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Topic/Chapter: [TOPIC]

BEFORE YOU WRITE: Solve the problem fully in your own working area first. Pick 3 to 4 methods that clearly differ in difficulty (for example: counting/listing, then a simple formula, then an algebraic technique, then a general/elegant theorem). Confirm every rung gives the SAME final answer; fix any error before writing.

OUTPUT REQUIREMENTS:

1. 'Final Answer' boxed at the top.
2. Present the ladder as 'Rung 1 (Easiest)' up to the top rung. For each rung give: a 3-5 word name, 'Who it suits' (one line, for example beginner / average / advanced), numbered steps, and 'What you need to know first' (prerequisites).
3. A one-line bridge between consecutive rungs explaining what extra idea the next rung adds.
4. A closing note: 'Teach bottom-up; expect students to reach Rung __ by [CLASS/GRADE]'.
5. Math in clean LaTeX or clear plain text; fully legible.

QUALITY BARS:

- Never invent theorems, formulas, or values; stay within [CLASS/GRADE] and [BOARD or EXAM] (the top rung may preview a slightly higher-level idea but must be correct).
- Verify the final answer with at least TWO independent checks and state them.
- Plain ASCII only: straight quotes ' and ", normal hyphens -, no fancy symbols.
- Keep the easiest rung truly accessible to a struggling student.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

3.10 Cross-Chapter Method (Borrow a Tool)

Text only

Any AI chat

What you get: The standard solution plus a surprise solution using a tool from a totally different chapter.

How to use: Paste a problem to show the usual method and a clever one borrowed from another chapter, so students see how topics connect.

COPY EVERYTHING BELOW

Act as a mathematics teacher who shows students how chapters connect by solving a problem with a tool borrowed from elsewhere.

TASK: Solve ONE problem TWO ways: the STANDARD METHOD from its own chapter, and a CROSS-CHAPTER METHOD that uses a technique from a different chapter or branch of mathematics.

INPUT FROM ME:

- Problem: [PASTE THE QUESTION HERE]
- Class/Grade: [CLASS/GRADE]
- Board/Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- The problem's own chapter: [TOPIC]
- (Optional) Chapter to borrow from: [TOPIC]

BEFORE YOU WRITE: Solve the problem fully in your own working area first. For the cross-chapter method, choose a genuinely different tool (for example: solve a geometry problem with coordinates or vectors or trigonometry; solve an algebra problem with a graph or a counting argument; solve a series with calculus). Confirm BOTH methods give the SAME final answer; fix any error before writing.

OUTPUT REQUIREMENTS:

1. 'Final Answer' boxed at the top.
2. Section A 'Standard Method (from [TOPIC])': numbered steps.
3. Section B 'Cross-Chapter Method': state clearly which chapter's tool you are borrowing and why it applies, then numbered steps.
4. A 'Connection note' (2-3 lines) explaining the link between the two chapters so students remember it.
5. A one-line caution on whether the cross-chapter method is acceptable in [BOARD or EXAM] answers.
6. Math in clean LaTeX or clear plain text; fully legible.

QUALITY BARS:

- Never invent theorems, formulas, or values; every borrowed tool must be standard and correctly applied for [CLASS/GRADE] and [BOARD or EXAM].
- Verify the final answer with at least TWO independent checks and state them.
- Plain ASCII only: straight quotes ' and ', normal hyphens -, no special unicode.
- Make the cross-chapter link genuine, not forced.

FOOTER: End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.



One-page formula posters and revision sheets for any chapter, ready to print.

4.1 One-Page Chapter Formula Poster

Text only

Any AI chat

What you get: A single-page, classroom-ready poster of every formula in one chapter.

How to use: Type your chapter name and grade, paste the prompt, and you will get a tidy one-page poster you can print and pin up.

COPY EVERYTHING BELOW

Act as an expert mathematics teacher and exam-paper designer compiling a printable one-page formula poster.

TASK: Create a clean, complete ONE-PAGE formula poster for the chapter [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: The chapter is [TOPIC]. If you need the exact syllabus scope, assume the standard [BOARD or EXAM] syllabus for [CLASS/GRADE]. I may also paste my chapter heading list here: [PASTE SUBTOPICS OR LEAVE BLANK].

OUTPUT REQUIREMENTS:

1. A title block at the very top: chapter name, class, and board, plus 'Compiled by Indrajeet Yadav'.
2. Group formulas under 4 to 8 clearly labelled sections (use bold headings). Inside each section list every standard formula as a numbered item.
3. Write each formula in clean LaTeX, and also give a plain-text version in brackets so a teacher with no LaTeX viewer can still read it.
4. Keep it to ONE printed page worth of content - be complete but not bloated. Prefer formulas over prose.
5. For each formula add a 3 to 6 word tag saying what it is for (e.g. 'area of sector').
6. Add a short 'Key constants / standard values' strip if the chapter has any.

QUALITY BARS:

- Solve and recall internally first, then VERIFY every formula by at least two independent checks (e.g. dimensional check, a known special case, or re-derivation). Do not print any formula you cannot verify.
- Never invent theorems, formulas, or values. If something is outside the [BOARD or EXAM] scope, omit it and say so in one line at the end.
- Keep all mathematics fully legible and consistent in notation; define any symbol that is not obvious.

FORMATTING: Use a calm, scannable layout with headings, numbered lists, and clear spacing so it prints well in black and white on A4.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.2 Last-Day Quick Revision Sheet

Text only

Any AI chat

What you get: A high-yield night-before-exam recap with only the must-know formulas.

How to use: Tell it the chapters and the exam, and it gives students a short, calm recap to read the night before.

COPY EVERYTHING BELOW

Act as a senior mathematics mentor preparing a last-day-before-exam quick revision sheet for stressed students.

TASK: Build a SHORT, high-yield revision sheet covering [TOPIC or LIST OF CHAPTERS] for [CLASS/GRADE] sitting [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. The goal is fast recall in the final hours, not full learning.

INPUT FROM ME: Chapters to cover: [TOPIC or LIST OF CHAPTERS]. Exam is in [TIME e.g. 12 hours].

Difficulty focus: [DIFFICULTY e.g. board-average / top-scorer].

OUTPUT REQUIREMENTS:

1. A header block with chapter list, exam name, and 'Compiled by Indrajeet Yadav'.
2. A 'Must-Know Formulas' list - ONLY the highest-frequency, most exam-critical formulas (aim for 15 to 25 items, no more). In LaTeX plus plain text in brackets.
3. A 'Quick Traps to Remember' list of 5 to 8 one-line reminders of where students lose marks.
4. A '60-Second Recall' box: the 6 to 8 single most important results to glance at right before walking in.
5. A calm one-line confidence note for the student at the end.

QUALITY BARS:

- Internally rank formulas by exam frequency before selecting; keep only the truly high-yield ones.
- VERIFY every formula by at least two independent checks (special case, dimensional check, or re-derivation). Never invent formulas or values.
- Keep all math fully legible and the whole sheet short enough to read in under 15 minutes.

FORMATTING: Tight, scannable, A4-friendly, black-and-white printable. Use bold headings and short lines.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.3 Formula + Worked Mini-Example Pairs

Text only

Any AI chat

What you get: Each formula paired with one tiny solved example showing it in action.

How to use: Name the chapter and the AI lists every formula with a one-line solved example beside it, so students see how each is used.

COPY EVERYTHING BELOW

Act as an expert mathematics teacher creating a formula sheet where every formula is taught through one tiny worked example.

TASK: For the chapter [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT], list each key formula and pair it with one SHORT solved mini-example that uses it.

INPUT FROM ME: Chapter is [TOPIC]. Number of formulas to cover: [NUMBER e.g. 15] (or 'all standard ones'). Difficulty of examples: [DIFFICULTY].

OUTPUT REQUIREMENTS:

1. Header block with chapter, class, board, and 'Compiled by Indrajeet Yadav'.
2. For each item, present in this exact 4-line structure:
 - Formula (name + LaTeX, with plain text in brackets)
 - When to use it (one short line)
 - Mini-example (a tiny, realistic problem)
 - Solution (2 to 4 lines, ending with the final answer in bold)
3. Keep numbers small and clean so the arithmetic is easy to follow.
4. Number the items so a teacher can refer to them in class.

QUALITY BARS:

- Solve every mini-example internally FIRST, then VERIFY each final answer by at least two independent methods (e.g. substitute back, alternate approach, or estimation). Only then write it down.
- Never invent formulas, theorems, or values. Stay within the [BOARD or EXAM] scope.
- Keep all mathematics fully legible with consistent notation; show enough steps that a weak student can follow.

FORMATTING: Clean numbered blocks, plenty of white space, A4 black-and-white printable.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.4 Formula Sheet With 'When To Use' Notes

Text only

Any AI chat

What you get: Every formula tagged with the exact situation that signals you to use it.

How to use: Great for students who know the formulas but pick the wrong one - this sheet tells them when each applies.

COPY EVERYTHING BELOW

Act as an experienced mathematics strategist who teaches students not just formulas but WHEN each one applies.

TASK: Build a decision-focused formula sheet for [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. The emphasis is on recognising which formula to reach for in a given problem.

INPUT FROM ME: Chapter is [TOPIC]. Student level: [CLASS/GRADE]. Focus: [DIFFICULTY or LEAVE BLANK].

OUTPUT REQUIREMENTS:

1. Header block with chapter, class, board, and 'Compiled by Indrajeet Yadav'.
2. Present every key formula in a three-column style (or three labelled lines if columns are hard):
 - FORMULA (LaTeX plus plain text in brackets)
 - USE WHEN (the exact trigger words or conditions in the problem that signal this formula)
 - WATCH OUT (one short caution, e.g. a required condition or common confusion)
3. Group related formulas so students see the choices side by side (e.g. which mean to use, which integration method, which probability rule).
4. Add a short 'Decision Flow' at the end: 4 to 6 if-then lines that guide formula choice for typical questions.

QUALITY BARS:

- VERIFY every formula by at least two independent checks before listing it. Never invent formulas, conditions, or values.
- State the domain or condition for any formula that has one; a formula listed without its condition is an error.
- Keep all mathematics fully legible with consistent notation.

FORMATTING: Clear columns or labelled blocks, scannable, A4 black-and-white printable.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.5 Step-By-Step Derivation Sheet

Text only

Any AI chat

What you get: Clean derivations of the chapter's key formulas, step by step.

How to use: Pick the formulas you want proved and the AI writes neat, verifiable derivations you can teach straight from.

COPY EVERYTHING BELOW

Act as a meticulous mathematics teacher writing a derivation reference sheet that shows WHERE each key formula comes from.

TASK: Produce clear, step-by-step derivations of the most important formulas in [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Chapter is [TOPIC]. Formulas to derive: [LIST THE FORMULAS or write 'the key derivable formulas in this chapter']. Depth: [DIFFICULTY e.g. board-level rigour / competitive rigour].

OUTPUT REQUIREMENTS:

1. Header block with chapter, class, board, and 'Compiled by Indrajeet Yadav'.
2. For each formula give:
 - The result being derived (LaTeX plus plain text in brackets)
 - The starting point / assumptions (one line)
 - A numbered derivation, one clean step per line, with a short reason beside each non-obvious step
 - The boxed final result
3. Keep each derivation self-contained and legible; a student should be able to reproduce it.
4. Where a shorter alternative derivation exists, add it as a brief 'Alternative' note.

QUALITY BARS:

- Derive each formula internally FIRST, then VERIFY the result by at least two independent checks (a known special case, dimensional analysis, or a second derivation route). Do not present a derivation whose result you cannot verify.
- Never invent theorems, lemmas, or steps. Every step must be mathematically valid; flag any assumption explicitly.
- Keep all mathematics fully legible with consistent notation and defined symbols.

FORMATTING: Numbered steps, generous spacing, A4 black-and-white printable.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.6 Visual Infographic Formula One-Pager

Text only

ChatGPT (image) / Gemini

What you get: A colourful infographic-style formula poster image, ready to print.

How to use: Ask an image-capable AI to design a bright one-page formula poster for your chapter, then download and print it.

COPY EVERYTHING BELOW

Act as a mathematics teacher and graphic designer creating a printable INFOGRAPHIC-STYLE formula poster image.

TASK: Design a single-page, visually attractive formula poster for [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. Output should be an IMAGE that looks like a polished classroom infographic.

INPUT FROM ME: Chapter is [TOPIC]. Preferred colour theme: [COLOUR e.g. blue and white]. Page size: [A4 portrait].

OUTPUT REQUIREMENTS:

1. A bold title banner with the chapter name, class, and board.
2. 4 to 7 clearly separated coloured sections (cards or boxes), each holding a group of related formulas with a small heading and a simple icon.
3. Every formula rendered large and crisp and mathematically correct; keep symbols clean and legible.
4. A small 'key values' or 'remember this' callout box.
5. Balanced layout with clear hierarchy, readable fonts, and enough contrast to print well.

QUALITY BARS:

- Before generating the image, solve and VERIFY every formula internally by at least two independent checks. Never invent formulas, theorems, or values, and never distort a formula to fit the design - mathematical accuracy outranks aesthetics.
- Double-check that every symbol, exponent, and subscript renders correctly in the final image; re-render if any formula looks garbled.
- Stay within the [BOARD or EXAM] syllabus scope.

FORMATTING: One single image, portrait, print-ready, clean and uncluttered.

Sign the bottom-right of every page with an elegant handwritten signature 'Indrajeet Yadav' with a small underline flourish.

4.7 Comparison Tables (Conics, Distributions, etc.)

Text only

Any AI chat

What you get: A side-by-side table comparing every member of a family of formulas.

How to use: Perfect for chapters with families like conics or probability distributions – the AI lines them all up in one neat table.

COPY EVERYTHING BELOW

Act as an expert mathematics teacher building a side-by-side COMPARISON TABLE that lines up a whole family of related results.

TASK: Create a comparison-table cheat sheet for [TOPIC e.g. all conic sections / all probability distributions / all trig functions] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Family to compare: [TOPIC]. The items in the family are: [LIST THE ITEMS or write 'list the standard ones for this syllabus']. Properties I care about: [LIST PROPERTIES or write 'choose the most useful comparison columns'].

OUTPUT REQUIREMENTS:

1. Header block with topic, class, board, and 'Compiled by Indrajeet Yadav'.
2. One MAIN comparison table: rows = each member of the family; columns = the key properties / formulas (e.g. standard form, key parameters, special formulas, conditions). Put every formula in LaTeX with plain text in brackets inside the cell.
3. Keep columns consistent so differences jump out; align notation across all rows.
4. Below the table, add a short 'How they differ at a glance' list of 4 to 6 contrast points.
5. Add a one-line 'easy way to tell them apart' tip if relevant.

QUALITY BARS:

- VERIFY every entry by at least two independent checks (special case, dimensional check, or re-derivation) before filling the cell. Never invent formulas, values, or conditions.
- Every cell must use identical notation conventions so the comparison is fair and legible.
- Stay strictly within the [BOARD or EXAM] scope; omit anything out of syllabus and note it in one line.

FORMATTING: A clean, well-aligned table that prints on A4 in black and white.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.8 Mnemonics & Memory Tricks Sheet

Text only

Any AI chat

What you get: Catchy mnemonics and tricks to lock the chapter's formulas into memory.

How to use: Name the chapter and the AI invents friendly memory hooks for the tricky formulas your students keep forgetting.

COPY EVERYTHING BELOW

Act as a creative mathematics teacher who is brilliant at memory tricks, turning hard formulas into things students never forget.

TASK: Create a MNEMONICS and memory-tricks sheet for [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Chapter is [TOPIC]. The formulas students most often forget are: [LIST THEM or write 'pick the commonly forgotten ones']. Language flavour: [English / simple English / Hinglish].

OUTPUT REQUIREMENTS:

1. Header block with chapter, class, board, and 'Compiled by Indrajeet Yadav'.
2. For each tricky formula give:
 - The formula (LaTeX plus plain text in brackets)
 - The mnemonic or memory trick (a short phrase, acronym, rhyme, story, or visual hook)
 - One line on how to USE the trick to reconstruct the formula
3. Keep mnemonics clean, respectful, and classroom-appropriate.
4. Add a short 'Quick acronyms' box collecting the best 4 to 6 in one place for fast revision.

QUALITY BARS:

- VERIFY every underlying formula by at least two independent checks before building a trick around it. A clever mnemonic for a WRONG formula is worse than none - accuracy first.
- Make sure each mnemonic actually maps correctly to the formula (test it by reconstructing the formula from the trick). Never invent formulas or values.
- Keep all mathematics fully legible.

FORMATTING: Friendly, scannable, A4 black-and-white printable, with the formula always shown next to its trick.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.9 Common Mistakes & Sign-Error Warnings Sheet

Text only

Any AI chat

What you get: A warning sheet pairing each common error with the correct version.

How to use: Hand this out to stop the silly sign slips and formula mix-ups that cost students easy marks.

COPY EVERYTHING BELOW

Act as a veteran mathematics examiner who has marked thousands of papers and knows exactly where students lose marks.

TASK: Create a COMMON MISTAKES and sign-error warning sheet for [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Chapter is [TOPIC]. If I have specific errors I keep seeing, here they are: [PASTE COMMON ERRORS or LEAVE BLANK so you choose them].

OUTPUT REQUIREMENTS:

1. Header block with chapter, class, board, and 'Compiled by Indrajeet Yadav'.
2. List 12 to 20 of the most frequent mistakes. For each, use this exact structure:
 - WRONG (the mistaken formula, sign, or step students write) in LaTeX
 - CORRECT (the right version) in LaTeX, with plain text in brackets
 - WHY it happens (one short line)
 - FIX (a one-line habit or check to avoid it)
3. Add a dedicated 'Sign-error checklist' box: 5 to 8 quick yes/no checks for negatives, ranges, and direction conventions.
4. Group mistakes by sub-topic so they are easy to scan.

QUALITY BARS:

- For every pair, VERIFY the CORRECT version by at least two independent checks before writing it; make sure the WRONG version is a realistic student error, not a strawman.
- Never invent formulas, theorems, or values. The corrected math must be exactly right - this sheet is the source of truth.
- Keep all mathematics fully legible with consistent notation.

FORMATTING: Clear WRONG-vs-CORRECT blocks, scannable, A4 black-and-white printable.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.10 Printable Formula Flashcards

Text only

Any AI chat

What you get: Front/back flashcards (name on front, formula + use on back) ready to cut out.

How to use: Name your chapter and the AI lays out cut-and-fold flashcards you can print double-sided for revision.

COPY EVERYTHING BELOW

Act as a mathematics teacher designing printable double-sided FLASHCARDS for active-recall revision.

TASK: Generate a set of formula flashcards for [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Chapter is [TOPIC]. Number of cards: [NUMBER e.g. 20] (or 'cover all key formulas'). Difficulty: [DIFFICULTY].

OUTPUT REQUIREMENTS:

1. A header block with chapter, class, board, and 'Compiled by Indrajeet Yadav'.
2. Present each flashcard clearly labelled, in this exact layout:
 - CARD n - FRONT: the formula NAME or a short prompt question only
 - CARD n - BACK: the formula (LaTeX plus plain text in brackets) + a one-line 'use when' note + any condition
3. Number the cards so front and back stay paired when printed double-sided.
4. Keep each back side compact enough to fit a small card.
5. After the cards, add a one-line printing tip (e.g. 'print, cut along lines, fold').

QUALITY BARS:

- VERIFY every formula on every back side by at least two independent checks before finalising. Never invent formulas, conditions, or values.
- Make sure each FRONT prompt uniquely identifies its BACK formula (no ambiguous fronts).
- Keep all mathematics fully legible with consistent notation; state the domain or condition wherever one exists.

FORMATTING: Cleanly separated, numbered card blocks, A4 black-and-white printable, easy to cut.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.11 Units, Conditions & Domains Reference

Text only

Any AI chat

What you get: Every formula listed with its valid domain, conditions, and units.

How to use: Use this when students apply formulas outside their valid range - it spells out where each one is allowed to be used.

COPY EVERYTHING BELOW

Act as a precise mathematics teacher who insists students always check the conditions, domain, and units before using a formula.

TASK: Create a UNITS, CONDITIONS and DOMAINS reference sheet for [TOPIC] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Chapter is [TOPIC]. Level: [CLASS/GRADE]. Any specific formulas to include: [LIST or write 'all standard ones'].

OUTPUT REQUIREMENTS:

1. Header block with chapter, class, board, and 'Compiled by Indrajeet Yadav'.
2. For each formula present in a four-part row:
 - FORMULA (LaTeX plus plain text in brackets)
 - VALID DOMAIN / RANGE (where the variables must lie)
 - CONDITIONS (any assumptions, e.g. 'denominator not zero', 'angle in radians', 'data normally distributed')
 - UNITS (if applicable; write 'dimensionless' or 'n/a' where relevant)
3. Group by sub-topic and keep notation uniform.
4. Add a short 'Most-missed conditions' box highlighting the 5 to 8 conditions students forget most.

QUALITY BARS:

- VERIFY each formula AND its stated domain/conditions by at least two independent checks (test a boundary case and an interior case). A formula listed without its true conditions is an error.
- Never invent formulas, conditions, units, or values. Where a condition is subtle, state it explicitly rather than glossing over it.
- Keep all mathematics fully legible with consistent, clearly defined notation.

FORMATTING: Clean four-column table or labelled blocks, A4 black-and-white printable.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

4.12 Full-Syllabus Master Formula Booklet Outline

Text only

Any AI chat

What you get: A structured chapter-by-chapter outline for a whole-year formula booklet.

How to use: Tell it the class and board and it plans an entire year's formula booklet you can fill chapter by chapter.

COPY EVERYTHING BELOW

Act as a head of the mathematics department planning a complete, year-long MASTER FORMULA BOOKLET for a whole class.

TASK: Produce a structured outline (and starter content) for a full-syllabus master formula booklet for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Class and board: [CLASS/GRADE] / [BOARD or EXAM]. If I have the official chapter list, here it is: [PASTE CHAPTER LIST or write 'use the standard syllabus']. Depth wanted: [DIFFICULTY e.g. essential only / comprehensive].

OUTPUT REQUIREMENTS:

1. A cover/title block: subject, class, board, academic context, and 'Compiled by Indrajeet Yadav'.
2. A clear TABLE OF CONTENTS listing every chapter in syllabus order with an estimated formula count per chapter.
3. For EACH chapter, a one-page-style section containing: the 5 to 12 most important formulas (LaTeX plus plain text in brackets), a one-line note per formula on its use, and any key constants. Keep each chapter to roughly one page of content so the whole thing stays a booklet, not a textbook.
4. A final 'Top 30 formulas of the year' quick-revision appendix pulling the highest-yield results across all chapters.
5. Clear page-break markers (e.g. '--- PAGE BREAK ---') between chapters so it prints cleanly.

QUALITY BARS:

- VERIFY every formula by at least two independent checks before including it. Never invent formulas, theorems, values, or chapter names.
- Keep the chapter list complete and in the correct [BOARD or EXAM] order; flag in one line if any chapter is uncertain rather than guessing.
- Keep all mathematics fully legible with consistent notation throughout the whole booklet.

FORMATTING: Booklet style with a title page, table of contents, one section per chapter, and an appendix; A4 black-and-white printable.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.



Full exam papers, sample papers and unit tests with blueprint, sections and answer keys.

5.1 Full Board-Pattern Paper (Blueprint + Sections + Answer Key)

Text only

Any AI chat

What you get: A complete board-style question paper with blueprint table, all sections and a full answer key.

How to use: Fill in your class, board, chapters and marks, paste it, and you get a ready-to-print paper.

COPY EVERYTHING BELOW

Act as a senior mathematics examiner and paper-setter for [BOARD or EXAM e.g. CBSE / ICSE / State Board].

TASK: Create ONE complete board-pattern mathematics question paper for Class [CLASS/GRADE] covering these chapters/topics: [LIST CHAPTERS or TOPICS]. Total marks: [TOTAL MARKS]. Time: [TIME e.g. 3 hours]. Difficulty mix: [e.g. 40% easy, 40% moderate, 20% hard].

INPUT INSTRUCTIONS: Use exactly the chapters I listed above. If I leave a field blank, assume the standard latest board pattern for that class and state your assumption in one line at the top. Do NOT go outside the listed syllabus.

OUTPUT REQUIREMENTS, in this order:

1. A header block with: school name line ' _____ ', 'Class: [CLASS/GRADE]', 'Subject: Mathematics', 'Board: [BOARD or EXAM]', 'Time: [TIME]', 'Maximum Marks: [TOTAL MARKS]', and the line 'Compiled by Indrajeet Yadav'.
2. General Instructions (numbered, 5 to 7 lines).
3. A BLUEPRINT TABLE before the questions: rows = chapters, columns = number of questions and marks per section, with a TOTAL row that must add up exactly to [TOTAL MARKS].
4. The question paper divided into clearly labelled sections with marks shown against every question.
5. After the paper, a complete ANSWER KEY with the final answer to every question, plus a 1 to 3 line working outline for long questions.

QUALITY BARS: Solve every question internally before printing it. Verify each final answer by at least two independent checks (for example, re-substitution, an alternate method, or a units/sign sanity check). Never invent theorems, formulas or values. Keep all mathematics fully legible using clean LaTeX or clear plain-text math. Ensure section marks sum exactly to the total; if they do not, fix the paper before showing it.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.2 Single-Chapter Unit Test

Text only

Any AI chat

What you get: A focused unit test on one chapter with mixed question types and an answer key.

How to use: Type the chapter, class and marks, and get a clean unit test you can print today.

COPY EVERYTHING BELOW

Act as an experienced mathematics teacher and unit-test designer.

TASK: Create ONE unit test on the single chapter [TOPIC] for Class [CLASS/GRADE] following the [BOARD or EXAM] style. Total marks: [TOTAL MARKS]. Time: [TIME e.g. 40 minutes]. Number of questions: [NUMBER OF QUESTIONS]. Difficulty: [DIFFICULTY e.g. easy / moderate / mixed].

INPUT INSTRUCTIONS: Restrict everything strictly to the chapter [TOPIC]. List the 4 to 6 sub-topics inside this chapter that you will test, and make sure the questions cover all of them in a balanced way. If a field is blank, use a sensible default and note it in one line.

OUTPUT REQUIREMENTS:

1. Header block: 'Unit Test', 'Chapter: [TOPIC]', 'Class: [CLASS/GRADE]', 'Time: [TIME]', 'Max Marks: [TOTAL MARKS]', and 'Compiled by Indrajeet Yadav'.
2. A short sub-topic coverage list showing which question tests which sub-topic.
3. The questions grouped as: a few 1-mark items, some short-answer items, and one or two longer items, with marks shown beside each.
4. A full ANSWER KEY with final answers and a brief working outline for the longer questions.

QUALITY BARS: Solve each question yourself before printing. Confirm every final answer with at least two independent checks. Do not invent any formula or value. Keep all math legible in clean LaTeX or clear plain text. The marks must add up exactly to [TOTAL MARKS]; correct the test if they do not.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.3 Sample / Model Paper in Latest Exam Pattern

Text only

Gemini / ChatGPT

What you get: A model paper mirroring the most recent official exam pattern, with answer key.

How to use: Give your class and board; it produces a realistic model paper students can practise on.

COPY EVERYTHING BELOW

Act as a mathematics paper-setter who closely tracks the latest official examination pattern.

TASK: Create ONE sample/model mathematics paper for Class [CLASS/GRADE] in the most recent [BOARD or EXAM] pattern. Total marks: [TOTAL MARKS]. Time: [TIME]. Syllabus to cover: [LIST CHAPTERS or 'full syllabus'].

INPUT INSTRUCTIONS: Match the current pattern as closely as you can: same section structure, same question-type mix, and the same internal-choice style commonly used. State clearly at the top, in 2 lines, which pattern/year you are modelling and any assumptions. Stay strictly inside the syllabus I gave.

OUTPUT REQUIREMENTS:

1. Header block with 'Model Paper', class, subject, board, time, max marks, and 'Compiled by Indrajeet Yadav'.
2. General Instructions matching the latest pattern (including internal choices where applicable).
3. The full paper in the correct sections with marks against each question and internal choices clearly marked using 'OR'.
4. A short note at the end: 'How this matches the latest pattern' (3 to 4 bullet points).
5. A complete ANSWER KEY with final answers and concise solution outlines.

QUALITY BARS: Solve everything internally first. Verify each final answer with at least two independent checks. Never invent theorems, formulas, marks distributions or values; if unsure about a pattern detail, say so rather than guessing silently. Keep all math fully legible in clean LaTeX or clear plain text. Ensure total marks equal [TOTAL MARKS] exactly.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.4 Sectioned Paper (A: MCQ, B: Short, C: Long, D: Case-Study)

Text only

Any AI chat

What you get: A four-section paper with MCQs, short, long and case-study questions and marks balanced.

How to use: Tell it the chapters and total marks; you get a neatly sectioned exam with a case study included.

COPY EVERYTHING BELOW

Act as a mathematics examination designer who builds modern sectioned papers.

TASK: Create ONE mathematics paper for Class [CLASS/GRADE], [BOARD or EXAM] style, with exactly four sections: Section A (MCQ), Section B (short answer), Section C (long answer), and Section D (case-study / passage-based). Total marks: [TOTAL MARKS]. Time: [TIME]. Topics: [LIST CHAPTERS or TOPICS].

INPUT INSTRUCTIONS: Distribute marks sensibly across the four sections and show that distribution before the questions. Keep all questions inside the listed topics. If a field is blank, choose a standard distribution and note it in one line.

OUTPUT REQUIREMENTS:

1. Header block: title, class, subject, board, time, max marks, and 'Compiled by Indrajeet Yadav'.
2. A small marks-distribution table for Sections A to D, with a total that equals [TOTAL MARKS] exactly.
3. Section A: multiple-choice questions with four options (a) to (d) each, one correct.
4. Section B: short-answer questions.
5. Section C: long-answer questions.
6. Section D: one or two case-study / passage questions, each with a short real-life context followed by sub-parts.
7. Marks shown beside every question.
8. A complete ANSWER KEY: correct option for each MCQ, final answers for B and C with outlines, and full solutions for the case-study sub-parts.

QUALITY BARS: Solve everything internally first and verify each final answer with at least two independent checks. For MCQs, confirm that exactly one option is correct and that the distractors are plausible but wrong. Never invent formulas or values. Keep all math legible in clean LaTeX or clear plain text. The four sections must sum exactly to [TOTAL MARKS].

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.5 Pure MCQ Test with OMR-Friendly Layout

Text only

Any AI chat

What you get: An all-MCQ test laid out for OMR, with a clean answer grid and verified single answers.

How to use: Pick the chapters and number of questions; you get an objective test plus a ready answer grid.

COPY EVERYTHING BELOW

Act as a mathematics objective-test setter who designs OMR-friendly MCQ papers.

TASK: Create ONE pure multiple-choice mathematics test for Class [CLASS/GRADE], [BOARD or EXAM] style. Number of questions: [NUMBER OF QUESTIONS]. Marks per question: [e.g. 1 mark each]. Total marks: [TOTAL MARKS]. Time: [TIME]. Topics: [LIST CHAPTERS or TOPICS]. Negative marking: [e.g. none / -0.25].

INPUT INSTRUCTIONS: Every question must be a single-correct MCQ with four options (A) to (D). Keep all questions strictly within the listed topics. Number questions sequentially 1, 2, 3, and keep the layout clean so it can be answered on an OMR sheet.

OUTPUT REQUIREMENTS:

1. Header block: title, class, subject, board, time, max marks, marking scheme, and 'Compiled by Indrajeet Yadav'.
2. Short instructions for OMR marking (use a dark pen/pencil, fill one bubble only, no overwriting).
3. The numbered MCQs, each with options labelled (A) (B) (C) (D) on clearly separated lines.
4. A blank OMR-style ANSWER GRID: a simple table listing question numbers with empty A/B/C/D cells for students to mark.
5. A separate ANSWER KEY table mapping each question number to its single correct option, plus a one-line reason or final value for any tricky item.

QUALITY BARS: Solve every question internally and verify each answer with at least two independent checks. Guarantee exactly one correct option per question and make distractors reflect common student errors, not random values. Never invent formulas or values. Keep all math legible in clean LaTeX or clear plain text. Confirm question count and total marks match what I asked for.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.6 Competency / Application-Based Question Paper

Text only

Gemini / ChatGPT

What you get: A real-world, competency-focused paper that tests reasoning and application, with rubric answers.

How to use: Name the chapters; it builds higher-order, real-life questions instead of plain textbook sums.

COPY EVERYTHING BELOW

Act as a mathematics assessment specialist who designs competency-based, application-focused papers.

TASK: Create ONE competency-based mathematics paper for Class [CLASS/GRADE], [BOARD or EXAM] style, that emphasises reasoning, real-life application, data interpretation and higher-order thinking rather than rote recall. Total marks: [TOTAL MARKS]. Time: [TIME]. Topics: [LIST CHAPTERS or TOPICS]. Target competency share: [e.g. at least 60% application/HOTS].

INPUT INSTRUCTIONS: Wrap questions in realistic Indian and everyday contexts (markets, travel, banking, sports, agriculture, science) while staying strictly inside the listed topics. Avoid copying standard textbook problems verbatim. If a field is blank, choose sensible defaults and note them in one line.

OUTPUT REQUIREMENTS:

1. Header block with title 'Competency-Based Mathematics Paper', class, subject, board, time, max marks, and 'Compiled by Indrajeet Yadav'.
2. A short tag on each question marking its competency type: 'Application', 'Analysis', 'Reasoning', or 'Data Interpretation'.
3. The questions with marks beside each, including at least one data/graph-based item and one multi-step real-life problem.
4. A complete ANSWER KEY with final answers, full reasoning steps, and a brief marking rubric showing how marks are split across reasoning, method and answer.

QUALITY BARS: Solve everything internally first and verify each final answer with at least two independent checks. Make sure every context is mathematically consistent (numbers, units and assumptions must hold up). Never invent formulas, data or values. Keep all math legible in clean LaTeX or clear plain text. Total marks must equal [TOTAL MARKS] exactly.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.7 Previous-Year-Style Paper on One Chapter

Text only

Any AI chat

What you get: A practice paper in the style of past exam questions for a single chapter, with answers.

How to use: Choose a chapter; it mimics the way that chapter is usually asked in real exams.

COPY EVERYTHING BELOW

Act as a mathematics teacher who has studied many years of [BOARD or EXAM] question papers.

TASK: Create ONE practice paper in the STYLE of previous-year exam questions, focused on the single chapter [TOPIC] for Class [CLASS/GRADE]. Total marks: [TOTAL MARKS]. Time: [TIME]. Number of questions: [NUMBER OF QUESTIONS].

INPUT INSTRUCTIONS: Write fresh, original questions that imitate the typical phrasing, difficulty curve and marks pattern that [BOARD or EXAM] commonly uses for [TOPIC]. Do NOT claim any question is from a specific real past paper, and do not copy copyrighted text; produce your own questions in that recognisable style. Stay strictly within [TOPIC].

OUTPUT REQUIREMENTS:

1. Header block: 'Previous-Year-Style Practice Paper', 'Chapter: [TOPIC]', class, board, time, max marks, and 'Compiled by Indrajeet Yadav'.
2. A 2-line note describing the typical exam pattern for this chapter (how it is usually asked and for how many marks).
3. The questions arranged from easier to harder, with marks beside each, including the common high-value question types for this chapter.
4. A complete ANSWER KEY with final answers and clear step-by-step solutions for the higher-mark questions.

QUALITY BARS: Solve each question internally and verify every final answer with at least two independent checks. Never invent theorems, formulas or values, and never falsely attribute a question to a real exam year. Keep all math fully legible in clean LaTeX or clear plain text. Confirm marks total exactly [TOTAL MARKS].

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.8 Two Parallel Versions (Set A / Set B) to Stop Copying

Text only

Gemini / ChatGPT

What you get: Two equal-difficulty papers, Set A and Set B, with matched blueprint and both answer keys.

How to use: Give chapters and marks; you get two fair, parallel papers so neighbours cannot copy.

COPY EVERYTHING BELOW

Act as a mathematics paper-setter who builds parallel exam sets to prevent copying.

TASK: Create TWO parallel mathematics papers, 'Set A' and 'Set B', for Class [CLASS/GRADE], [BOARD or EXAM] style. They must test the same concepts at the same difficulty and carry the same marks per question, but use different numbers, contexts and question order. Total marks each: [TOTAL MARKS]. Time: [TIME]. Topics: [LIST CHAPTERS or TOPICS].

INPUT INSTRUCTIONS: For every Set A question there must be a matching Set B question on the same concept and marks, but not identical. Keep both sets strictly within the listed topics. If a field is blank, use sensible defaults and note them in one line.

OUTPUT REQUIREMENTS:

1. A shared header block: title, class, subject, board, time, max marks, and 'Compiled by Indrajeet Yadav'.
2. A short MAPPING TABLE showing 'Q-number → concept tested → marks', proving Set A and Set B match item-for-item.
3. The full 'SET A' paper with marks beside each question.
4. The full 'SET B' paper with marks beside each question.
5. Two separate ANSWER KEYS clearly labelled 'Answer Key - Set A' and 'Answer Key - Set B', each with final answers and short solution outlines.

QUALITY BARS: Solve every question in both sets internally and verify each final answer with at least two independent checks. Confirm the two sets are genuinely equal in difficulty and that no question is accidentally identical. Never invent formulas or values. Keep all math legible in clean LaTeX or clear plain text. Each set must total exactly [TOTAL MARKS].

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.9 Detailed Marking Scheme / Step-Wise Marking Key

Text only

Any AI chat

What you get: A step-by-step marking scheme that splits every question's marks across stages.

How to use: Paste your existing question paper and get a fair, examiner-style marking key for it.

COPY EVERYTHING BELOW

Act as a senior mathematics examiner who writes official-style step-wise marking schemes.

TASK: For the question paper I provide below, create a DETAILED MARKING SCHEME that shows exactly how each question's marks are awarded step by step, in the [BOARD or EXAM] examiner style, for Class [CLASS/GRADE].

INPUT INSTRUCTIONS: Here is my paper (with the marks for each question shown): [PASTE THE QUESTION PAPER HERE]. If any answer or mark value is missing or ambiguous, list your assumption clearly before solving. Do not change the questions themselves.

OUTPUT REQUIREMENTS, question by question:

1. Restate the question number and its total marks.
2. Give the correct final answer.
3. Provide a step-wise mark split, for example: correct formula/setup - 1 mark, correct substitution - 1 mark, correct simplification - 1 mark, final answer - 1 mark, so that the parts add up exactly to that question's marks.
4. Add 'Acceptable alternative methods' where a student could legitimately solve it another way and still earn full marks.
5. Add 'Common errors and partial credit' notes (what to deduct and what to still reward).
6. At the end, a 1-line check confirming the sum of all question marks equals the paper total.

QUALITY BARS: Solve each question internally first and verify every final answer with at least two independent checks. Ensure the step marks for each question add up exactly to that question's allotted marks; fix any mismatch. Never invent theorems, formulas or values. Keep all math fully legible in clean LaTeX or clear plain text.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.10 Blueprint-Only (Weightage Table) for a Syllabus

Text only

Any AI chat

What you get: A clean blueprint/weightage table mapping chapters to marks and question types.

How to use: List your chapters and total marks; you get a planning blueprint before writing any paper.

COPY EVERYTHING BELOW

Act as a mathematics curriculum planner and chief paper-setter.

TASK: Create a BLUEPRINT (weightage table) only - no questions - for a Class [CLASS/GRADE] mathematics paper in the [BOARD or EXAM] style. Total marks: [TOTAL MARKS]. Time: [TIME]. Chapters/units with their relative importance: [LIST CHAPTERS, and weightage if you know it].

INPUT INSTRUCTIONS: Spread the marks across the listed chapters in proportion to their importance and typical exam weight. Also spread marks across question types and cognitive levels. If I do not give weightage, propose a fair one and explain it in one line. Stay strictly within the chapters I listed.

OUTPUT REQUIREMENTS:

1. A header line: 'Blueprint - Class [CLASS/GRADE] Mathematics - [BOARD or EXAM]', and 'Compiled by Indrajeet Yadav'.
2. A main BLUEPRINT TABLE with rows = chapters and columns for each question type (for example MCQ, Short, Long, Case-Study), showing number of questions and marks in each cell, plus a TOTAL row and TOTAL column.
3. A second small table: marks by cognitive level (Remember/Understand, Apply, Analyse/Evaluate) with percentages.
4. A 3 to 4 line summary explaining the design choices.

QUALITY BARS: Make every row total, column total and the grand total add up exactly to [TOTAL MARKS]; re-check the arithmetic by adding rows and columns independently (two checks) and fix any mismatch. Never invent official weightages; if you assume a distribution, clearly label it as a suggested distribution. Keep the tables clean and legible in clear plain text or simple markdown tables.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.11 Surprise Revision Test (Multiple Chapters Mixed)

Text only

Any AI chat

What you get: A mixed-chapter surprise test that spirals revision across several topics, with answers.

How to use: List a few chapters to revise; it blends them into one quick surprise test with a key.

COPY EVERYTHING BELOW

Act as a mathematics teacher who designs quick mixed-revision surprise tests.

TASK: Create ONE surprise revision test for Class [CLASS/GRADE], [BOARD or EXAM] style, that MIXES these chapters together: [LIST 3 to 6 CHAPTERS]. Total marks: [TOTAL MARKS]. Time: [TIME e.g. 30 minutes]. Number of questions: [NUMBER OF QUESTIONS]. Difficulty: [DIFFICULTY].

INPUT INSTRUCTIONS: Shuffle the chapters so questions are not grouped by topic - the student should not know which chapter is coming next. Cover all the listed chapters fairly. Stay strictly within those chapters. If a field is blank, use a sensible default and note it in one line.

OUTPUT REQUIREMENTS:

1. Header block: 'Surprise Revision Test', class, board, time, max marks, the list of chapters covered, and 'Compiled by Indrajeet Yadav'.
2. A short coverage line: how many marks come from each chapter (must total [TOTAL MARKS]).
3. The mixed questions in shuffled order with marks beside each, ranging from quick recall to one slightly tricky item.
4. A complete ANSWER KEY with final answers and brief working for the harder questions, and a tag showing which chapter each question belongs to.

QUALITY BARS: Solve every question internally and verify each final answer with at least two independent checks. Make sure each listed chapter is genuinely represented and the per-chapter marks add to [TOTAL MARKS]. Never invent formulas or values. Keep all math legible in clean LaTeX or clear plain text.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.

5.12 Timed Mock-Test Series Plan for the Term

Text only

Gemini / ChatGPT

What you get: A full term schedule of timed mock tests with dates, syllabus per test and marks plan.

How to use: Give your syllabus and number of weeks; it plans a whole series of mock tests for the term.

COPY EVERYTHING BELOW

Act as a mathematics exam coach who plans a full term's timed mock-test series.

TASK: Create a complete TIMED MOCK-TEST SERIES PLAN for Class [CLASS/GRADE], [BOARD or EXAM] style, spread over [NUMBER OF WEEKS or MONTHS]. The final exam pattern to prepare for: total marks [TOTAL MARKS], time [TIME]. Full syllabus / chapter list: [LIST CHAPTERS].

INPUT INSTRUCTIONS: Build a spiral schedule where early mocks cover smaller portions and later mocks become full-syllabus, exam-length tests. Tell me, for each mock, the suggested date/week, the portion of syllabus, the marks, the duration, and the goal. Stay strictly within the listed syllabus. If a field is blank, assume a sensible default and note it in one line.

OUTPUT REQUIREMENTS:

1. A header line: 'Mock-Test Series Plan - Class [CLASS/GRADE] - [BOARD or EXAM]', and 'Compiled by Indrajeet Yadav'.
2. A SCHEDULE TABLE: columns = Mock No., Week/Date, Syllabus covered, Marks, Time, Difficulty, Goal of this mock.
3. A short 'pattern progression' note explaining how portion and difficulty grow toward the final exam.
4. For each mock, a 2 to 3 line BLUEPRINT (how many marks per section/chapter) so the teacher can build the actual paper later.
5. A revision-and-feedback plan: what to do after each mock (review weak chapters, re-test, etc.).

QUALITY BARS: Make sure the schedule is realistic and that the cumulative coverage actually finishes the full syllabus before the final exam; double-check by listing any chapter that is never tested and fixing the plan if so (two passes). Every per-mock blueprint's marks must add to that mock's total. Never invent official exam dates or values; label suggested dates clearly as suggestions. Keep everything legible in clear plain text or simple markdown tables.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title or header block.



Graded worksheets, homework and practice sets tailored to your class.

6.1 Graded-Difficulty Worksheet (Easy to Hard)

Text only

Any AI chat

What you get: A worksheet that rises smoothly from warm-up to challenge, with answer key.

How to use: Just fill in your topic, class and number of questions, then paste it into any AI chat.

COPY EVERYTHING BELOW

Act as an experienced mathematics teacher and assessment designer who builds carefully graded practice sets.

TASK: Create a single worksheet on [TOPIC] for [CLASS/GRADE] students following the [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT] syllabus. The worksheet must contain [NUMBER OF QUESTIONS] questions whose difficulty rises smoothly from very easy to genuinely hard.

INPUT FROM ME: My topic is [TOPIC]. My class is [CLASS/GRADE]. My board/exam is [BOARD or EXAM]. I want [NUMBER OF QUESTIONS] questions and the total worksheet should take about [TIME] minutes.

OUTPUT REQUIREMENTS:

1. Split the questions into four clearly labelled bands: Warm-up (recall), Foundation (single-step), Application (multi-step word problems), and Challenge (proof, reasoning or tricky cases). Distribute the questions sensibly across the four bands.
2. Number every question. Beside each, show a small difficulty star rating from 1 star (easiest) to 5 stars (hardest) and the marks for that question.
3. Use clean, fully legible mathematics. Write all formulas and steps in clear LaTeX, or if LaTeX is not supported, in tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$.
4. After the questions, add a separate ANSWER KEY section with the final answer to every question plus a one-line method hint. Do NOT mix answers into the question section.
5. Keep language simple for [CLASS/GRADE] and use Indian/real-world contexts where natural.

QUALITY BARS: Solve every question yourself internally before writing it. Verify each final answer by at least two independent checks (for example, substitute the answer back, or solve by a second method) and silently fix any mismatch. Never invent formulas, theorems or values; use only standard, correct mathematics. Ensure the difficulty genuinely increases and no two questions are near-duplicates.

SIGNATURE: Place 'Compiled by Indrajeet Yadav' in the title or header block at the top, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.2 Themed Worksheet on One Sub-Skill

Text only

Any AI chat

What you get: A focused drill on a single sub-skill so students master one thing deeply.

How to use: Name the exact micro-skill students keep getting wrong and let the AI drill just that.

COPY EVERYTHING BELOW

Act as a mathematics teacher who designs laser-focused practice on a single sub-skill so students overlearn one specific technique.

TASK: Build a themed worksheet that drills ONLY the sub-skill [SUB-SKILL e.g. factorising quadratics by splitting the middle term] within the topic [TOPIC], for [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. The whole worksheet must stay on this one sub-skill - do not drift into other techniques.

INPUT FROM ME: Sub-skill to drill: [SUB-SKILL]. Topic: [TOPIC]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM]. Number of questions: [NUMBER OF QUESTIONS]. Difficulty: [DIFFICULTY].

OUTPUT REQUIREMENTS:

1. Begin with a short 'Method Reminder' box (3 to 5 lines) stating the single technique and one fully worked model example, step by step.
2. Then give [NUMBER OF QUESTIONS] practice questions, all targeting the same sub-skill, ordered from gentle to demanding. Number them clearly.
3. Deliberately vary the surface features (signs, coefficients, fractions, decimals, contexts) so students cannot rely on a memorised pattern, but keep the underlying skill identical.
4. Include 2 or 3 'spot-the-error' items where a wrong worked solution is shown and students must find and fix the mistake.
5. End with a clearly separated ANSWER KEY giving final answers and a one-line key step for each.
6. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Solve each item yourself first. Verify every final answer by at least two independent checks and correct any error before presenting it. Never invent formulas or values; use only standard mathematics. Make sure no two practice questions are near-duplicates.

SIGNATURE: Put 'Compiled by Indrajeet Yadav' in the header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.3 Holiday Homework Pack

Text only

Gemini / ChatGPT

What you get: A multi-day holiday pack with daily sets, a project and self-check answers.

How to use: Tell it how many days of holidays and which chapters to revise, and it builds the whole pack.

COPY EVERYTHING BELOW

Act as a mathematics teacher preparing a complete, engaging holiday homework pack that keeps students in practice without overwhelming them.

TASK: Create a holiday homework pack covering the chapters [LIST OF CHAPTERS/TOPICS] for [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT], spread across [NUMBER OF DAYS] days of holidays.

INPUT FROM ME: Chapters/topics: [LIST OF CHAPTERS/TOPICS]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM]. Number of holiday days: [NUMBER OF DAYS]. Roughly [TIME] minutes of work per day.

OUTPUT REQUIREMENTS:

1. Open with a friendly one-paragraph note to students and parents explaining the goal and how to use the pack.
2. Provide a 'Daily Plan' table mapping each day to a chapter/sub-topic and the work for that day.
3. For each day, give a short revision recap (3 to 5 key points) followed by a small set of graded questions (mix of skill drills and 1 to 2 word problems), numbered and with marks shown.
4. Include ONE fun applied mini-project or real-life maths activity (for example, budgeting, measuring at home, or a data-collection task) that uses the chapters.
5. Add a final 'Holiday Challenge' section with 3 to 5 higher-order thinking questions.
6. Provide a separate SELF-CHECK ANSWER KEY at the very end so students can mark their own work, with final answers and one-line hints.
7. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Solve everything internally first. Verify every final answer by at least two independent checks and fix any mismatch silently. Never invent formulas, theorems or values. Avoid near-duplicate questions across days. Keep the daily load realistic for [CLASS/GRADE].

SIGNATURE: Place 'Compiled by Indrajeet Yadav' in the title block at the top, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.4 Remedial Worksheet for Weak Students

Text only

Any AI chat

What you get: A confidence-building worksheet that rebuilds basics in tiny supported steps.

How to use: Use this for students who are behind; it starts from the very basics with lots of support.

COPY EVERYTHING BELOW

Act as a patient, encouraging mathematics teacher who specialises in helping struggling students rebuild confidence from the basics.

TASK: Create a gentle remedial worksheet on [TOPIC] for weaker [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT] who have gaps in the fundamentals. The goal is success and confidence, not difficulty.

INPUT FROM ME: Topic students are struggling with: [TOPIC]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM]. Common mistakes my students make: [DESCRIBE COMMON MISTAKES, or write 'not sure']. Number of questions: [NUMBER OF QUESTIONS].

OUTPUT REQUIREMENTS:

1. Start with a 'Back to Basics' refresher: the prerequisite ideas needed for [TOPIC], explained in very simple language with one tiny worked example each.
2. Give a 'Watch Out' box listing the 3 to 4 most common mistakes (use the ones I described if provided) and how to avoid each.
3. Provide [NUMBER OF QUESTIONS] questions that stay easy and increase only slightly. Begin the first 2 or 3 questions as guided fill-in-the-blank steps, then let later ones be done fully by the student.
4. Keep numbers friendly (small whole numbers where possible), one idea per question, and add a short encouraging line at section breaks.
5. End with a clearly separated ANSWER KEY showing the full worked solution (not just the final answer) for every question, in small clear steps.
6. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Solve every item yourself first. Verify each final answer by at least two independent checks and correct any error. Never invent formulas or values; use only standard, correct mathematics. Keep the cognitive load low and the steps small and explicit.

SIGNATURE: Put 'Compiled by Indrajeet Yadav' in the header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.5 Enrichment / Challenge Worksheet for Toppers

Text only

Any AI chat

What you get: A high-ceiling worksheet of tricky, olympiad-flavoured problems for your best students.

How to use: Give this to fast finishers and toppers who need harder, more interesting problems.

COPY EVERYTHING BELOW

Act as a mathematics olympiad coach who designs stretching enrichment problems for high-achieving students.

TASK: Create a challenge worksheet on [TOPIC] for top-performing [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. These students already know the standard syllabus, so push their reasoning, not just their speed.

INPUT FROM ME: Topic: [TOPIC]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM]. Number of questions: [NUMBER OF QUESTIONS]. Optional flavour: [e.g. olympiad / competitive-exam / proof-based].

OUTPUT REQUIREMENTS:

1. Provide [NUMBER OF QUESTIONS] non-routine problems that demand insight: multi-step reasoning, clever substitutions, proofs, generalisations, or unusual real-world modelling. Avoid plug-and-chug.
2. Tag each problem with a difficulty star rating (3 to 5 stars) and an estimated solve time.
3. Include at least one 'prove or disprove' item and at least one 'find all values' or 'find the general pattern' item.
4. After all problems, give a separate HINTS section (one gentle nudge per problem) so a student can try again before seeing the full solution.
5. After the hints, give a separate FULL SOLUTIONS section with elegant, complete reasoning and the verified final answer for each.
6. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Solve every problem yourself first and make sure each is actually solvable at [CLASS/GRADE] level with effort. Verify every final answer and every key claim by at least two independent checks. Never invent theorems, formulas or values; state any non-obvious result you rely on. Ensure no two problems are near-duplicates and that they are genuinely challenging.

SIGNATURE: Place 'Compiled by Indrajeet Yadav' in the header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.6 Spiral-Review Worksheet (Mixes Past Chapters)

Text only

Gemini / ChatGPT

What you get: A mixed-revision sheet that interleaves several past chapters to keep them fresh.

How to use: List the chapters done so far and it cleverly mixes them so old topics are not forgotten.

COPY EVERYTHING BELOW

Act as a mathematics teacher who uses spiral revision (interleaving) to keep previously taught chapters fresh in students' memory.

TASK: Create a spiral-review worksheet that mixes questions from several past chapters for [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. The point is that questions from different chapters are interleaved, not grouped, so students must first decide which method applies.

INPUT FROM ME: Chapters to revise: [LIST OF CHAPTERS/TOPICS]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM]. Number of questions: [NUMBER OF QUESTIONS]. Suggested time: [TIME] minutes.

OUTPUT REQUIREMENTS:

1. Pull questions from across all the chapters I listed and INTERLEAVE them (do not put all questions from one chapter together). Number every question.
2. Aim for a balanced spread so each listed chapter is represented; vary difficulty across the sheet.
3. Beside each question, hide a small chapter tag in the answer key (not next to the question) so students cannot tell the topic in advance.
4. Include a few 'mixed-skill' questions that combine ideas from two chapters at once.
5. Provide a separate ANSWER KEY at the end giving, for each question: the chapter tag, the final answer, and a one-line method hint.
6. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Solve every question yourself first. Verify each final answer by at least two independent checks and fix any mismatch. Never invent formulas, theorems or values; use only standard mathematics. Ensure genuine variety across chapters and difficulties, with no near-duplicate questions.

SIGNATURE: Put 'Compiled by Indrajeet Yadav' in the title block at the top, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.7 Exit-Ticket (3 to 5 Quick End-of-Class Questions)

Text only

Any AI chat

What you get: A tiny 5-minute checkpoint that tells you who understood today's lesson.

How to use: Use the last 5 minutes of class; paste your lesson topic and get a quick understanding check.

COPY EVERYTHING BELOW

Act as a mathematics teacher who uses quick exit-tickets to check, in the last five minutes of class, whether students understood today's lesson.

TASK: Create a short exit-ticket of 3 to 5 quick questions on TODAY'S lesson [TODAY'S LESSON/TOPIC] for [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. It must be solvable in about [TIME e.g. 5] minutes.

INPUT FROM ME: Today's lesson/topic: [TODAY'S LESSON/TOPIC]. The single most important idea I want to check: [KEY IDEA]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM]. Time available: [TIME] minutes.

OUTPUT REQUIREMENTS:

1. Give 3 to 5 short, targeted questions that directly test whether the [KEY IDEA] was understood. Order them easy to slightly harder. Number them.
2. Make at least one question a quick conceptual or 'explain in one line' question, not just calculation.
3. Keep each question short enough to answer in roughly one minute.
4. Add a small 'Self-rating' line at the bottom for the student: 'How confident are you, 1 to 5?'
5. Provide a compact ANSWER KEY with the final answer and a one-line teacher note on what a wrong answer to each question reveals (the likely misconception).
6. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Solve each question yourself first. Verify every final answer by at least two independent checks and correct any error. Never invent formulas or values; use only standard mathematics. Keep it short, sharp and clearly tied to the [KEY IDEA].

SIGNATURE: Put 'Compiled by Indrajeet Yadav' in the header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.8 Fill-in-the-Blank Guided Derivation Worksheet

Text only

Any AI chat

What you get: A scaffolded derivation where students fill missing steps to reach the formula or proof.

How to use: Great for teaching a formula or proof; students complete the blanks to build it themselves.

COPY EVERYTHING BELOW

Act as a mathematics teacher who teaches formulas and proofs through guided, fill-in-the-blank derivations so students construct the result themselves.

TASK: Create a fill-in-the-blank guided derivation worksheet for the result/formula/proof [FORMULA OR DERIVATION OR PROOF e.g. derivation of the quadratic formula] within topic [TOPIC], for [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Result/formula/proof to derive: [FORMULA OR DERIVATION OR PROOF]. Topic: [TOPIC]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM].

OUTPUT REQUIREMENTS:

1. Present the full derivation as a numbered sequence of logical steps, but REMOVE a key part of selected steps and replace it with a blank like _____ (1), _____ (2), and so on. Blank out roughly every second step so students must actively complete it.
2. Beside or above each step, write a short 'Reason' prompt asking why this step is valid (these reasons can also have blanks).
3. Start with a clear statement of what is given and what is to be proved or derived.
4. After the scaffolded version, add a short 'Your Turn' task: ask students to redo the entire derivation on a blank sheet with no help.
5. Provide a separate COMPLETED ANSWER KEY filling in every blank and every reason, showing the full correct derivation.
6. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Work the entire derivation yourself first and make sure it is mathematically valid end to end. Verify the final result by at least two independent checks (for example, test the derived formula on a known case). Never invent steps, theorems or values; every step must follow logically. Ensure the blanks are meaningful, not trivial.

SIGNATURE: Put 'Compiled by Indrajeet Yadav' in the header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.9 Match-the-Column / Sorting Worksheet

Text only

Any AI chat

What you get: A matching and sorting activity that tests connections, not just calculation.

How to use: Use this for a lively revision activity where students match or sort items into groups.

COPY EVERYTHING BELOW

Act as a mathematics teacher who designs matching and sorting activities to test whether students understand connections between concepts, not just calculations.

TASK: Create a match-the-column and sorting worksheet on [TOPIC] for [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Topic: [TOPIC]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM]. Number of matching pairs: [NUMBER e.g. 8]. Difficulty: [DIFFICULTY].

OUTPUT REQUIREMENTS:

1. SECTION A - Match the Columns: present Column A (for example, expressions, problems, statements or terms) and Column B (their results, equivalents, definitions or graphs) as two clearly numbered/lettered lists. Deliberately scramble the order so they are NOT already aligned. Include 1 or 2 extra distractor entries in Column B that match nothing, to prevent guessing.
2. SECTION B - Sort into Groups: give a list of 8 to 12 items and 3 or 4 named categories, and ask students to sort each item into the correct category.
3. SECTION C - Odd One Out: give 3 small groups of 4 items each where one item does not belong; students circle it and explain why in one line.
4. Keep all items unambiguous so each has exactly one correct placement.
5. Provide a separate ANSWER KEY giving the correct matches (e.g. 1-c, 2-a), the correct sorting, the odd ones out, and a one-line reason for each.
6. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Solve and check the matching, sorting and odd-one-out yourself first. Verify every intended answer by at least two independent checks and remove any item with more than one defensible answer. Never invent formulas or values; use only standard mathematics. Avoid near-duplicate pairs.

SIGNATURE: Put 'Compiled by Indrajeet Yadav' in the header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

6.10 Scaffolded Worksheet (Each Question Hints the Next)

Text only

Any AI chat

What you get: A chain of questions where each answer becomes the stepping stone to the next.

How to use: Pick one big problem and the AI breaks it into a guided ladder of smaller linked questions.

COPY EVERYTHING BELOW

Act as a mathematics teacher who builds scaffolded question chains, where solving one part gives students exactly what they need for the next, leading them to a big result step by step.

TASK: Create a scaffolded worksheet on [TOPIC] for [CLASS/GRADE] students of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT], built around the central goal/big problem [BIG PROBLEM OR GOAL e.g. find the area between two curves]. Break that goal into a connected ladder of smaller questions.

INPUT FROM ME: Topic: [TOPIC]. Central big problem or goal: [BIG PROBLEM OR GOAL]. Class: [CLASS/GRADE]. Board/exam: [BOARD or EXAM]. Number of steps: [NUMBER e.g. 6].

OUTPUT REQUIREMENTS:

1. Decompose the big goal into a numbered ladder of [NUMBER] linked sub-questions where the answer (or insight) from each question feeds directly into the next. Make the dependency explicit, for example, 'Using your answer to Q2, now find...'.
2. Start the ladder very easy and make the final 1 or 2 rungs combine everything to solve the original big problem.
3. Add brief 'Why this step matters' notes between rungs so students see the strategy.
4. Finish with a 'Now do it alone' transfer question: a similar big problem with no scaffolding, so students prove they can build the whole chain themselves.
5. Provide a separate ANSWER KEY with the worked final answer to every rung and to the transfer question, showing how each links to the next.
6. Write all mathematics in clean LaTeX or tidy plain text such as x^2 , $\sqrt{x}$, $(a+b)/c$, fully legible.

QUALITY BARS: Solve the entire chain yourself first and confirm that each rung's output truly feeds the next. Verify every final answer by at least two independent checks and fix any break in the chain. Never invent formulas, theorems or values; use only standard mathematics. Ensure the scaffolding is genuine and each rung is necessary.

SIGNATURE: Put 'Compiled by Indrajeet Yadav' in the header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

Daily and weekly practice sets with hints and answer keys to build momentum.

7.1 Daily 10-Question DPP with Hints + Answers

Text only

Any AI chat

What you get: A ready-to-print daily sheet of 10 questions, with hints and a full answer key.

How to use: Just fill in your topic, class and board, paste the prompt, and you get a complete daily sheet you can print or share on WhatsApp.

COPY EVERYTHING BELOW

Act as an experienced mathematics teacher and question-paper setter for Indian and international school exams.

TASK: Create one Daily Practice Problem (DPP) sheet of exactly 10 questions on the topic [TOPIC] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. The sheet should take a focused student about [TIME e.g. 30 minutes] to finish.

INPUT FROM ME: Topic = [TOPIC]. Class = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. Overall difficulty = [DIFFICULTY e.g. easy / moderate / mixed]. If anything is unclear, make a sensible assumption and state it in one line at the top.

QUESTION MIX: Make question 1-3 easy (warm-up), 4-8 moderate (core practice), and 9-10 challenging (stretch). Use a variety of forms: some short-answer, at least 2 word problems set in real Indian contexts (rupees, local examples), and at least 1 conceptual or 'find the mistake' question. Number every question clearly.

OUTPUT REQUIREMENTS, in this exact order:

1. A title block: 'Daily Practice Problem (DPP) - [TOPIC]' with Class, Board/Exam, suggested time, and the line 'Compiled by Indrajeet Yadav'.
2. SECTION A - QUESTIONS: the 10 numbered questions only, no answers shown here, so a teacher can print this part for students.
3. SECTION B - HINTS: one short one-line hint per question (the first nudge, not the full method).
4. SECTION C - ANSWER KEY: the final answer to every question.
5. SECTION D - FULL SOLUTIONS: clean step-by-step working for each question.

CORRECTNESS RULES (very important): Solve every problem yourself internally BEFORE writing it. Verify each final answer by at least two independent checks (for example substitute back into the original equation, use an alternate method, or estimate to confirm the magnitude). Never invent formulas, theorems or values; use only standard, correct mathematics appropriate for [CLASS/GRADE]. Keep all mathematics fully legible using clean LaTeX or clear plain-text math (write fractions, powers and roots unambiguously). Make sure every question actually has a clean, solvable answer suitable for daily practice.

FORMAT: Use clear headings and numbering so it can be copied straight into a Word document or printed. End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title block.

7.2 7-Day Weekly DPP Series on One Chapter

Text only

Any AI chat

What you get: Seven connected daily sheets that walk one chapter from basics to mastery over a week.

How to use: Name the chapter and your class, and you get a full week of daily sheets that you can release one per day.

COPY EVERYTHING BELOW

Act as a senior mathematics teacher who designs structured weekly practice plans for school and coaching students.

TASK: Build a 7-day Daily Practice Problem (DPP) series on the single chapter [CHAPTER/TOPIC] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. Each day must have its own DPP sheet so I can release one per day across a week.

INPUT FROM ME: Chapter = [CHAPTER/TOPIC]. Class = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. Questions per day = [NUMBER OF QUESTIONS e.g. 6]. Daily time = [TIME e.g. 25 minutes]. If the chapter is large, choose the most exam-relevant sub-topics and tell me your plan in one line.

PROGRESSION: Spread the chapter across the week so difficulty and depth build day by day. Day 1 = foundational definitions and basic problems. Days 2-4 = core skills and standard exam questions, one sub-topic focus per day. Day 5 = mixed problems from everything so far. Day 6 = higher-order and tricky problems. Day 7 = a mini self-test covering the whole chapter. At the start of each day, write a one-line 'Today's focus' and a one-line 'Yesterday's recap'.

OUTPUT REQUIREMENTS: For each of the 7 days produce, clearly separated: (a) a day header 'DAY n - [sub-topic]' with the focus and recap lines; (b) the numbered questions for that day; (c) a short hint for each question; (d) the answer key; (e) brief step-by-step solutions. Put a single title block at the very top: 'Weekly DPP Series - [CHAPTER/TOPIC]' with Class, Board/Exam, and 'Compiled by Indrajeet Yadav'.

CORRECTNESS RULES: Solve every question yourself first. Verify each final answer with at least two independent checks (back-substitution, an alternate method, or a sanity estimate). Never invent formulas, theorems or values; stay within the correct syllabus level for [CLASS/GRADE]. Keep all math fully legible in clean LaTeX or clear plain-text. Ensure no question repeats across days unless it is a deliberate recap, and label recaps as such.

FORMAT: Use clean headings and numbering so each day can be copied out and printed separately. End the whole output with a footer line: 'Prepared by Indrajeet Yadav'.

7.3 Timed Sprint DPP (Beat the Clock)

Text only

Any AI chat

What you get: A fast-paced timed challenge sheet with per-question time limits and a scoring scale.

How to use: Pick your topic and a total time, paste, and you get a speed-drill sheet students can race against the clock.

COPY EVERYTHING BELOW

Act as a mathematics coach who trains students to solve accurately AND quickly under time pressure.

TASK: Create a timed 'sprint' Daily Practice Problem (DPP) on [TOPIC] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. The whole sprint must fit inside [TIME e.g. 15 minutes].

INPUT FROM ME: Topic = [TOPIC]. Class = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. Number of questions = [NUMBER OF QUESTIONS e.g. 12]. Total time = [TIME]. Question style = [short objective / MCQ / one-line answer]. If I leave the style blank, use short objective questions that can each be solved quickly.

DESIGN: Favour short, crisp questions that reward speed and mental-math tricks rather than long derivations. Assign a suggested time to each question (for example 'aim: 60 sec') so the total matches [TIME]. Order questions from quickest to slightly harder so students warm up fast. Include 2-3 questions that have a smart shortcut, and reveal that shortcut in the solution.

OUTPUT REQUIREMENTS, in this order:

1. Title block: 'Timed Sprint DPP - [TOPIC]' with Class, Board/Exam, total time, total marks, and 'Compiled by Indrajeet Yadav'.
2. RULES line: time allowed, how to score (for example +1 correct, 0 wrong, and note if guessing is discouraged).
3. SPRINT QUESTIONS: numbered questions each with its target time in brackets.
4. QUICK ANSWER KEY: a compact list of final answers for instant self-checking.
5. SPEED SOLUTIONS: the fastest correct method for each, with shortcuts highlighted.
6. SCORING SCALE: a simple 'how did you do' table (for example 'all correct = excellent; explanation of what each band means').

CORRECTNESS RULES: Solve everything yourself first and verify each final answer with at least two independent checks (alternate method, back-substitution, or estimate). Never invent formulas, values or 'tricks' that are not mathematically valid; every shortcut must be genuinely correct. Keep all math fully legible in clean LaTeX or clear plain-text. Make sure each question is genuinely solvable within its suggested time.

FORMAT: Clean headings and numbering, copy-paste ready. End with a footer line: 'Prepared by Indrajeet Yadav'.

7.4 Mixed-Revision DPP Across Chapters

Text only

Any AI chat

What you get: A single mixed sheet that pulls questions from several chapters, tagged by topic.

How to use: List the chapters you have already taught and you get one blended revision sheet that keeps everything fresh.

COPY EVERYTHING BELOW

Act as a mathematics teacher who builds spaced-revision practice so students do not forget earlier chapters.

TASK: Create one Mixed-Revision Daily Practice Problem (DPP) that blends questions from several chapters for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. This sheet is for revising old chapters, not learning new ones.

INPUT FROM ME: Chapters to revise = [LIST THE CHAPTERS HERE, e.g. Linear Equations, Triangles, Probability]. Class = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. Total questions = [NUMBER OF QUESTIONS e.g. 12]. Difficulty = [DIFFICULTY e.g. mixed]. If I forget to list chapters, ask me once, then proceed with sensible common chapters for that class.

DESIGN: Distribute the questions fairly across all the listed chapters so each gets covered. Deliberately shuffle the order so two consecutive questions are usually from different chapters (this is what makes revision effective). Next to each question, put a small tag in brackets showing which chapter it tests, for example '(Probability)'. Include at least 2 questions that combine ideas from two chapters at once.

OUTPUT REQUIREMENTS, in this order:

1. Title block: 'Mixed-Revision DPP' with the chapter list, Class, Board/Exam, total marks, suggested time, and 'Compiled by Indrajeet Yadav'.
2. QUESTIONS: numbered, each ending with its chapter tag in brackets.
3. ANSWER KEY: final answers in a compact list.
4. SOLUTIONS: clear step-by-step working for each.
5. COVERAGE TABLE: a short table showing how many questions came from each chapter, so I can see the balance.

CORRECTNESS RULES: Solve every problem yourself first and verify each final answer with at least two independent checks (alternate method, back-substitution, or sanity estimate). Never invent formulas, theorems or values; keep everything at the correct level for [CLASS/GRADE]. Keep all mathematics fully legible in clean LaTeX or clear plain-text, with fractions, powers and roots written unambiguously. Ensure every chapter tag is accurate.

FORMAT: Clean headings and numbering, copy-paste ready for printing. End with a footer line: 'Prepared by Indrajeet Yadav'.

7.5 Level-Up DPP That Gets Harder Each Day

Text only

Any AI chat

What you get: A five-level daily ladder where each day's questions are noticeably tougher than the last.

How to use: Choose a topic and you get a step-by-step difficulty ladder to push students from comfortable to challenged.

COPY EVERYTHING BELOW

Act as a mathematics mentor who designs progressive 'level-up' practice that gradually raises difficulty.

TASK: Build a Level-Up Daily Practice Problem (DPP) ladder of 5 daily levels on [TOPIC] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. Each level is one day's sheet and must be clearly harder than the day before.

INPUT FROM ME: Topic = [TOPIC]. Class = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. Questions per level = [NUMBER OF QUESTIONS e.g. 5]. Daily time = [TIME e.g. 20 minutes]. If the topic is too small for 5 distinct levels, tell me and adjust sensibly.

LEVEL DESIGN: Level 1 = Beginner (direct application of one idea). Level 2 = Easy (two simple steps). Level 3 = Intermediate (standard exam-level). Level 4 = Advanced (multi-step or unusual framing). Level 5 = Mastery (challenge problems, higher-order thinking, exam-topper level). At the top of each level write a one-line 'What changes at this level' note describing why it is harder, and a 'Pass mark to advance' suggestion (for example 'get at least 4 of 5 right before moving up').

OUTPUT REQUIREMENTS: For each of the 5 levels, clearly separated, give: (a) a header 'LEVEL n - [name]' with the 'what changes' note and pass mark; (b) numbered questions; (c) one hint per question; (d) the answer key; (e) full step-by-step solutions. Put a single title block at the very top: 'Level-Up DPP - [TOPIC]' with Class, Board/Exam, and 'Compiled by Indrajeet Yadav'.

CORRECTNESS RULES: Solve every question yourself first and verify each final answer with at least two independent checks (alternate method, back-substitution, or estimate). Make sure each level is genuinely harder than the previous one, not just longer. Never invent formulas, theorems or values; stay within correct mathematics for [CLASS/GRADE] while stretching difficulty honestly. Keep all math fully legible in clean LaTeX or clear plain-text.

FORMAT: Clean headings and numbering so each level can be printed on its own. End the whole output with a footer line: 'Prepared by Indrajeet Yadav'.

7.6 Exam-Countdown DPP (Last 30 Days Plan)

Text only

Gemini / ChatGPT

What you get: A 30-day day-by-day DPP calendar with daily topics, question counts and revision days built in.

How to use: Tell it the exam and the chapters and you get a full month of dated daily practice leading right up to exam day.

COPY EVERYTHING BELOW

Act as a mathematics exam strategist who builds final-month revision schedules for students.

TASK: Create a 30-day Exam-Countdown Daily Practice Problem (DPP) plan for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. This is a last-30-days plan with a small daily practice set assigned to each day.

INPUT FROM ME: Exam = [BOARD or EXAM]. Class = [CLASS/GRADE]. Chapters/syllabus to cover = [LIST THE CHAPTERS OR SYLLABUS]. Questions per day = [NUMBER OF QUESTIONS e.g. 8]. Daily study time = [TIME e.g. 45 minutes]. Weak areas to emphasise = [OPTIONAL: LIST WEAK CHAPTERS]. If I leave the chapter list blank, use the standard syllabus for that exam and class and say so.

PLANNING: Spread all chapters across Days 1-22 with the heaviest or weak chapters getting more days. Make Days 23-26 mixed full-syllabus revision. Make Days 27-29 timed mock-style mixed practice. Make Day 30 a light confidence sheet plus exam-day reminders. Every day must have: a date label 'Day n', a clear topic focus, the day's question count, and a one-line goal.

OUTPUT REQUIREMENTS, in this order:

1. Title block: '30-Day Exam-Countdown DPP - [BOARD or EXAM]' with Class, the syllabus list, and 'Compiled by Indrajeet Yadav'.
2. CALENDAR TABLE: a 30-row table with columns Day, Topic Focus, Number of Questions, Goal.
3. SAMPLE DPP SHEETS: write out the FULL DPP (questions + hints + answer key + solutions) for Day 1, Day 15 and Day 30 as ready-to-use examples, and clearly tell me 'reply with a day number to generate that day's full sheet'.

CORRECTNESS RULES: For every sample sheet, solve each question yourself first and verify each final answer with at least two independent checks (alternate method, back-substitution, or estimate). Never invent formulas, theorems or values; keep everything at the correct level for [CLASS/GRADE] and [BOARD or EXAM]. Keep all math fully legible in clean LaTeX or clear plain-text. Make the calendar realistic for the stated daily time.

FORMAT: Clean headings, a proper table, copy-paste ready. End the output with a footer line: 'Prepared by Indrajeet Yadav'.

7.7 Single-Concept Mastery DPP

Text only

Any AI chat

What you get: A laser-focused sheet that drills one narrow concept from every possible angle until it sticks.

How to use: Name one small concept students keep getting wrong, and you get a deep drill sheet that hammers just that idea.

COPY EVERYTHING BELOW

Act as a mathematics teacher who specialises in fixing one specific concept that students repeatedly get wrong.

TASK: Create a Single-Concept Mastery Daily Practice Problem (DPP) that drills ONLY the one narrow concept [CONCEPT e.g. sign rules in inequalities / area of a triangle by coordinates / converting word problems to equations] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Concept = [CONCEPT]. Class = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. Number of questions = [NUMBER OF QUESTIONS e.g. 10]. Difficulty = [DIFFICULTY e.g. easy-to-moderate]. If the concept is broad, narrow it to the single most error-prone sub-skill and tell me which one you chose.

DESIGN: Stay tightly on this one concept; do NOT drift into other topics. Approach the concept from many angles: direct questions, reversed/'work backwards' questions, true-or-false statements, a 'spot the common mistake' question, a real-life application, and at least one 'why does this work' conceptual question. Start with a 1-2 line 'Key idea' refresher box at the top so students recall the rule before practising.

OUTPUT REQUIREMENTS, in this order:

1. Title block: 'Single-Concept Mastery DPP - [CONCEPT]' with Class, Board/Exam, and 'Compiled by Indrajeet Yadav'.
2. KEY IDEA: the short refresher box.
3. COMMON MISTAKES: 2-3 bullet points listing the typical errors students make with this concept.
4. QUESTIONS: numbered, all on this single concept.
5. HINTS: one short hint per question.
6. ANSWER KEY and then FULL SOLUTIONS, with each solution pointing out where students usually slip.

CORRECTNESS RULES: Solve every question yourself first and verify each final answer with at least two independent checks (alternate method, back-substitution, or estimate). Never invent formulas, theorems or values; the 'key idea' and 'common mistakes' must be genuinely accurate for [CLASS/GRADE]. Keep all math fully legible in clean LaTeX or clear plain-text. Make sure every question truly isolates this one concept.

FORMAT: Clean headings and numbering, copy-paste ready for printing. End with a footer line: 'Prepared by Indrajeet Yadav'.

7.8 One Tricky Problem a Day (with Full Reasoning)

Text only

Any AI chat

What you get: A set of standalone brain-teaser problems, each with hints and a full guided solution.

How to use: Pick a topic and how many days you want, and you get one share-ready tricky problem per day with deep reasoning.

COPY EVERYTHING BELOW

Act as a mathematics olympiad-style mentor who picks one beautiful, thought-provoking problem per day to sharpen reasoning.

TASK: Create a 'One Tricky Problem a Day' Daily Practice Problem (DPP) set on [TOPIC] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. Produce one carefully chosen tricky problem for each of [NUMBER OF DAYS e.g. 7] days.

INPUT FROM ME: Topic = [TOPIC]. Class = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. Number of days = [NUMBER OF DAYS e.g. 7]. Difficulty = [DIFFICULTY e.g. challenging but fair]. If the topic is too narrow for that many distinct good problems, tell me and reduce the number sensibly.

DESIGN: Each problem must be genuinely interesting - a clever twist, a hidden shortcut, a common trap, or an elegant idea - not just a long routine sum. Vary the flavour across days (one geometric, one algebraic twist, one word-problem trap, and so on, as the topic allows). For each day, structure it so a student can attempt it first, then peek at staged help.

OUTPUT REQUIREMENTS: For each day, clearly separated, give in this order: (a) 'DAY n - THE PROBLEM' with the problem stated cleanly; (b) 'WHY IT IS TRICKY' - one line on the trap or twist (no spoilers of the answer); (c) 'HINT 1' and 'HINT 2' - two progressive nudges; (d) 'FULL REASONING' - a clear, friendly step-by-step solution that explains the thinking, not just the steps; (e) 'FINAL ANSWER' on its own line; (f) 'TAKEAWAY' - one line on the lesson or technique to remember. Put a single title block at the top: 'One Tricky Problem a Day - [TOPIC]' with Class, Board/Exam, and 'Compiled by Indrajeet Yadav'.

CORRECTNESS RULES: Solve every problem yourself first and verify each final answer with at least two independent checks (alternate method, back-substitution, or estimate). Never invent formulas, theorems, values or fake 'tricks'; every clever step must be genuinely valid for [CLASS/GRADE]. Keep all math fully legible in clean LaTeX or clear plain-text. Make sure each problem has a clean, well-defined answer.

FORMAT: Clean headings and numbering so each day can be shared on its own. End the whole output with a footer line: 'Prepared by Indrajeet Yadav'.



Crystal-clear explanations, intuition and visual mind maps for any topic.

8.1 One-Page Concept Explainer

Text only

Any AI chat

What you get: A single, printable page that explains any topic clearly.

How to use: Just fill in the topic and class, paste it in, and print the result for your students.

COPY EVERYTHING BELOW

Act as an expert mathematics teacher and instructional designer who is brilliant at making hard ideas simple.

TASK: Create a clear ONE-PAGE concept explainer on the topic [TOPIC] for [CLASS/GRADE] students following [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. It must fit on a single printed A4 page when copied into a document.

INPUT FROM ME: Topic = [TOPIC]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. (Optional) Specific sub-points to include = [PASTE ANY SUB-POINTS OR LEAVE BLANK].

OUTPUT REQUIREMENTS - produce these sections in this exact order:

1. Title block (topic name + 'Compiled by Indrajeet Yadav').
2. 'In one sentence' - the whole idea in a single plain-English line.
3. 'The core idea' - 3 to 5 short paragraphs in simple language a [CLASS/GRADE] student can read alone.
4. 'Key definitions' - each term defined in one line, with the symbol if any.
5. 'Key formulas' - listed in clean LaTeX (or clear plain-text math), each with a one-line note on when to use it. State the conditions/assumptions for each formula.
6. 'Worked mini-example' - one short solved example, every step shown and legible.
7. 'Remember this' - 3 to 5 bullet takeaways students must not forget.
8. 'One quick self-check question' with the answer hidden one line below.

QUALITY BARS: Solve every example internally first, then VERIFY the final answer by at least two independent methods (for example, re-derive it a second way and also sanity-check with numbers) before writing it. Never invent theorems, formulas, or values - if something is not standard for [BOARD or EXAM], say so plainly. Keep all mathematics fully legible with clean LaTeX or clear plain-text math. Use simple words; avoid jargon unless you define it. Plain ASCII only, straight quotes.

FINISH: End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title block at the top.

8.2 Chapter Mind Map (Text Tree)

Text only

Any AI chat

What you get: A neat branching tree map of a whole chapter you can copy to the board.

How to use: Type your chapter name and grade, paste this, and you get a ready tree map to draw on the board.

COPY EVERYTHING BELOW

Act as an expert mathematics teacher and a master of visual note-making and mind maps.

TASK: Build a complete VISUAL MIND MAP of the chapter [TOPIC/CHAPTER] for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT], drawn as an indented text tree using simple characters (- and spaces and arrows →) so I can copy it straight onto a whiteboard or into notes.

INPUT FROM ME: Chapter = [TOPIC/CHAPTER]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM].

OUTPUT REQUIREMENTS:

1. Put the chapter name as the central node at the top.
2. Create 4 to 7 MAIN BRANCHES (the big sub-topics). Under each, add SUB-BRANCHES, and under those add leaf nodes (key formulas, definitions, or facts).
3. Beside each formula, show it in clean plain-text math and note its condition in brackets.
4. Use indentation and dashes to show levels clearly, like:
 - MAIN BRANCH
 - sub-branch
 - leaf (formula or fact)
5. After the tree, add a short 'How the branches connect' note (3 to 5 lines) explaining the links between branches with arrows, e.g. 'Branch A → feeds into → Branch C'.
6. Keep the whole map balanced - no branch should be far larger than the others.

QUALITY BARS: Before writing, list every formula internally and VERIFY each one by two independent checks (re-derive it, and test it on one simple number) so nothing is wrong. Never invent theorems, formulas, or values; include only content standard for [BOARD or EXAM]. Keep all mathematics fully legible in clean LaTeX or clear plain-text math. Plain ASCII only, straight quotes, normal hyphens.

FINISH: Put 'Compiled by Indrajeet Yadav' in the title/header block above the central node, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

8.3 Real-World Analogy Explainer

Text only

Any AI chat

What you get: Everyday analogies that make an abstract topic click instantly.

How to use: Drop in your tricky topic and class; the AI gives you relatable real-life comparisons to use in class.

COPY EVERYTHING BELOW

Act as a beloved mathematics teacher famous for explaining abstract ideas using everyday real-world analogies.

TASK: Explain the topic [TOPIC] to [CLASS/GRADE] students (following [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]) using clear, relatable REAL-WORLD ANALOGIES from daily Indian and global life (kitchen, cricket, money, travel, shopping, water tanks, traffic, etc.).

INPUT FROM ME: Topic = [TOPIC]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. (Optional) Context my students relate to = [PASTE ANY CONTEXT OR LEAVE BLANK].

OUTPUT REQUIREMENTS:

1. 'The idea in plain words' - 2 to 3 lines, no jargon.
2. 'Main analogy' - one strong everyday comparison, explained step by step, mapping each part of the analogy to the exact part of the math (use a small two-column 'Real life → Maths' table).
3. 'Two more quick analogies' - two shorter alternative comparisons for variety.
4. 'Where the analogy breaks' - 2 to 3 honest lines on where the comparison stops being accurate, so students do not over-trust it.
5. 'Back to the real maths' - now state the proper definition and any key formula in clean LaTeX or plain-text math, with conditions.
6. 'Try it' - one tiny question that uses the analogy, with the answer one line below.

QUALITY BARS: Solve any numbers internally first and VERIFY the final answer by at least two independent methods before writing it. Never invent theorems, formulas, or values; keep everything correct and standard for [BOARD or EXAM]. Analogies must illuminate, never mislead. Keep all mathematics fully legible with clean LaTeX or clear plain-text math. Use simple, warm language. Plain ASCII only, straight quotes, normal hyphens.

FINISH: Put 'Compiled by Indrajeet Yadav' in the title/header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

8.4 Story-Based Concept Explanation

Text only

Any AI chat

What you get: A short memorable story that teaches the concept end to end.

How to use: Give your topic and class, and the AI writes a fun story your students will actually remember the concept by.

COPY EVERYTHING BELOW

Act as a gifted mathematics storyteller who teaches concepts through short, memorable narratives.

TASK: Teach the topic [TOPIC] to [CLASS/GRADE] students (board/exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]) by writing a short STORY or narrative in which the mathematics is the hero of the plot.

INPUT FROM ME: Topic = [TOPIC]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. (Optional) Setting or characters I want = [PASTE OR LEAVE BLANK].

OUTPUT REQUIREMENTS:

1. 'The story' - 200 to 350 words, with one or two relatable characters facing a problem that can only be solved by understanding [TOPIC]. The maths must appear naturally inside the plot, not bolted on.
2. As the story solves the problem, show the actual steps and any formula in clean LaTeX or plain-text math, with conditions clearly stated.
3. 'What just happened (the maths)' - after the story, a clear 5 to 8 line plain explanation translating the story back into the real concept and definitions.
4. 'The one formula to keep' - the single most important formula, boxed in text with a one-line note on when to use it.
5. 'Story recap question' - one question tied to the story, with the answer one line below.

QUALITY BARS: Work out all maths internally first and VERIFY every final value by at least two independent checks before putting it in the story. Never invent theorems, formulas, or values; keep all content accurate and standard for [BOARD or EXAM]. The story may be playful but the mathematics must be exact and fully legible in clean LaTeX or clear plain-text math. Plain ASCII only, straight quotes, normal hyphens.

FINISH: Put 'Compiled by Indrajeet Yadav' in the title/header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

8.5 Intuition First, Formula Later

Text only

Any AI chat

What you get: Deep intuition built up before any formula is introduced.

How to use: Enter your topic and class to get an explanation that builds the gut feeling first, then reveals the formula.

COPY EVERYTHING BELOW

Act as a master mathematics teacher who always builds deep intuition BEFORE showing any formula.

TASK: Explain [TOPIC] to [CLASS/GRADE] students (board/exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]) using an 'intuition first, formula later' approach. The formula must only appear AFTER the student already feels why it must be true.

INPUT FROM ME: Topic = [TOPIC]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM].

OUTPUT REQUIREMENTS, in this exact order:

1. 'Start with a feeling' - a question or simple situation that makes the student curious and gives a gut sense of the idea, with NO formula yet.
2. 'Build the picture' - 3 to 5 short steps (using small numbers, a pattern, or a simple diagram described in words) that grow the intuition. Let the student almost guess the rule.
3. 'Now name it' - reveal the actual concept name and what the student was sensing all along.
4. 'And here is the formula' - only now, present the formula in clean LaTeX or plain-text math, and show clearly how it matches the intuition just built. State all conditions.
5. 'Quick derivation or justification' - a short, legible reason the formula holds.
6. 'Test your intuition' - one question, answer one line below.

QUALITY BARS: Solve all numeric work internally first and VERIFY each final answer by at least two independent methods before writing it. Never invent theorems, formulas, or values; keep everything standard for [BOARD or EXAM]. The intuition steps must lead honestly to the correct formula. Keep all mathematics fully legible in clean LaTeX or clear plain-text math. Plain ASCII only, straight quotes, normal hyphens.

FINISH: Put 'Compiled by Indrajeet Yadav' in the title/header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

8.6 Topic Connection Map

Text only

Any AI chat

What you get: A map showing how topics link to and depend on each other.

How to use: List the topics or chapter and class; the AI shows how everything connects so students see the bigger picture.

COPY EVERYTHING BELOW

Act as an expert mathematics teacher who helps students see how everything in maths connects, not as isolated chapters.

TASK: Build a CONNECTION MAP showing how [TOPIC] links to other topics in the [CLASS/GRADE] syllabus of [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT] - what it depends on, what depends on it, and what it sits beside.

INPUT FROM ME: Central topic = [TOPIC]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM].
(Optional) Specific topics to relate = [PASTE A LIST OR LEAVE BLANK].

OUTPUT REQUIREMENTS:

1. 'Central topic' stated clearly at the top.
2. 'Comes from (prerequisites)' - 3 to 6 topics that feed INTO this one, each with a one-line 'why it matters here'.
3. 'Leads to (what it unlocks)' - 3 to 6 later topics that USE this one, each with a one-line link.
4. 'Sits beside (siblings)' - related topics at the same level and how they differ.
5. 'Connection diagram' - an arrow map in plain text, e.g. 'Prerequisite → [TOPIC] → Advanced use', covering the most important links.
6. 'The big-picture sentence' - one line a student can memorise to see where this topic fits in all of mathematics.
7. 'One cross-over question' that needs two of these connected topics together, with the answer one line below.

QUALITY BARS: Verify any maths internally by at least two independent checks before writing it. Never invent theorems, formulas, values, or fake connections; every link must be genuine and standard for [BOARD or EXAM]. Keep all mathematics fully legible in clean LaTeX or clear plain-text math. Plain ASCII only, straight quotes, normal hyphens.

FINISH: Put 'Compiled by Indrajeet Yadav' in the title/header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

8.7 Misconceptions vs Correct Idea

Text only

Any AI chat

What you get: Common student mistakes paired with the correct understanding.

How to use: Type the topic and class to get the exact wrong ideas students hold and how to fix each one.

COPY EVERYTHING BELOW

Act as an experienced mathematics teacher who knows exactly where students go wrong and how to correct their thinking.

TASK: For the topic [TOPIC] at [CLASS/GRADE] level (board/exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]), list the most COMMON MISCONCEPTIONS students hold and pair each one with the CORRECT idea and a clear fix.

INPUT FROM ME: Topic = [TOPIC]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. (Optional) Mistakes I keep seeing = [PASTE OR LEAVE BLANK].

OUTPUT REQUIREMENTS:

1. A short intro line: why this topic trips students up.
2. A table or clear list of 6 to 10 entries, each with these four parts:
 - 'The wrong belief' (stated the way a student would actually say it).
 - 'Why it feels right' (the trap that makes it tempting).
 - 'The correct idea' (the accurate concept, with formula in clean LaTeX or plain-text math and its conditions).
 - 'Quick fix / memory hook' (a one-line correction or trick to never repeat the error).
3. 'Spot-the-error drill' - 2 short questions containing a typical mistake; ask the student to find and fix the error. Put the corrected solutions a few lines below under a 'Solutions' heading.

QUALITY BARS: For every 'correct idea', solve internally first and VERIFY by at least two independent methods so your correction is definitely right. Never invent theorems, formulas, or values; keep everything standard for [BOARD or EXAM]. Be careful that the WRONG beliefs are clearly labelled as wrong so no student copies them by mistake. Keep all mathematics fully legible in clean LaTeX or clear plain-text math. Plain ASCII only, straight quotes, normal hyphens.

FINISH: Put 'Compiled by Indrajeet Yadav' in the title/header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

8.8 Prerequisite Roadmap

Text only

Any AI chat

What you get: An ordered list of exactly what to learn before this topic.

How to use: Enter the target topic and class to get a step-by-step 'learn these first' roadmap for weak students.

COPY EVERYTHING BELOW

Act as an expert mathematics mentor who designs the perfect learning order so no student is ever lost.

TASK: Build a PREREQUISITE MAP for the topic [TOPIC] at [CLASS/GRADE] level (board/exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]) - the exact things a student must already know BEFORE they can learn [TOPIC], in the right order.

INPUT FROM ME: Target topic = [TOPIC]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM].
(Optional) What my students already struggle with = [PASTE OR LEAVE BLANK].

OUTPUT REQUIREMENTS:

1. 'Target' - state [TOPIC] and, in one line, what mastering it looks like.
2. 'Prerequisite ladder' - an ordered, numbered list (earliest first) of 5 to 9 prerequisite skills. For each rung give: the skill name, a one-line 'why you need it for [TOPIC]', and one tiny check-question the student should be able to answer.
3. 'Readiness checklist' - a short tick-box list a student can self-assess before starting [TOPIC].
4. 'Fast diagnostic' - 5 quick questions (one per key prerequisite) that reveal gaps. Put a 'Diagnostic answers' section a few lines below.
5. 'If a gap is found' - one line per prerequisite telling the teacher what to revise first.
6. A simple text arrow path at the end: 'Skill 1 → Skill 2 → ... → [TOPIC]'.

QUALITY BARS: Verify any maths internally by at least two independent checks before writing it. Never invent theorems, formulas, or values; every prerequisite must be genuinely required and standard for [BOARD or EXAM]. Order must be truly logical - nothing should require something listed later. Keep all mathematics fully legible in clean LaTeX or clear plain-text math. Plain ASCII only, straight quotes, normal hyphens.

FINISH: Put 'Compiled by Indrajeet Yadav' in the title/header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.

8.9 Mind Map Poster (Image)

Text only

ChatGPT (image) / Gemini

What you get: A colourful printable mind-map poster image of the topic.

How to use: Use an AI that makes images, type your topic and class, and download a ready-to-print classroom poster.

COPY EVERYTHING BELOW

Act as a professional educational designer creating a beautiful, classroom-ready MIND MAP POSTER as an IMAGE.

TASK: Design and generate a single colourful, well-organised mind-map POSTER IMAGE for the topic [TOPIC] for [CLASS/GRADE] students under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT], suitable for printing and putting on a classroom wall.

INPUT FROM ME: Topic = [TOPIC]. Class/Grade = [CLASS/GRADE]. Board/Exam = [BOARD or EXAM]. (Optional) Colour theme = [PASTE OR LEAVE BLANK].

DESIGN AND CONTENT REQUIREMENTS:

1. Put [TOPIC] in a bold central bubble in the middle of the poster.
2. Draw 5 to 7 curved branches radiating out, each a different colour, for the main sub-topics; add smaller twigs for key formulas, definitions, and facts.
3. Use simple icons or small illustrations on branches to aid memory.
4. Show the most important formulas clearly and CORRECTLY on the branches; every symbol must be legible and accurate.
5. Use a clean, readable font, generous spacing, and a layout that is balanced and not cramped. Portrait A4 or A3 proportions.
6. Keep the wording short on the poster (it is a visual aid, not paragraphs).

ACCURACY: Before generating, work out and VERIFY internally that every formula and fact on the poster is correct, checking each by at least two independent methods. Never invent theorems, formulas, or values; include only content standard for [BOARD or EXAM]. Double-check that no symbol is garbled or misspelt in the image.

FINISH: Place the title text 'Compiled by Indrajeet Yadav' clearly in the header area of the poster. Sign the bottom-right of every page with an elegant handwritten signature 'Indrajeet Yadav' with a small underline flourish.

8.10 5 Levels of the Same Idea

Text only

Any AI chat

What you get: One concept explained from child level up to deep level.

How to use: Pick a topic and the AI explains it five ways, so you can pitch it to any student in the room.

COPY EVERYTHING BELOW

Act as a versatile mathematics teacher who can explain one idea at five different depths for five different audiences.

TASK: Explain the topic [TOPIC] at FIVE escalating levels of depth, so the same idea is made clear to a child, a teenager, a board student, a competitive aspirant, and a deep learner.

INPUT FROM ME: Topic = [TOPIC]. Main exam I care about = [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. (Optional) My students' usual class = [CLASS/GRADE OR LEAVE BLANK].

OUTPUT REQUIREMENTS - write all five levels, clearly labelled, each building on the last:

1. 'Level 1 - Child (around age 8)': pure intuition, everyday words, no symbols, maybe one tiny picture described in words.
2. 'Level 2 - Teen (around age 14)': introduce the idea properly with the first simple formula in plain-text math and one small example.
3. 'Level 3 - Board student': the full syllabus treatment for [BOARD or EXAM], proper definition, key formulas in clean LaTeX or plain-text math with conditions, and one solved board-style example.
4. 'Level 4 - Competitive aspirant (JEE/NEET/Olympiad level)': sharper insight, a clever trick or shortcut, a tougher worked example, and common traps.
5. 'Level 5 - Deep learner': the why behind it - a short rigorous justification, links to higher mathematics, and any elegant generalisation.

Finish with a one-line 'Through-line' showing it is the SAME idea growing across all five levels.

QUALITY BARS: Solve every example internally first and VERIFY each final answer by at least two independent methods before writing it. Never invent theorems, formulas, or values; keep all content accurate and standard. Each level must be honest and not contradict the levels around it. Keep all mathematics fully legible in clean LaTeX or clear plain-text math. Plain ASCII only, straight quotes, normal hyphens.

FINISH: Put 'Compiled by Indrajeet Yadav' in the title/header block, and end the output with a footer line: 'Prepared by Indrajeet Yadav'.



Clean, correctly labelled diagrams, graphs and geometry figures as images.

9.1 Clean Labelled Geometry Figure from a Description

Text only

ChatGPT (image) / Gemini

What you get: One crisp, correctly labelled geometry diagram ready for the board or a worksheet.

How to use: Just describe the shape in plain words and the AI draws a neat, labelled figure you can save and print.

COPY EVERYTHING BELOW

Act as a senior mathematics teacher and technical illustrator who draws exam-quality geometry diagrams.

TASK: Create ONE clean, correctly labelled geometry figure as an image, based on my description below.

INPUT I AM GIVING YOU:

- Figure to draw: [DESCRIBE THE FIGURE, e.g. triangle ABC with a perpendicular from A to BC meeting at D]
- Given measurements / properties: [e.g. $AB = 5$ cm, angle B = 90 degrees, $BC = 12$ cm]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE]
- Topic this belongs to: [TOPIC e.g. Pythagoras theorem]

BEFORE DRAWING (do this silently): Work out the figure on paper internally. Confirm the shape is geometrically possible with the given data. Verify any computed value (lengths, angles) by at least two independent checks (for example, recompute a side using the Pythagoras theorem and again using trigonometry). Do NOT invent measurements, theorems, or properties I did not give; if something is impossible, say so clearly and stop.

OUTPUT REQUIREMENTS:

1. A single, neat, high-contrast figure on a plain white background, large enough to read.
2. Label every vertex with capital letters, mark all given lengths and angles exactly as provided, and use standard notation (small square for right angles, tick marks for equal sides, arc marks for equal angles).
3. Keep proportions roughly to scale so the figure looks correct, not distorted.
4. Use clean, fully legible text; no overlapping labels; no clutter.
5. Below the image, list in plain text every label and value you used, so I can verify nothing was changed.
6. Sign the bottom-right of every page with an elegant handwritten signature "Indrajeet Yadav" with a small underline flourish.

QUALITY BAR: This figure must be classroom-ready and printable in black and white without losing any label. If you are unsure a value is correct, recompute before drawing.

9.2 Accurate Function Graph with Axes, Intercepts and Asymptotes

Text only

ChatGPT (image) / Gemini

What you get: A precise graph of any function with labelled axes, key points and asymptotes.

How to use: Type the function you want graphed and the AI returns a tidy, labelled graph you can drop into notes.

COPY EVERYTHING BELOW

Act as a mathematics teacher and precise graph illustrator who produces exam-accurate function graphs.

TASK: Draw ONE accurate graph of the function I give, as a clean labelled image.

INPUT I AM GIVING YOU:

- Function: [PASTE THE FUNCTION HERE e.g. $y = (x^2 - 4) / (x - 1)$]
- Domain / x-range to show: [e.g. -10 to 10]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / JEE / SAT]

BEFORE DRAWING (do this silently): Compute the key features yourself: x-intercepts, y-intercept, vertical and horizontal (or slant) asymptotes, turning points, and behaviour at the ends. Verify each critical value by at least two independent methods (for example, find roots by factoring and again by substitution; confirm an asymptote by limits and again by long division). Never invent points or asymptotes; if a feature does not exist, say so.

OUTPUT REQUIREMENTS:

1. One clean graph on a white background with clearly labelled x-axis and y-axis, an even numbered scale, and visible gridlines.
2. Plot and label the y-intercept, every x-intercept, and any turning points with their coordinates.
3. Draw asymptotes as dashed lines and label their equations.
4. The curve must be smooth and correct in shape; mark the function name on the curve.
5. Keep all numbers and labels fully legible and non-overlapping.
6. Below the image, give a short plain-text table listing intercepts, asymptotes and turning points so I can check them.
7. End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

QUALITY BAR: The graph must be mathematically accurate and printable in black and white. Recompute any value you are unsure about before drawing.

9.3 Step-by-Step Geometric Construction (Ruler and Compass Style)

Text only

ChatGPT (image) / Gemini

What you get: A multi-step construction figure showing arcs and lines exactly as a student would draw them.

How to use: Name the construction you teach and the AI shows each step with arcs, so students copy it correctly.

COPY EVERYTHING BELOW

Act as a mathematics teacher who specialises in ruler-and-compass constructions and draws them step by step.

TASK: Produce a clear step-by-step construction figure (as an image) for the construction I name, in the classic ruler-and-compass style with visible arcs.

INPUT I AM GIVING YOU:

- Construction required: [DESCRIBE e.g. bisect a 70 degree angle / draw a perpendicular bisector of a 6 cm line / construct a triangle with sides 5, 6, 7 cm]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board]
- Topic: [TOPIC e.g. Constructions]

BEFORE DRAWING (do this silently): Plan the exact sequence of valid construction steps. Confirm the construction is possible with the given data and that the standard method is correct. Verify the final result by at least two independent checks (for example, confirm an angle bisector splits the angle into two equal parts by measurement and by symmetry). Do NOT skip steps or invent shortcuts that are not valid constructions.

OUTPUT REQUIREMENTS:

1. Show the construction as a numbered series of small frames OR one clear figure with each step labelled Step 1, Step 2, and so on.
2. Draw the compass arcs as light curved lines, the straightedge lines as solid lines, and keep the final answer line bold.
3. Label all points with capital letters and write the radius or measurement used at each step.
4. Below the image, write the construction steps in plain numbered text ("Step 1: ...") that a student can follow with a real compass.
5. Keep everything fully legible, uncluttered, and printable in black and white.
6. Sign the bottom-right of every page with an elegant handwritten signature "Indrajeet Yadav" with a small underline flourish.

QUALITY BAR: A student must be able to reproduce this exactly on paper. If any step is not a valid ruler-and-compass step, fix it before drawing.

9.4 Number Line / Inequality Illustration

Text only

ChatGPT (image) / Gemini

What you get: A tidy number line showing the solution of an inequality with correct open/closed circles.

How to use: Paste the inequality and the AI draws a clean number line with shading and the right circle types.

COPY EVERYTHING BELOW

Act as a mathematics teacher who draws clean number-line illustrations for inequalities and intervals.

TASK: Draw ONE clear number line (as an image) that shows the solution of the inequality or interval I give.

INPUT I AM GIVING YOU:

- Inequality or interval: [PASTE THE INEQUALITY HERE e.g. $-3 \leq x < 5$, or $2x + 1 > 7$]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / State Board / SAT]
- Topic: [TOPIC e.g. Linear inequalities]

BEFORE DRAWING (do this silently): Solve the inequality fully yourself. Decide which endpoints are included (closed/filled circle) and which are excluded (open/hollow circle), and which region to shade. Verify the solution by at least two independent checks (for example, test a value inside the region and a value outside it to confirm the direction). Never guess the direction of shading; confirm it by substitution.

OUTPUT REQUIREMENTS:

1. One horizontal number line on a white background with evenly spaced, clearly labelled integer marks covering the relevant range plus a little margin on each side.
2. Use a filled (closed) circle for included endpoints and a hollow (open) circle for excluded endpoints, and write the endpoint value below each circle.
3. Shade or thicken the solution region clearly, with an arrow if it extends to infinity.
4. Write the solution in interval notation and in inequality notation above the line.
5. Keep all marks and numbers fully legible and printable in black and white.
6. Below the image, show the solving steps in plain text and state the final solution set.
7. End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

QUALITY BAR: The open/closed circles and shading direction must be 100 percent correct. Re-test the boundary before drawing.

9.5 Neat 3D Solid Figure with Labels

Text only

ChatGPT (image) / Gemini

What you get: A clear 3D drawing of a solid with hidden edges dashed and all dimensions labelled.

How to use: Tell the AI which solid and its measurements, and it draws a neat 3D figure you can use in mensuration class.

COPY EVERYTHING BELOW

Act as a mathematics teacher and 3D figure illustrator who draws clear solids for mensuration lessons.

TASK: Draw ONE neat 3D solid figure (as an image), correctly labelled, based on the solid I describe.

INPUT I AM GIVING YOU:

- Solid: [DESCRIBE e.g. cylinder, cone, cuboid, sphere, frustum, prism]
- Dimensions: [e.g. radius 7 cm, height 10 cm]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board]
- Topic: [TOPIC e.g. Surface area and volume]

BEFORE DRAWING (do this silently): Picture the solid in correct proportion. If I ask for surface area or volume, compute it yourself and verify the result by at least two independent checks (for example, compute volume using the standard formula and again by a unit-consistency check, confirming the units come out as cubic units). Never invent dimensions or formulas; use only the values I gave.

OUTPUT REQUIREMENTS:

1. One clean 3D drawing on a white background with correct perspective and realistic proportions for the given dimensions.
2. Draw visible edges as solid lines and hidden edges as dashed lines.
3. Label every key dimension (radius, height, length, breadth, slant height) with its value and unit, using arrows or leader lines so labels do not touch the solid.
4. If relevant, lightly shade one curved or top face so the 3D shape reads clearly.
5. Keep all labels fully legible, non-overlapping, and printable in black and white.
6. Below the image, if I asked for area or volume, show the formula, the substitution, and the final answer with units in plain text.
7. Sign the bottom-right of every page with an elegant handwritten signature "Indrajeet Yadav" with a small underline flourish.

QUALITY BAR: The solid must look correct in 3D and every dimension must match what I provided. Recompute any area or volume before showing it.

9.6 Statistical Chart from Data (Bar, Pie or Histogram)

Text only

ChatGPT (image) / Gemini

What you get: A clean, correctly scaled chart from your data with title, labels and values.

How to use: Paste your data table and choose a chart type; the AI returns a tidy, labelled chart for your statistics lesson.

COPY EVERYTHING BELOW

Act as a mathematics teacher and data-visualisation specialist who builds accurate statistical charts for class.

TASK: Create ONE clean statistical chart (as an image) from the data I paste below.

INPUT I AM GIVING YOU:

- Chart type wanted: [bar graph / pie chart / histogram]
- Data: [PASTE THE DATA HERE as label and value pairs, e.g. Maths 40, Science 25, English 20, Hindi 15]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board]
- Topic: [TOPIC e.g. Data handling / Statistics]

BEFORE DRAWING (do this silently): For a pie chart, compute each sector angle (value divided by total, times 360 degrees) yourself; for a bar graph or histogram, fix a sensible even scale. Verify by at least two independent checks (for example, confirm all pie angles add to 360 degrees, and confirm all percentages add to 100). Use only the numbers I gave; do NOT round in a way that breaks the totals, and do NOT invent extra categories.

OUTPUT REQUIREMENTS:

1. One clean chart on a white background with a clear title, labelled axes (for bar/histogram) or labelled sectors (for pie), and an even, numbered scale.
2. Show the exact value (and percentage for a pie chart) on or beside each bar or sector.
3. Use clearly distinguishable shading or patterns that still read well in black and white, plus a small legend if needed.
4. Keep all text fully legible and non-overlapping.
5. Below the image, give a plain-text table of every category, its value, its percentage, and (for pie charts) its angle, and confirm the totals.
6. End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

QUALITY BAR: Every value must match my data exactly and all totals must add up correctly. Recheck the arithmetic before drawing.

9.7 Coordinate-Geometry Plot with Points and Lines

Text only

ChatGPT (image) / Gemini

What you get: A labelled coordinate plane showing your points, lines and shapes plotted accurately.

How to use: List your points or line equations and the AI plots them on a clean, labelled grid for coordinate geometry.

COPY EVERYTHING BELOW

Act as a mathematics teacher who draws accurate coordinate-geometry plots on a labelled grid.

TASK: Plot the points, lines or shapes I give onto ONE clean coordinate plane (as an image).

INPUT I AM GIVING YOU:

- Points: [e.g. A(2,3), B(-1,4), C(0,-2)]
- Lines or equations (if any): [e.g. $y = 2x + 1$, or the line joining A and B]
- What to show: [e.g. plot the points and join them to form a triangle / draw both lines and mark their intersection]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / JEE / SAT]

BEFORE DRAWING (do this silently): Place each point correctly on the plane. If I ask for a length, slope, midpoint or intersection, compute it yourself and verify by at least two independent methods (for example, find an intersection by solving the equations and again by reading the plotted graph; find a distance by the distance formula and again by counting grid steps where possible). Do NOT misplace points or invent coordinates.

OUTPUT REQUIREMENTS:

1. One coordinate plane on a white background with a labelled x-axis and y-axis, the origin marked 0, an even integer scale, and light gridlines.
2. Plot each point as a clear dot, label it with its name and coordinates (for example, A(2,3)).
3. Draw any requested lines or shapes accurately and label them; mark and label any intersection point.
4. Keep all labels fully legible and non-overlapping; ensure the chosen range fits every point with a small margin.
5. Below the image, show in plain text any computed values (slope, length, midpoint, intersection) with the working.
6. Sign the bottom-right of every page with an elegant handwritten signature "Indrajeet Yadav" with a small underline flourish.

QUALITY BAR: Every point must sit on its exact grid location and all computed values must be verified. Recheck before drawing.

9.8 Vector / Force Diagram

Text only

ChatGPT (image) / Gemini

What you get: A clean vector diagram with correctly directed, labelled arrows and the resultant shown.

How to use: Describe your vectors or forces and the AI draws arrows in the right directions with the resultant marked.

COPY EVERYTHING BELOW

Act as a mathematics and physics teacher who draws precise vector and force diagrams.

TASK: Draw ONE clear vector or force diagram (as an image) from the vectors I describe, with correct directions and a labelled resultant if asked.

INPUT I AM GIVING YOU:

- Vectors / forces: [e.g. $F_1 = 10\text{ N}$ to the east, $F_2 = 6\text{ N}$ to the north, acting at a point]
- What to show: [e.g. add them and show the resultant / show the triangle law / resolve into components]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / JEE / NEET]
- Topic: [TOPIC e.g. Vectors / Forces in equilibrium]

BEFORE DRAWING (do this silently): Work out the resultant magnitude and direction yourself. Verify by at least two independent methods (for example, compute the resultant using components and again using the parallelogram or triangle law; check the direction with the tangent of the angle). Keep arrow lengths proportional to magnitudes. Never invent magnitudes, directions, or angles I did not give.

OUTPUT REQUIREMENTS:

1. One clean diagram on a white background, with each vector drawn as a straight arrow whose length is proportional to its magnitude and whose direction is exactly correct.
2. Label each arrow with its name, magnitude and unit (for example, $F_1 = 10\text{ N}$), and mark any given or computed angle.
3. If a resultant is requested, draw it in a distinct style (for example, a bold or dashed arrow) and label its magnitude and direction.
4. Include reference axes or a compass (N, E, S, W) if directions are given that way.
5. Keep all labels fully legible, non-overlapping and printable in black and white.
6. Below the image, show the resultant calculation in plain text with the final magnitude and direction.
7. End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

QUALITY BAR: Directions and proportional lengths must be accurate and the resultant must be verified two ways before drawing.

9.9 Venn Diagram / Set Illustration

Text only

ChatGPT (image) / Gemini

What you get: A clean Venn diagram with correctly placed elements and the right region shaded.

How to use: Give your sets and what to show, and the AI draws a tidy Venn diagram with the correct region shaded.

COPY EVERYTHING BELOW

Act as a mathematics teacher who draws clear, correctly shaded Venn diagrams for set theory lessons.

TASK: Draw ONE clean Venn diagram (as an image) from the sets I give, with elements placed correctly and the requested region shaded.

INPUT I AM GIVING YOU:

- Number of sets: [2 or 3]
- Sets and their elements (or sizes): [e.g. $A = \{1,2,3,4\}$, $B = \{3,4,5,6\}$, universal set $U = \{1..8\}$]
- What to show or shade: [e.g. $A \cap B$ / $A \cup B$ / $A - B$ / $(A \cup B)^c$]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board]
- Topic: [TOPIC e.g. Sets]

BEFORE DRAWING (do this silently): Work out exactly which elements fall in each region (only-A, only-B, both, neither, and the three-set regions if applicable). Determine precisely which region the requested operation covers. Verify by at least two independent checks (for example, list the elements of the result set directly, and confirm the count using the inclusion-exclusion principle). Never misplace an element or shade the wrong region.

OUTPUT REQUIREMENTS:

1. One Venn diagram on a white background, with a rectangle for the universal set U and clearly labelled circles for each set (A , B , and C if used).
2. Place every element in its correct region, or show the size of each region if I gave sizes instead of elements.
3. Shade exactly the requested region with a clearly distinguishable fill that still reads in black and white, and write the operation (for example, $A \cap B$) as a caption.
4. Keep all labels and elements fully legible and non-overlapping.
5. Below the image, list the resulting set in plain text and confirm its element count.
6. Sign the bottom-right of every page with an elegant handwritten signature "Indrajeet Yadav" with a small underline flourish.

QUALITY BAR: Element placement and the shaded region must be exactly correct. Recheck both before drawing.

9.10 Transformation Storyboard (Before to After Frames)

Text only

ChatGPT (image) / Gemini

What you get: A side-by-side before-and-after storyboard showing a shape under a transformation on a grid.

How to use: Describe a shape and a transformation, and the AI draws labelled before and after frames on a grid.

COPY EVERYTHING BELOW

Act as a mathematics teacher who illustrates geometric transformations as clear before-and-after storyboards on a grid.

TASK: Create ONE storyboard image showing a shape before and after the transformation I describe, on a labelled coordinate grid.

INPUT I AM GIVING YOU:

- Original shape and its vertices: [e.g. triangle with vertices A(1,1), B(4,1), C(1,3)]
- Transformation: [DESCRIBE e.g. reflect in the y-axis / rotate 90 degrees anticlockwise about the origin / translate by (3,-2) / enlarge by scale factor 2 about the origin]
- Class / Grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM e.g. CBSE / ICSE / SAT]
- Topic: [TOPIC e.g. Transformations / Symmetry]

BEFORE DRAWING (do this silently): Compute the image coordinates of every vertex under the transformation yourself. Verify by at least two independent methods (for example, apply the transformation rule to each point and again check using a property such as distance preserved for reflections/rotations, or distances scaled by the factor for enlargements). Never invent image coordinates; derive them correctly.

OUTPUT REQUIREMENTS:

1. Show two clear frames side by side (or original and image on one grid in different styles): Frame 1 = original shape, Frame 2 = transformed shape, each on a labelled coordinate grid with an even integer scale.
2. Label every original vertex (A, B, C) with its coordinates and every image vertex (A', B', C') with its new coordinates.
3. Caption the transformation clearly (for example, "Reflection in the y-axis") and, if helpful, draw the mirror line, centre of rotation, or translation arrow.
4. Keep both frames the same scale so the change is easy to compare; ensure all labels are fully legible and non-overlapping.
5. Below the image, list each original point and its image point in plain text and state the transformation rule used.
6. End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

QUALITY BAR: Every image coordinate must be mathematically correct and verified two ways before drawing, and both frames must share one consistent scale.



Lecture decks, YouTube scripts and short-form reels to teach beyond the classroom.

10.1 Slide-by-Slide Lecture Deck Outline

Text only

Any AI chat

What you get: A full slide-by-slide deck plan with titles, bullets and board work for one lecture

How to use: Just type your topic, class and how long the lecture is, then paste the deck text into PowerPoint, Google Slides or Canva.

COPY EVERYTHING BELOW

Act as an expert mathematics lecturer and instructional designer who builds clear, exam-focused slide decks.

TASK: Design a complete slide-by-slide outline for a single lecture on [TOPIC] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. The lecture must fit in [TIME e.g. 40 minutes].

INPUT FROM ME: My topic is [TOPIC]. My students are [CLASS/GRADE] and their level is [DIFFICULTY e.g. beginner / average / advanced]. The exam focus is [BOARD or EXAM]. The lecture length is [TIME]. (Optional: I want to emphasise [SUBTOPIC or COMMON MISTAKE].)

BEFORE YOU WRITE: Solve every example internally first. Verify each final answer by at least two independent methods (for example, a different algebraic route, substitution of values, or a quick estimation check) and only show examples whose answers you have confirmed. Never invent theorems, formulas, or values; use only standard, correct mathematics. Keep all math fully legible in clean LaTeX, and also give a plain-text version of each formula in brackets so a non-technical teacher can read it without rendering.

OUTPUT REQUIREMENTS:

1. A title slide block with the lecture title, the class/exam, and the line 'Compiled by Indrajeet Yadav'.
2. One numbered block per slide (aim for 12 to 18 slides). For EACH slide give: a short slide TITLE, 3 to 5 concise BULLET points (the exact text to put on the slide), a SPEAKER NOTE of 2 to 3 sentences telling me what to say, and a BOARD/VISUAL CUE describing any diagram, graph, or worked step to show.
3. Include at least: 1 hook/why-it-matters slide, 2 to 3 fully worked example slides, 1 common-mistakes slide, 1 quick-practice slide with 2 questions and answers hidden at the end, and 1 recap slide.
4. Mark the natural mid-lecture pause point.
5. Keep language simple and student-friendly; define every new term the first time it appears.

QUALITY BAR: Logical flow from easy to hard, no slide overcrowded, every example correct and double-checked, and timings noted per section so the whole deck fits [TIME].

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title slide block.

10.2 60-Second Reel Script (Instagram / YouTube Shorts)

Text only

Any AI chat

What you get: A punchy 60-second vertical reel script with on-screen text, voiceover and captions

How to use: Type the one idea or trick you want to teach and your class, then read the voiceover into your phone while showing the on-screen text.

COPY EVERYTHING BELOW

Act as a viral mathematics content creator who makes addictive, accurate 60-second teaching reels for students.

TASK: Write a 60-second vertical reel script that teaches ONE single idea: [TOPIC or TRICK] for [CLASS/GRADE] students studying [BOARD or EXAM e.g. CBSE / JEE / NEET / SAT].

INPUT FROM ME: The one idea is [TOPIC or TRICK]. My audience is [CLASS/GRADE]. Exam context: [BOARD or EXAM]. Tone I want: [TONE e.g. energetic / calm / funny]. (Optional: hook angle [PASTE ANY ANGLE OR LEAVE BLANK].)

BEFORE YOU WRITE: Work out the math internally and verify the final answer two independent ways (for example, recompute by a second method and also plug the answer back in). Use only correct, standard formulas; never invent shortcuts that are not mathematically valid. Show every formula in clean LaTeX with a plain-text version in brackets.

OUTPUT REQUIREMENTS - present as a table with one row per beat and these columns: TIME (e.g. 0-3s), VOICEOVER (exact words to speak, conversational), ON-SCREEN TEXT (short, large, the words to display), and VISUAL/ACTION (what to show or write). Structure the 60 seconds as: 0-3s scroll-stopping HOOK, 3-10s the problem or promise, 10-45s the method in 2 to 3 crisp steps with one worked number, 45-55s the satisfying payoff/answer, 55-60s a call to action ('Follow for more [SUBJECT] tricks'). Also provide: a 1-line CAPTION for the post, 8 relevant HASHTAGS, and a SUGGESTED ON-SCREEN TITLE.

QUALITY BAR: Total spoken words must fit comfortably in 60 seconds (about 130-150 words), every number correct and double-checked, on-screen text short enough to read on a phone, and the hook must create curiosity in the first 3 seconds.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title/header block of the script.

10.3 Full YouTube Lesson Script with Timestamps

Text only

Any AI chat

What you get: A complete narrated YouTube lesson script with chapter timestamps and B-roll cues

How to use: Tell it the topic and how long you want the video, then record straight from the script section by section.

COPY EVERYTHING BELOW

Act as an expert mathematics teacher and YouTube educator who writes complete, well-paced lesson scripts that students actually finish watching.

TASK: Write a full YouTube lesson script on [TOPIC] for [CLASS/GRADE] preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. Target video length: [TIME e.g. 12 minutes].

INPUT FROM ME: Topic is [TOPIC]. Audience: [CLASS/GRADE], level [DIFFICULTY]. Exam: [BOARD or EXAM]. Video length: [TIME]. (Optional: must-cover subtopics or question types: [PASTE HERE].)

BEFORE YOU WRITE: Solve all examples internally and verify each final answer by at least two independent checks (a second solving method plus a substitution or estimation check). Do not invent any theorem, identity, or value; use only standard, correct mathematics. Present every formula in clean LaTeX and add a plain-text version in brackets so I can read it aloud without confusion.

OUTPUT REQUIREMENTS:

1. A VIDEO TITLE (clickable, honest) and a 3-line DESCRIPTION with 6 keywords.
2. The script in clearly labelled CHAPTERS, each with an estimated TIMESTAMP (e.g. 00:00 Intro, 01:30 Concept...). Include: a 0-15 second HOOK, an INTRO that states what the viewer will be able to do by the end, the CONCEPT explained simply, 2 to 3 fully WORKED EXAMPLES read step by step, a COMMON MISTAKES section, a QUICK RECAP, and an OUTRO with a call to subscribe and one practice question for the comments.
3. For each chapter give the exact SPOKEN NARRATION plus, in square brackets, a B-ROLL / ON-SCREEN CUE (what to write on the board or show).
4. A copy-paste list of YouTube CHAPTER TIMESTAMPS for the description box.

QUALITY BAR: Word count roughly matches [TIME] at about 140 words per minute, smooth transitions between chapters, every example verified correct, and language friendly for [CLASS/GRADE].

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the title/description header block.

10.4 Animation Storyboard (Manim / Whiteboard Style)

Text only

Any AI chat

What you get: A scene-by-scene animation storyboard with exact visual cues, motion and narration

How to use: Describe the concept you want to animate and hand the storyboard to an animator or use the cues yourself in a whiteboard/animation tool.

COPY EVERYTHING BELOW

Act as a mathematics animation director who designs clear, beautiful concept animations in the style of 3Blue1Brown / Manim or hand-drawn whiteboard videos.

TASK: Create a scene-by-scene animation STORYBOARD that visually explains [TOPIC] for [CLASS/GRADE] studying [BOARD or EXAM e.g. CBSE / JEE / NEET / SAT]. Target runtime: [TIME e.g. 4 minutes].

INPUT FROM ME: Concept to animate: [TOPIC]. Audience: [CLASS/GRADE], level [DIFFICULTY]. Exam: [BOARD or EXAM]. Runtime: [TIME]. Visual style: [STYLE e.g. Manim-clean / hand-drawn whiteboard]. (Optional: the single 'aha' moment I most want to land: [PASTE HERE].)

BEFORE YOU WRITE: Work through the underlying math internally and verify every value, coordinate, and final result two independent ways. Use only correct, standard mathematics; never invent a formula or a graph that is not accurate. Give all formulas in clean LaTeX with a plain-text version in brackets, and make sure any coordinates or graph features you describe are mathematically exact.

OUTPUT REQUIREMENTS - present as a numbered SCENE list. For EACH scene give: SCENE NUMBER and a short title, DURATION (seconds), VISUAL CUE (precisely what appears on screen - objects, axes, shapes, colours, labels, exact values), MOTION/ANIMATION (what moves, transforms, fades, or highlights, and in what order), NARRATION (the exact words spoken over the scene), and ON-SCREEN TEXT/EQUATION (in LaTeX plus plain text). Build the sequence as: a curiosity-raising opening visual, a build-up of the idea one element at a time, the key transformation that delivers the insight, a worked numeric example shown visually, and a clean summary frame.

Also include a SHOT LIST SUMMARY and a note on colour/labelling consistency across scenes.

QUALITY BAR: Every scene advances understanding, nothing on screen is cluttered, all values and graphs are exact and verified, and durations add up to [TIME].

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the storyboard title/header block.

10.5 Whiteboard Explainer Talk-Track

Text only

Any AI chat

What you get: A word-for-word talking script synced to exactly what you write on the board

How to use: Pick a topic, then follow the script line by line, writing on the board exactly when it tells you to.

COPY EVERYTHING BELOW

Act as a master mathematics teacher who narrates while writing on a whiteboard, keeping the talk perfectly in sync with the board work.

TASK: Write a word-for-word whiteboard explainer TALK-TRACK for [TOPIC], aimed at [CLASS/GRADE] students studying [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET]. Target length: [TIME e.g. 8 minutes].

INPUT FROM ME: Topic: [TOPIC]. Audience: [CLASS/GRADE], level [DIFFICULTY]. Exam: [BOARD or EXAM]. Length: [TIME]. (Optional: the exact worked problem I want to centre it on: [PASTE THE QUESTION HERE].)

BEFORE YOU WRITE: Solve everything internally first and verify each final answer by at least two independent methods (for example a second solving approach plus plugging the answer back in). Use only correct, standard mathematics; never invent steps, identities, or values. Write every formula in clean LaTeX and also in plain text in brackets so I can speak and write it without ambiguity.

OUTPUT REQUIREMENTS - present as a two-column flow. LEFT column = SAY (the exact words I speak, natural and warm). RIGHT column = WRITE/DRAW (exactly what to put on the board at that moment - each line, symbol, arrow, box, or underline, in the order it should appear). Structure it as: a warm greeting and the goal of the lesson, the concept introduced in plain words, one problem solved live step by step (with me narrating each step as I write it), a pause-and-think moment with a question to the viewer, the common error called out and corrected, and a short verbal recap.

Mark every place I should PAUSE, point, or circle something for emphasis. Note the approximate running time at each major step.

QUALITY BAR: The SAY and WRITE columns must always match the same moment, every board line must be legible and in logical order, all math verified correct, and the tone encouraging for [CLASS/GRADE].

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the talk-track title/header block.

10.6 Hook + Payoff Short ('You've Been Doing This Wrong')

Text only

Any AI chat

What you get: A myth-busting short script that flips a common mistake into the correct method

How to use: Tell it the common mistake students make, and it writes a 'you've been doing this wrong' reel that ends with the right way.

COPY EVERYTHING BELOW

Act as a high-energy mathematics creator who makes myth-busting shorts that correct a common student mistake without being condescending.

TASK: Write a 45 to 60 second 'You have been doing this wrong' style short about the topic [TOPIC] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / JEE / NEET / SAT]. The whole short hinges on exposing and fixing one specific common mistake.

INPUT FROM ME: Topic: [TOPIC]. The common mistake students make is: [PASTE THE COMMON MISTAKE - or write 'you choose the most common one']. Audience: [CLASS/GRADE]. Exam: [BOARD or EXAM]. Tone: [TONE e.g. playful / serious].

BEFORE YOU WRITE: Internally solve a clear example the WRONG way and the RIGHT way, and verify the correct final answer by at least two independent checks (a second method plus substitution or estimation). Be absolutely sure the 'wrong' method really is wrong and the 'right' method really is right - never misrepresent valid mathematics. Show all formulas in clean LaTeX with a plain-text version in brackets.

OUTPUT REQUIREMENTS - present as a beat-by-beat table with columns TIME, VOICEOVER, ON-SCREEN TEXT, and VISUAL/ACTION. Structure: 0-3s a bold hook ('Stop doing [MISTAKE] - here is why'), 3-15s show the tempting wrong approach and the wrong answer it gives, 15-20s the 'plot twist' that reveals the error, 20-50s the correct method in 2 to 3 clean steps with the verified right answer, 50-60s a memorable one-line takeaway plus a call to action.

Also give: a scroll-stopping CAPTION, 8 HASHTAGS, and a PINNED-COMMENT idea that re-explains the fix in one sentence.

QUALITY BAR: The contrast between wrong and right must be crystal clear, the correct answer double-checked, spoken words fit 60 seconds, and the tone corrects the mistake kindly, never mocking students.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the script title/header block.

10.7 Exam-Tip Carousel (8 Slides)

Text only

Any AI chat

What you get: Ready-to-design text for an 8-slide swipe carousel of exam tips

How to use: Type your topic and exam, then paste each slide's text into Canva or Instagram's carousel one box at a time.

COPY EVERYTHING BELOW

Act as an exam-strategy mentor and mathematics teacher who creates clean, high-saving Instagram/LinkedIn carousels for students.

TASK: Write the full text for an 8-slide swipe carousel of exam tips on [TOPIC] for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: Topic: [TOPIC]. Audience: [CLASS/GRADE], level [DIFFICULTY]. Exam: [BOARD or EXAM]. Focus of the tips: [FOCUS e.g. speed / accuracy / avoiding silly mistakes / last-minute revision]. (Optional: a specific question type to centre tips on: [PASTE HERE].)

BEFORE YOU WRITE: For any tip that involves a calculation or shortcut, work it out internally and verify the result two independent ways before including it. Never present an invalid shortcut as valid; use only correct, standard mathematics. Show any formula in clean LaTeX plus a plain-text version in brackets, kept short enough to fit on a slide.

OUTPUT REQUIREMENTS - give exactly 8 SLIDES, numbered, with the exact text to display on each:

Slide 1 = COVER: a bold scroll-stopping title + a subtitle promising the value (and the line 'Compiled by Indrajeet Yadav' in small text).

Slides 2 to 7 = ONE actionable tip each: a short punchy HEADLINE (max 6 words), 2 to 3 lines explaining it simply, and where helpful one tiny worked example or before/after.

Slide 8 = RECAP + CALL TO ACTION: a 1-line summary of all tips and a prompt to save, share, and follow. Keep each slide to roughly 25 to 40 words so it is readable on a phone. Use a consistent style. Also provide a CAPTION for the post and 8 HASHTAGS.

QUALITY BAR: Every tip is genuinely useful and exam-relevant, any math is verified correct, no slide is overcrowded, and the tips progress logically.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' on the cover slide header block.

10.8 Concept in 60 Seconds (Rapid Script)

Text only

Any AI chat

What you get: A tight rapid-fire script that explains one whole concept clearly in 60 seconds

How to use: Name the concept and your class, then read the fast-paced script straight into your camera in one take.

COPY EVERYTHING BELOW

Act as a mathematics explainer who can compress a whole concept into a crisp, accurate 60-second rapid script without losing clarity.

TASK: Write a 'Concept in 60 seconds' rapid script that fully explains [TOPIC] for [CLASS/GRADE] students studying [BOARD or EXAM e.g. CBSE / JEE / NEET / SAT]. It must feel fast but still make complete sense.

INPUT FROM ME: Concept: [TOPIC]. Audience: [CLASS/GRADE], level [DIFFICULTY]. Exam: [BOARD or EXAM]. Energy: [TONE e.g. fast and punchy / calm and clear]. (Optional: the one definition or formula that MUST appear: [PASTE HERE].)

BEFORE YOU WRITE: Solve any example internally and verify the final answer two independent ways (a second method plus a substitution or estimation check). Use only correct, standard mathematics and never invent a formula or value. Provide each formula in clean LaTeX with a plain-text version in brackets, and keep it short enough to flash on screen.

OUTPUT REQUIREMENTS - present as a tight beat table with columns TIME, VOICEOVER (exact words), ON-SCREEN TEXT, and VISUAL/ACTION. Cover the concept in this arc inside 60 seconds: 0-5s a one-line hook stating what they will understand by the end, 5-20s the core definition in the simplest possible words plus the key formula, 20-45s ONE quick worked example showing the concept in action with the verified answer, 45-55s the single most important thing to remember, 55-60s a sign-off and call to action. Keep total spoken words to about 140-150 so it fits 60 seconds. Also add a TITLE, a 1-line CAPTION, and 6 HASHTAGS.

QUALITY BAR: Even though it is fast, a [CLASS/GRADE] student must be able to follow it; the example must be verified correct; on-screen text must be short and readable; and nothing important is skipped.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the script title/header block.

10.9 Doubt-Clearing Reel from a Student Question

Photo needed

ChatGPT (image) / Gemini

What you get: A short reel script that answers one real student doubt step by step

How to use: Attach a photo of the question a student got stuck on (or type it), and it writes a quick reel that clears the doubt.

COPY EVERYTHING BELOW

Act as a patient mathematics teacher who turns one real student doubt into a clear, satisfying short reel.

TASK: Create a 60 to 90 second doubt-clearing reel script that answers a specific student question on [TOPIC] for [CLASS/GRADE] students studying [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

INPUT FROM ME: The student doubt or question is: [PASTE THE QUESTION HERE] OR [ATTACH A PHOTO OF THE PROBLEM]. If a photo is attached, first read it carefully and restate the exact question in text before solving. Topic area: [TOPIC]. Audience: [CLASS/GRADE]. Exam: [BOARD or EXAM]. The specific point students get confused about (if I know it): [PASTE THE CONFUSION OR LEAVE BLANK].

BEFORE YOU WRITE: Solve the problem internally and verify the final answer by at least two independent methods (for example a second solving route plus substituting the answer back in). If the photo or text is unclear or has a typo, state your assumption explicitly rather than guessing silently. Use only correct, standard mathematics; never invent a formula or value. Show every step in clean LaTeX with a plain-text version in brackets.

OUTPUT REQUIREMENTS - present as a beat table with columns TIME, VOICEOVER, ON-SCREEN TEXT, and VISUAL/ACTION. Structure: 0-5s a hook that names the exact doubt ('Stuck on this? Here is the clean way'), 5-15s show the question clearly on screen, 15-70s solve it step by step with each step narrated and the reason for each step stated, then 70-90s the verified final answer plus the one insight that removes the confusion and a call to send more doubts.

Also provide a CAPTION inviting students to comment their doubts and 6 HASHTAGS.

QUALITY BAR: The restated question must match what was given, every step must have a reason, the answer must be double-checked, and the confusion must be directly addressed in plain words.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the script title/header block.

10.10 Story-Driven Lesson (Open with a Puzzle)

Text only

Any AI chat

What you get: A narrative lesson script that opens with a hook puzzle and resolves it with the concept

How to use: Give it the concept and your class, and it wraps the whole lesson around an intriguing story or puzzle students want solved.

COPY EVERYTHING BELOW

Act as a master storyteller and mathematics teacher who teaches concepts by opening with an irresistible puzzle and resolving it with the math.

TASK: Write a story-driven lesson script on [TOPIC] for [CLASS/GRADE] students studying [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. Target length: [TIME e.g. 10 minutes]. The lesson must OPEN with a puzzle or real-life mystery and pay it off at the end using the concept.

INPUT FROM ME: Concept: [TOPIC]. Audience: [CLASS/GRADE], level [DIFFICULTY]. Exam: [BOARD or EXAM]. Length: [TIME]. (Optional: a real-life context or theme I like: [PASTE HERE e.g. cricket, money, travel, games].)

BEFORE YOU WRITE: Design the opening puzzle so its answer genuinely depends on [TOPIC]. Solve the puzzle and every example internally and verify each final answer two independent ways (a second method plus substitution or estimation). Use only correct, standard mathematics; never invent a fact, formula, or value to make the story work. Show all formulas in clean LaTeX with a plain-text version in brackets.

OUTPUT REQUIREMENTS - write the script in labelled SECTIONS with approximate timings:

1. THE HOOK PUZZLE: pose an intriguing puzzle or short story and promise that by the end they will be able to crack it. Give the exact words to say.
 2. WHY IT IS TRICKY: show why intuition or guessing fails here.
 3. THE TOOL: introduce the concept [TOPIC] in plain words as the key that unlocks the puzzle, with the core formula.
 4. BUILD-UP: 1 to 2 small worked examples that teach the method step by step (each verified).
 5. THE PAYOFF: return to the opening puzzle and solve it fully using the concept, revealing the satisfying answer.
 6. RECAP + CHALLENGE: summarise the idea in 2 lines and leave the viewer one challenge puzzle (with the answer hidden at the very end).
- For each section give the SPOKEN NARRATION and, in square brackets, a BOARD/VISUAL CUE.

QUALITY BAR: The puzzle must truly require [TOPIC] to solve, the narrative must stay clear and age-appropriate for [CLASS/GRADE], every answer must be verified correct, and the payoff must feel earned.

End the output with a footer line: 'Prepared by Indrajeet Yadav', and place 'Compiled by Indrajeet Yadav' in the script title/header block.



Find the exact mistake, fix misconceptions and turn errors into learning.

11.1 Find My Mistake (from a photo)

Photo needed

ChatGPT (image) / Gemini

What you get: The exact line where the student went wrong, plus the corrected solution.

How to use: Snap a clear photo of the student's handwritten work and attach it, then paste this prompt below the photo.

COPY EVERYTHING BELOW

Act as a meticulous mathematics examiner and error-detective for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

TASK: I am attaching a photo of a student's handwritten solution to a [TOPIC] problem. Find the EXACT step where the mistake happens, explain why it is wrong, and show the correct path.

INPUT INSTRUCTIONS:

- [ATTACH A PHOTO OF THE PROBLEM AND THE STUDENT'S FULL WORKING]
- The original question is: [PASTE THE QUESTION HERE]
- If any handwriting is unclear, tell me exactly which symbol or line you cannot read and state your best assumption before continuing.

WHAT TO DO INTERNALLY FIRST (do not skip):

1. Read the question and solve it yourself, cleanly, from scratch.
2. Then read the student's work line by line and compare against your own solution.
3. Identify the FIRST line where the student deviates from a correct method (the root error), and separate it from later errors that are only consequences of the first.

OUTPUT REQUIREMENTS (use these exact headings):

1. What the student was trying to do - one or two sentences in plain language.
2. The first mistake - quote the exact line as written, label it 'Line X', and explain in simple words why it is wrong (e.g. sign error, wrong formula, dropped term, invalid cancellation).
3. Why students commonly make this error - the underlying misconception in one short paragraph.
4. The correct solution - rewrite from the point of the error to the final answer, with each step on its own line in clean LaTeX or clear plain-text math.
5. Final answer - boxed or clearly stated.
6. One quick tip so the student never repeats this mistake.

QUALITY BARS:

- Verify your final answer by at least two independent checks (e.g. substitute back into the original equation AND re-derive by a second method). State both checks briefly.
- Never invent formulas, theorems, or values. If a step needs a standard result, name it correctly.
- Keep all mathematics fully legible and aligned. Use plain ASCII only, with straight quotes.
- Be encouraging but precise; address the student's reasoning, not their intelligence.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

11.2 Common-Error Sheet for a Chapter

Text only

Any AI chat

What you get: A ready-to-print sheet of the top mistakes students make in one chapter, with fixes.

How to use: Just type your chapter and class, and you will get a printable list of the traps your students keep falling into.

COPY EVERYTHING BELOW

Act as an experienced [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT] mathematics teacher who has graded thousands of papers and knows exactly where students lose marks.

TASK: Create a 'Common Errors and How to Avoid Them' reference sheet for the chapter [TOPIC] for [CLASS/GRADE].

INPUT INSTRUCTIONS:

- Chapter / topic: [TOPIC]
- Class / grade: [CLASS/GRADE]
- Board or exam: [BOARD or EXAM]
- Optional focus: [ANY SUBTOPIC YOU WANT EMPHASISED, e.g. word problems / proofs / sign rules - leave blank for full chapter]

WHAT TO PRODUCE: A table of the [NUMBER OF QUESTIONS e.g. 12] most frequent and most costly mistakes in this chapter. For EACH error give the following columns:

1. The mistake - stated as a short, recognisable error a student actually writes.
2. Wrong example - one concrete line of wrong work in clean math.
3. Correct version - the same line done correctly, side by side.
4. Why it happens - the misconception in one sentence.
5. Fix / rule to remember - a one-line takeaway.

After the table, add:

- A 'Top 3 Killers' box highlighting the three errors that cost the most marks in exams.
- A short 'Teacher's note' (3-4 sentences) on how to address these in class.

QUALITY BARS:

- Solve every example internally first and verify each correct version by an independent check before writing it; never show a 'correct' line that is actually wrong.
- Do not invent formulas, theorems, or fake exam statistics. Order errors from most to least common based on genuine teaching experience.
- Cover real, distinct mistakes - no two rows should be near-duplicates.
- Plain ASCII only, straight quotes, clean and legible math. Format so it prints neatly on one or two pages.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

11.3 Misconception Buster (Myth vs Truth)

Text only

Any AI chat

What you get: A clear myth-vs-truth breakdown that kills a stubborn misconception for good.

How to use: Name the wrong idea your students keep believing, and get a simple proof of why it is false plus a memorable correction.

COPY EVERYTHING BELOW

Act as a patient mathematics teacher and conceptual-clarity specialist for [CLASS/GRADE] students following [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

TASK: Bust a specific mathematical misconception that my students keep believing, in a way that is impossible to forget.

INPUT INSTRUCTIONS:

- The misconception / wrong belief (the 'myth') is: [PASTE THE WRONG IDEA HERE, e.g. " $(a+b)^2 = a^2 + b^2$ " or "a fraction is always smaller than 1" or "square root of a number is always positive and negative"]
- Topic area: [TOPIC]
- Class / grade: [CLASS/GRADE]

WHAT TO PRODUCE (use these exact headings):

1. THE MYTH - restate the wrong belief clearly in the student's own likely words.
2. WHY IT FEELS TRUE - the seductive reasoning that makes students fall for it (2-3 sentences).
3. THE TRUTH - the correct statement, precisely worded.
4. PROOF IT IS WRONG - a clean counter-example with actual numbers that any student can verify on their own, shown step by step. If helpful, add a quick algebraic or visual argument too.
5. THE CORRECT RULE - stated fully, with the conditions under which it holds.
6. MEMORY HOOK - one short, sticky phrase or analogy so the truth replaces the myth permanently.
7. CHECK-YOURSELF QUESTION - one quick question (with the answer hidden below a line marked 'ANSWER:') so students can test whether they have unlearned the myth.

QUALITY BARS:

- Solve the counter-example internally first and verify it by at least two independent checks (e.g. direct substitution AND a second numerical case). Confirm the myth genuinely fails.
- Never invent formulas, theorems, or values; state every condition and restriction accurately.
- Keep the language simple and warm - the goal is to correct the idea without making the student feel foolish.
- Plain ASCII only, straight quotes, clean legible math.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

11.4 Stuck? Step-by-Step Solve (from a photo)

Photo needed

ChatGPT (image) / Gemini

What you get: A full, beginner-friendly solution to a problem you or a student got stuck on.

How to use: Photograph the problem your student is stuck on, attach it, and paste this to get a clear walked-through solution.

COPY EVERYTHING BELOW

Act as a calm, encouraging mathematics tutor for a [CLASS/GRADE] student who is stuck and needs a clear, complete walkthrough for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

TASK: A student is stuck on the attached problem. Solve it fully, explaining the thinking at every step as if teaching someone seeing this type of problem for the first time.

INPUT INSTRUCTIONS:

- [ATTACH A PHOTO OF THE PROBLEM]
- Also type the question here if the photo is unclear: [PASTE THE QUESTION HERE]
- Difficulty / level: [DIFFICULTY e.g. basic / board level / advanced]
- If the photo shows partial work the student already did, take it into account and tell them which part was fine.

WHAT TO DO INTERNALLY FIRST: Read and solve the problem completely on your own before writing anything. Decide the cleanest method a student at this level can actually follow.

OUTPUT REQUIREMENTS (use these exact headings):

1. What the question is asking - in plain words.
2. What we know and what we want - list the given information and the goal.
3. The plan - one or two sentences naming the strategy and why it fits.
4. Step-by-step solution - number every step; on each step show the math on its own line AND a short 'why' in brackets. Do not skip steps a beginner would need.
5. Final answer - clearly boxed or stated, with correct units if any.
6. Check - verify the final answer by two independent methods (e.g. substitute back AND estimate/approximate) and state both checks.
7. Where students usually get stuck on this type - one short note so the teacher can pre-empt it.

QUALITY BARS:

- Never invent formulas, theorems, or values; if a standard result is used, name it correctly.
- Show every step legibly in clean LaTeX or clear plain-text math; no jumps.
- Plain ASCII only, straight quotes. Keep the tone supportive and confidence-building.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

11.5 Where Did Marks Get Cut? (Examiner View)

Photo needed

ChatGPT (image) / Gemini

What you get: An examiner-style mark-by-mark breakdown showing exactly where marks were lost.

How to use: Attach a photo of the student's answer and tell the AI the total marks, then get the step-marking an examiner would actually apply.

COPY EVERYTHING BELOW

Act as a strict but fair board examiner for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT] who awards step marks exactly as per official marking schemes.

TASK: I am attaching a student's answer to a [TOPIC] question carrying [TOTAL MARKS] marks. Tell me precisely where marks were lost and what the realistic awarded score is.

INPUT INSTRUCTIONS:

- [ATTACH A PHOTO OF THE STUDENT'S WRITTEN ANSWER]
- The exact question is: [PASTE THE QUESTION HERE]
- Total marks for this question: [TOTAL MARKS]
- Class / grade: [CLASS/GRADE]
- If the marking scheme for this board allocates method marks vs accuracy marks, apply that convention.

WHAT TO DO INTERNALLY FIRST: Solve the question yourself and build a step-wise marking scheme (which step earns which mark) before you grade the student.

OUTPUT REQUIREMENTS (use these exact headings):

1. Model marking scheme - list each scoring step and the marks it carries, totalling [TOTAL MARKS].
2. Step-by-step grading of the student - go through the student's work against the scheme; for each step say MARK AWARDED or MARK LOST and one line of reason.
3. Marks table - a tidy table: Step | Marks available | Marks awarded | Reason.
4. Total awarded - state the realistic score out of [TOTAL MARKS].
5. Exactly where marks were lost - bullet the specific lapses (e.g. final step missing, no units, wrong sign, conclusion not stated).
6. How to recover those marks next time - concrete, exam-focused advice.

QUALITY BARS:

- Verify your own model solution by two independent checks before grading. Do not be harsher or softer than a real examiner.
- Award method marks even when the final answer is wrong, if the method earns them, exactly as boards do.
- Never invent marks, formulas, or values. If handwriting is unclear, state your assumption.
- Plain ASCII only, straight quotes, clean legible math.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

11.6 Detailed Grader Feedback on a Written Answer

Text only

Any AI chat

What you get: Warm, specific feedback on a student's reasoning with clear next steps.

How to use: Paste or photograph the student's full answer and get teacher-style feedback you can write straight onto their copy.

COPY EVERYTHING BELOW

Act as a supportive mathematics teacher writing personalised feedback on a student's answer for [CLASS/GRADE] under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

TASK: Give detailed, constructive feedback on the student's written solution below - not just right/wrong, but how the thinking can improve.

INPUT INSTRUCTIONS:

- The question: [PASTE THE QUESTION HERE]
- The student's full answer: [PASTE THE STUDENT'S ANSWER HERE - or ATTACH A PHOTO OF THE PROBLEM AND WORKING]
- Topic: [TOPIC]
- Optional: feedback tone [GENTLE for a struggling student / DIRECT for a strong student - default balanced].

WHAT TO DO INTERNALLY FIRST: Solve the question correctly yourself, then evaluate the student's reasoning, presentation, and accuracy against it.

OUTPUT REQUIREMENTS (use these exact headings):

1. Overall impression - 2-3 honest, encouraging sentences.
2. What the student did well - specific strengths (correct method choice, neat steps, good notation), quoting their work.
3. What went wrong and where - point to exact steps; separate genuine errors from things that are merely unclear.
4. Presentation and communication - comment on whether steps are justified, units shown, and conclusion stated - things that lose easy marks.
5. Corrected key steps - rewrite only the steps that need fixing, in clean math.
6. Two targeted practice suggestions - the kind of problem the student should try next to fix the specific gap.
7. One-line grade-style summary the teacher can copy onto the copy (e.g. 'Good method, watch your signs in the second line').

QUALITY BARS:

- Verify your own solution by two independent checks before judging the student. Comment on reasoning, never on the student's ability.
- Never invent formulas, theorems, or values; name any standard result correctly.
- Be specific - every comment should point to an actual line, not vague praise or criticism.
- Plain ASCII only, straight quotes, clean legible math.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

11.7 What To Try When Stuck (Strategy List)

Text only

Any AI chat

What you get: A go-to list of strategies students can run through whenever they freeze on a problem type.

How to use: Tell the AI the type of problem your students freeze on, and get a simple 'try this, then this' rescue list.

COPY EVERYTHING BELOW

Act as a master problem-solving coach for [CLASS/GRADE] mathematics students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

TASK: Create a 'What To Try When You Are Stuck' strategy list for a specific TYPE of problem, so a student who freezes has clear, ordered moves to attempt.

INPUT INSTRUCTIONS:

- Problem type / topic: [TOPIC e.g. trigonometric identities / quadratic word problems / definite integrals / coordinate geometry proofs]
- Class / grade: [CLASS/GRADE]
- Optional sample problem to ground the advice: [PASTE A REPRESENTATIVE QUESTION HERE - or leave blank]

WHAT TO PRODUCE (use these exact headings):

1. First, understand - 3 quick questions the student should ask before doing any math (What is given? What is asked? What does the answer look like?).
2. The strategy ladder - an ordered list of [NUMBER OF QUESTIONS e.g. 8] concrete moves to try for this problem type, from most-likely-to-work to last resort. For each move give: the move (one line), when to use it, and a tiny illustrative snippet of math.
3. Common dead-ends - 3 things students try that usually waste time on this type, and what to do instead.
4. The 'unstick' checklist - 5 fast prompts a student can read when totally blank (e.g. 'try a substitution', 'draw a figure', 'test a small case', 'work backwards from the answer').
5. Worked mini-demo - if a sample problem was given, apply the ladder to it in a few clean steps to show the strategy in action; otherwise create one simple example and do the same.

QUALITY BARS:

- Solve any demo internally first and verify the result by two independent checks before presenting it.
- Never invent formulas, theorems, or values; name standard techniques correctly.
- Keep each strategy genuinely distinct and actionable - no vague advice like 'think harder'.
- Plain ASCII only, straight quotes, clean legible math. Format so it can be printed as a one-page rescue card.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

11.8 Self-Check Checklist Before Submitting

Text only

Any AI chat

What you get: A short checklist students tick off to catch their own errors before handing in.

How to use: Give your topic and class to get a printable checklist students run through to catch silly mistakes before submitting.

COPY EVERYTHING BELOW

Act as an exam-strategy mentor for [CLASS/GRADE] mathematics students appearing in [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT].

TASK: Create a short, practical SELF-CHECK CHECKLIST that a student runs through on every answer before submitting, so they catch their own mistakes and stop losing easy marks.

INPUT INSTRUCTIONS:

- Topic or chapter the checklist is for: [TOPIC]
- Class / grade: [CLASS/GRADE]
- Exam time pressure: [TIME e.g. 3-hour board paper / 1-hour class test - so the checks stay realistic]

WHAT TO PRODUCE (use these exact headings):

1. The 60-second final check - a numbered list of [NUMBER OF QUESTIONS e.g. 10] yes/no checks the student must be able to answer YES to before moving on (e.g. 'Did I answer what was actually asked?', 'Did I write units?', 'Did I check signs?', 'Did I substitute my answer back?'). Make several checks specific to [TOPIC], not just generic.
2. Topic-specific traps to verify - 4-5 checks unique to common errors in [TOPIC].
3. The 'is my answer sane?' test - 3 quick sense-checks (size of the answer, sign, units/range) the student can do in seconds.
4. If something is wrong - a 3-step rescue routine for what to do if a check fails and time is short.
5. Tear-off version - repeat the checklist as a compact, no-explanation tick-box list the student can copy onto the corner of rough paper.

QUALITY BARS:

- Every check must be genuinely actionable in seconds; no vague items.
- Ensure all topic-specific checks reflect real, correct mathematics - do not invent formulas, theorems, or values.
- Keep items distinct and non-overlapping; order them by how often they catch real mistakes.
- Plain ASCII only, straight quotes, clean legible text. Format so it prints neatly as a single page.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.



MCQ banks with smart distractors, flashcards and rapid-fire quizzes.

12.1 MCQ Bank with Smart Distractors + Answer Key

[Text only](#)[Any AI chat](#)

What you get: A ready MCQ test where every wrong option mirrors a real student mistake, with a full answer key.

How to use: Just type your topic and class, paste this in, and you get a printable multiple-choice test in minutes.

COPY EVERYTHING BELOW

Act as an expert mathematics examiner who designs board-quality multiple-choice questions for Indian and international classrooms.

TASK: Create a multiple-choice question (MCQ) bank on the topic below, where the wrong options (distractors) are NOT random - each distractor must be the answer a student would get from one specific, common mistake.

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]
- Number of questions: [NUMBER OF QUESTIONS e.g. 15]
- Difficulty: [DIFFICULTY e.g. easy / medium / hard / mixed]

HOW TO BUILD EACH QUESTION:

1. Write a clear single-correct-answer question with 4 options labelled (a), (b), (c), (d).
2. Make exactly ONE option correct.
3. For the 3 wrong options, base each on a named, realistic student error - for example: sign error, forgetting to apply a formula fully, swapping numerator and denominator, dropping a factor, wrong unit, off-by-one, using degrees instead of radians, mis-applying a rule. The wrong options must be tempting, not silly.

OUTPUT REQUIREMENTS:

- Section 1: The clean question paper (questions + 4 options each only), suitable for printing.
- Section 2: ANSWER KEY as a simple table: Q.No | Correct Option | One-line reason it is correct.
- Section 3: DISTRACTOR ANALYSIS table: Q.No | Wrong option letter | The exact student mistake that produces it. This helps me teach.
- Use clean LaTeX for all maths, or clear plain-text math if LaTeX is not supported. Keep every symbol legible.

QUALITY BARS - follow strictly:

- Solve every question yourself FIRST, before writing options.
- Verify each final answer by at least two independent methods (for example: direct calculation and back-substitution, or a different formula). If two methods disagree, redo the question.
- Never invent formulas, theorems, or values. Use only standard, correct mathematics for the stated board.
- Ensure exactly one option is correct per question; no duplicate or overlapping options.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.2 Assertion-Reason Question Set

Text only

Any AI chat

What you get: A set of CBSE-style Assertion-Reason questions with the standard 4 options and clear explanations.

How to use: Pick your chapter, paste this, and get exam-pattern Assertion-Reason questions ready to print.

COPY EVERYTHING BELOW

Act as a senior CBSE/ICSE mathematics paper-setter who specialises in Assertion-Reason (A-R) questions.

TASK: Create a set of Assertion-Reason questions on the topic below, using the exact standard format used in board and competitive exams.

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE]
- Number of questions: [NUMBER OF QUESTIONS e.g. 10]
- Difficulty: [DIFFICULTY e.g. medium]

FORMAT FOR EVERY QUESTION (use these exact four options):

Give an Assertion (A) and a Reason (R), then ask the student to choose:

- (a) Both A and R are true, and R is the correct explanation of A.
- (b) Both A and R are true, but R is NOT the correct explanation of A.
- (c) A is true but R is false.
- (d) A is false but R is true.

DESIGN RULES:

- Spread the questions across all four answer types - do not make them all option (a). Aim for a balanced mix.
- Make the Reason a genuine mathematical statement (a definition, theorem, property, or formula), not a vague sentence.
- Include a few subtle cases where R is true but does not actually explain A, to test real understanding.

OUTPUT REQUIREMENTS:

- Section 1: The clean question set (Assertion, Reason, and the four options) for printing.
- Section 2: ANSWER KEY table: Q.No | Correct option | 2-3 line justification stating whether A is true, whether R is true, and whether R explains A.
- Use clean LaTeX for all maths, or clear plain-text math otherwise. Keep all mathematics fully legible.

QUALITY BARS:

- Decide the truth of A and of R by working each out yourself FIRST.
- Verify every classification by two independent checks (for example, test R as a general statement AND check whether it logically forces A). If unsure, rework the item.
- Never invent theorems, properties, or values; use only correct, standard mathematics for the stated board.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.3 True/False Quiz with Justification

Text only

Any AI chat

What you get: A True/False quiz where each statement must be justified - perfect for catching misconceptions.

How to use: Tell it the topic and class, and it builds True/False statements plus the reason each one is right or wrong.

COPY EVERYTHING BELOW

Act as a mathematics teacher and item-writer who designs True/False questions that expose hidden misconceptions.

TASK: Create a True/False quiz on the topic below. Each statement must require real reasoning, and every answer must come with a short justification.

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / SAT]
- Number of statements: [NUMBER OF QUESTIONS e.g. 15]
- Difficulty: [DIFFICULTY e.g. mixed]

DESIGN RULES:

- Write clear declarative statements that are each definitely True or definitely False - no ambiguous wording.
- Make roughly half True and half False, shuffled (do not put all True first).
- For FALSE statements, build in a common misconception (for example: 'squaring both sides never adds extra roots', 'all functions are continuous', ' $(a+b)^2 = a^2 + b^2$ '). The statement should look believable.
- Include 2-3 'almost true' statements that fail only in a special or edge case, to reward careful thinkers.

OUTPUT REQUIREMENTS:

- Section 1: The numbered list of statements (statements only) for students to mark T or F.
- Section 2: ANSWER KEY table: No. | True/False | One or two line justification. For a False statement, also give a quick counterexample or the corrected version.
- Use clean LaTeX for all maths, or clear plain-text math if LaTeX is unavailable. Keep all symbols legible.

QUALITY BARS:

- Determine the truth value of each statement by working it out yourself FIRST.
- Verify each verdict with two independent checks (for example, a general argument AND a concrete numerical example). If they conflict, fix the statement.
- Never invent theorems, rules, or values; use only correct standard mathematics for the stated board.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.4 Flashcard Set (Question Front, Full Answer Back)

Text only

Any AI chat

What you get: A printable flashcard deck - short prompt on the front, full worked answer on the back.

How to use: Name your topic and how many cards you want, and get a ready-to-cut revision flashcard set.

COPY EVERYTHING BELOW

Act as a mathematics revision specialist who builds high-quality flashcard decks for quick recall and exam prep.

TASK: Create a flashcard set on the topic below. Each card has a short prompt on the FRONT and the complete answer on the BACK.

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET]
- Number of cards: [NUMBER OF QUESTIONS e.g. 20]
- Card mix: include a balance of (a) formula/definition recall, (b) one-step concept checks, and (c) short worked problems.

FORMAT FOR EVERY CARD:

CARD n

FRONT: [a short question, formula prompt, or 'State / Define / Find ...']

BACK: [the full correct answer; for worked problems show the key steps briefly, not just the final value; for formulas give the formula plus when to use it]

DESIGN RULES:

- Keep FRONT text short enough to fit a small card; keep BACK complete but tight.
- Order cards from basic recall to short application.
- Avoid duplicate cards; each card should teach something distinct.

OUTPUT REQUIREMENTS:

- Present all cards in a clean two-column table: Card No. | FRONT | BACK, so I can print and cut them.
- After the table, add a one-line 'How to revise' tip for students (for example, cover the back and self-test).
- Use clean LaTeX for all maths, or clear plain-text math if LaTeX is not supported. Keep every formula legible.

QUALITY BARS:

- Work out every BACK answer yourself FIRST.
- Verify each numerical or algebraic answer by two independent methods (for example, direct solving and substitution). Correct any mismatch.
- Never invent formulas, theorems, or values; use only correct standard mathematics for the stated board.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.5 Fill-in-the-Blank Quiz

Text only

Any AI chat

What you get: A fill-in-the-blank worksheet on key formulas, terms and steps, with a clean answer key.

How to use: Give your chapter and class; it produces blanks that test the exact words and formulas students forget.

COPY EVERYTHING BELOW

Act as a mathematics teacher who designs fill-in-the-blank quizzes that lock in vocabulary, formulas, and key steps.

TASK: Create a fill-in-the-blank quiz on the topic below. Each item is a sentence or equation with one or more blanks marked as `_____` (a clear underline).

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE]
- Number of items: [NUMBER OF QUESTIONS e.g. 15]
- Difficulty: [DIFFICULTY e.g. easy to medium]

DESIGN RULES:

- Mix three kinds of blanks: (a) a missing keyword or definition, (b) a missing part of a formula or identity, and (c) a missing value or step in a short calculation.
- Each blank must have exactly ONE clearly correct answer - avoid blanks that could be filled many different ways.
- Where helpful, give a small word bank of options at the top (a few more words than blanks) so weaker students can attempt it; mark this section as optional.

OUTPUT REQUIREMENTS:

- Section 1: The clean quiz with numbered items and visible `_____` blanks, ready to print.
- Section 2 (optional): A shuffled word bank.
- Section 3: ANSWER KEY table: Item No. | Correct fill(s) | one-line note if the answer is commonly confused.
- Use clean LaTeX for all maths, or clear plain-text math otherwise. Keep all symbols legible.

QUALITY BARS:

- Solve and confirm each blank yourself FIRST.
- Verify every formula and value by two independent checks (for example, recall the standard form AND test it on a simple number). Fix any mismatch.
- Never invent formulas, definitions, or values; use only correct standard mathematics for the stated board.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.6 Rapid-Fire 1-Mark Oral Quiz

Text only

Any AI chat

What you get: A fast oral quiz of short 1-mark questions with instant answers for classroom drilling.

How to use: Choose a topic and class; read the questions aloud in class and check answers on the spot.

COPY EVERYTHING BELOW

Act as an energetic mathematics teacher running a fast oral 'rapid-fire' round to drill core skills.

TASK: Create a rapid-fire 1-mark oral quiz on the topic below. Questions must be short enough to ask aloud and answer quickly (within 15-30 seconds each).

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE foundation]
- Number of questions: [NUMBER OF QUESTIONS e.g. 25]
- Difficulty: [DIFFICULTY e.g. easy / medium / mixed]

DESIGN RULES:

- Every question must be answerable mentally or with minimal scratch work - short numbers, clean values, quick recall.
- Include a mix: quick mental calculation, formula recall, true/false on the spot, 'name the term', and 'what comes next'.
- Keep wording very short and spoken-style, since the teacher will read them aloud.
- Arrange from warm-up to slightly harder, so the round builds momentum.

OUTPUT REQUIREMENTS:

- A two-column table: Q.No | Question (read aloud) | Answer (for the teacher only).
- At the end, suggest a simple scoring/timing rule (for example, 30 seconds per question, +1 for correct, fastest team wins).
- Use clear plain-text math or clean LaTeX. Keep numbers tidy so they are easy to say aloud.

QUALITY BARS:

- Solve every question yourself FIRST and keep answers short and unambiguous.
- Verify each answer by two quick independent checks (for example, mental method AND reverse check). Fix any error.
- Never invent formulas or values; use only correct standard mathematics for the stated board.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.7 Kahoot / Game-Style Quiz with Timer Suggestions

Text only

Any AI chat

What you get: A game-show style 4-option quiz with per-question timers, ready to load into Kahoot or Quizizz.

How to use: Say your topic and class; copy the questions and timers straight into Kahoot or read them out for a class game.

COPY EVERYTHING BELOW

Act as a fun, sharp mathematics teacher who designs game-show style classroom quizzes (Kahoot/Quizizz style) that are exciting but still rigorous.

TASK: Create a game-style quiz on the topic below. Every question has 4 short answer options and a suggested countdown timer, formatted so I can quickly paste it into Kahoot, Quizizz, or read it on screen.

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE foundation]
- Number of questions: [NUMBER OF QUESTIONS e.g. 20]
- Difficulty: [DIFFICULTY e.g. mixed, building up]

DESIGN RULES:

- Keep each question and each option SHORT (they must fit on a game screen).
- Exactly one correct option per question; the other three should be believable wrong answers based on common slips.
- Suggest a timer per question based on difficulty: 20 seconds for quick recall, 30 seconds for one-step, 45-60 seconds for multi-step.
- Order questions to build excitement: start easy, add a couple of harder 'bonus' questions near the end.

OUTPUT REQUIREMENTS:

- A table: Q.No | Question | Option A | Option B | Option C | Option D | Correct Option | Suggested Timer.
- A short separate ANSWER KEY line list (Q.No → correct letter) for fast checking.
- One line of game tips (for example, award bonus points for speed, do a quick recap after each tricky question).
- Use clear plain-text math or clean LaTeX; keep everything short and legible on screen.

QUALITY BARS:

- Solve every question yourself FIRST.
- Verify each correct answer by two independent methods before finalising. Fix any mismatch.
- Never invent formulas or values; use only correct standard mathematics for the stated board. Ensure exactly one correct option per question.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.8 Match-the-Column Quiz

Text only

Any AI chat

What you get: A two-column matching quiz (terms, formulas or problems matched to answers) with the matching key.

How to use: Give your topic and class; you get a ready match-the-following exercise plus the correct pairings.

COPY EVERYTHING BELOW

Act as a mathematics paper-setter who designs 'match the column' (match the following) questions for exams and revision.

TASK: Create a match-the-column quiz on the topic below, with Column A and Column B, where students draw lines or write pairings.

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE]
- Number of sets: [NUMBER OF QUESTIONS e.g. 3 separate matching sets]
- Items per set: [e.g. 5 to 6 pairs each]
- Difficulty: [DIFFICULTY e.g. medium]

DESIGN RULES:

- Vary the type of matching across sets, for example: term to definition, problem to its answer, formula to its name, graph description to its function, or expression to its simplified form.
- Make Column A items unique and Column B items unique, with exactly one correct match each.
- Add 1 extra 'distractor' item in Column B in at least one set (more options than questions) so students cannot match by elimination alone. Clearly note which set has the extra option.
- Label Column A with numbers (1, 2, 3 ...) and Column B with letters (P, Q, R ...).

OUTPUT REQUIREMENTS:

- For each set: a neat two-column layout (Column A | Column B) ready to print.
- ANSWER KEY for each set as pairings, for example: 1-Q, 2-P, 3-S ... with a one-line reason for any tricky pair.
- Use clean LaTeX for all maths, or clear plain-text math otherwise. Keep all symbols legible.

QUALITY BARS:

- Work out every correct pairing yourself FIRST.
 - Verify each match by two independent checks (for example, solve the problem AND confirm it equals the listed answer). Fix any error.
 - Never invent formulas, names, or values; use only correct standard mathematics for the stated board.
- Ensure each item has exactly one correct match.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.9 Case-Study / Passage-Based MCQ Set

Text only

Any AI chat

What you get: A real-life passage followed by linked MCQs - the new CBSE competency-based pattern, with key.

How to use: Pick a topic and class; get a story-based passage with 4-5 multiple-choice questions and answers attached.

COPY EVERYTHING BELOW

Act as a CBSE competency-based assessment expert who writes case-study (passage-based) mathematics questions.

TASK: Write a short real-life CASE STUDY passage on the topic below, then attach a set of multiple-choice questions that all draw on that passage.

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board]
- Number of MCQs per case study: [NUMBER OF QUESTIONS e.g. 5]
- Number of case studies: [e.g. 2]
- Difficulty: [DIFFICULTY e.g. medium]

DESIGN RULES:

- Set each passage in a realistic, relatable Indian or everyday context (for example shopping, sports, travel, farming, savings, a school event) so the maths feels useful.
- Give the passage enough data (numbers, a diagram description, or a small table) for all the questions to be answered.
- Make the MCQs increase in difficulty: start by reading data, then a one-step calculation, then a multi-step or reasoning question.
- Each MCQ has 4 options, exactly one correct, with wrong options based on realistic mistakes.

OUTPUT REQUIREMENTS:

- For each case study: the PASSAGE (with any data table) followed by the numbered MCQs and their 4 options, ready to print.
- ANSWER KEY table: Q.No | Correct option | one-line reason.
- Use clean LaTeX for all maths, or clear plain-text math otherwise. Keep all numbers and symbols legible.

QUALITY BARS:

- Solve every question from the passage data yourself FIRST.
- Verify each answer by two independent methods (for example, calculate directly AND check it is consistent with the passage data). Fix any mismatch.
- Ensure the passage actually contains the data needed; do not require outside numbers.
- Never invent formulas or values; use only correct standard mathematics for the stated board. Exactly one correct option per question.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

12.10 Difficulty-Tiered Quiz (Bronze / Silver / Gold)

Text only

Any AI chat

What you get: One quiz in three rising levels - Bronze, Silver, Gold - so every student is challenged just right.

How to use: Enter your topic and class; get three graded levels in one go to handle mixed-ability classes.

COPY EVERYTHING BELOW

Act as a differentiation-focused mathematics teacher who designs tiered quizzes for mixed-ability classrooms.

TASK: Create ONE quiz on the topic below in three rising difficulty tiers - BRONZE (easy/foundation), SILVER (medium/standard), and GOLD (hard/challenge) - so every student is appropriately stretched.

INPUT FROM ME:

- Topic: [TOPIC]
- Class/Grade: [CLASS/GRADE]
- Board or Exam: [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE]
- Questions per tier: [NUMBER OF QUESTIONS e.g. 5 per tier]
- Question style: [e.g. all MCQ, or short-answer, or a mix - your choice]

TIER DESIGN RULES:

- BRONZE: direct recall and one-step application; builds confidence; clean numbers.
- SILVER: two-step problems, light reasoning, mild twists; the standard exam level.
- GOLD: multi-step, multi-concept, or 'trap' questions that reward deep understanding; suitable for toppers.
- Keep all three tiers on the SAME topic so a student can climb from Bronze to Gold.

OUTPUT REQUIREMENTS:

- Three clearly labelled sections: BRONZE, SILVER, GOLD, each with its numbered questions (and options if MCQ), ready to print.
- A combined ANSWER KEY table: Tier | Q.No | Answer | brief solution outline (2-3 lines for Silver and Gold).
- A short note suggesting which students each tier suits and how to score across tiers.
- Use clean LaTeX for all maths, or clear plain-text math otherwise. Keep all symbols legible.

QUALITY BARS:

- Solve every question in every tier yourself FIRST.
- Verify each final answer by two independent methods before finalising; redo any question where the methods disagree.
- Ensure the difficulty truly rises across tiers (Gold must be clearly harder than Bronze).
- Never invent formulas, theorems, or values; use only correct standard mathematics for the stated board.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.



Real-life problems, projects, history and games that make maths stick.

13.1 Real-Life Word Problems From Students' World

[Text only](#)[Any AI chat](#)

What you get: A fresh set of relatable word problems with full worked solutions and answer key.

How to use: Just type your topic, class and how many problems you want, then paste this in.

COPY EVERYTHING BELOW

Act as an experienced mathematics teacher and curriculum writer who specialises in making abstract topics relatable for everyday students.

TASK: Create [NUMBER OF QUESTIONS] original real-life word problems on the topic [TOPIC] for [CLASS/GRADE] students following [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET / SAT]. Difficulty: [DIFFICULTY e.g. easy / medium / hard / mixed].

INPUT FROM ME: I have given the topic, class, board and count above. If I attach nothing else, set the problems in contexts Indian and global students actually live in: pocket money and savings, mobile data packs, cricket and football scores, cooking and recipes, bus and train travel, electricity bills, online shopping discounts, festival shopping, and sharing food among friends. Use realistic numbers and the rupee symbol Rs where money appears, with a second currency note in brackets where useful.

OUTPUT REQUIREMENTS:

1. Number every problem. Keep the language simple and the scenario believable for a teenager.
2. After all the questions, add a clearly separated section titled SOLUTIONS with full step-by-step working for each, the formula or concept named, and the final answer in a box like [Answer: ...].
3. Add a one-line ANSWER KEY summary at the very end listing just the final answers.
4. Tag each problem with the exact sub-skill it tests (for example: setting up an equation, unit conversion, percentage).

QUALITY BARS: Solve every problem internally before writing it. Verify each final answer by at least two independent methods (for example, re-substitution and a quick estimate) and confirm both agree. Never invent formulas, theorems or fake data. Keep all mathematics fully legible using clean LaTeX or clear plain-text math. If any problem cannot be verified cleanly, replace it with one that can. Make sure no two problems are near-duplicates.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

13.2 Hands-On Maths Project With Marking Rubric

Text only

Any AI chat

What you get: Three project ideas plus a step-by-step plan and a ready marking rubric.

How to use: Tell it your topic and class, and it designs a full project students can actually build.

COPY EVERYTHING BELOW

Act as a creative mathematics teacher and project-based-learning designer who has run dozens of successful classroom projects.

TASK: Design hands-on maths project ideas on the topic [TOPIC] for [CLASS/GRADE] students under [BOARD or EXAM e.g. CBSE / ICSE / State Board]. The project should take about [TIME e.g. 1 week / 2 weeks] and be done [individually / in groups of 3-4].

INPUT FROM ME: I have given the topic, class, board and timeframe above. Assume a normal classroom with low-cost materials only (paper, cardboard, string, measuring tape, calculator, free phone apps). If I add any constraints, follow them.

OUTPUT REQUIREMENTS:

1. Propose 3 distinct project ideas, each with a catchy title, the real-world question it answers, and the exact maths skills it builds.
2. Pick the strongest idea and expand it into a full plan: learning objectives, materials list, a day-by-day or stage-by-stage student task breakdown, and a sample expected result with the actual maths worked out.
3. Provide a clean assessment RUBRIC as a table with 4-5 criteria (for example: mathematical accuracy, real-world connection, presentation, teamwork, creativity), each scored across 4 levels with clear descriptors and marks adding to [TOTAL MARKS].
4. Add a short list of guiding questions a teacher can ask to keep students unstuck, and 3 extension challenges for fast finishers.

QUALITY BARS: Work out the sample result internally and verify the numbers by at least two independent checks before presenting them. Never invent formulas or fake measurements; if you use sample data, label it clearly as illustrative. Keep all mathematics fully legible in clean LaTeX or plain-text math. Make the rubric fair and genuinely usable by a busy teacher.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

13.3 The Story Behind a Maths Concept

Text only

Any AI chat

What you get: A vivid, classroom-ready history story of a concept with a discussion plan.

How to use: Type the concept you are about to teach and it writes the human story behind it.

COPY EVERYTHING BELOW

Act as a mathematics teacher and historian of mathematics who can turn dry topics into gripping true stories that hook students.

TASK: Write the history and story of the concept [TOPIC] for [CLASS/GRADE] students studying [BOARD or EXAM e.g. CBSE / ICSE / State Board]. The goal is a 5-7 minute read-aloud story I can use at the start of a lesson to spark curiosity.

INPUT FROM ME: I have given the topic and class above. Pitch the language at this age group. If the concept has Indian mathematical roots (for example, zero, the decimal system, trigonometry, infinite series), include them accurately and with pride, alongside global contributions.

OUTPUT REQUIREMENTS:

1. Open with a hook question or surprising fact that connects to students' lives.
2. Tell the chronological story: who struggled with the problem, why it mattered, the key people and cultures involved, and the moment of breakthrough. Use real names, places and approximate dates.
3. Include one short, simple worked example showing the idea in action, with clean step-by-step maths.
4. End with a STARTER DISCUSSION block: 5 thought-provoking questions to ask the class, and a one-line link from the story to today's lesson objective.
5. Add a CLAIMS I VERIFIED note listing the historical facts you are confident are accurate, and flag anything that is commonly debated so I do not state a myth as fact.

QUALITY BARS: Do not invent people, dates, theorems or quotes. If a popular anecdote is legend rather than fact, say so clearly. Solve and verify the worked example by at least two independent checks. Keep all mathematics fully legible in clean LaTeX or plain-text math. Keep the tone warm, accurate and age-appropriate.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

13.4 Cross-Subject STEM Links (Physics, Finance, Coding)

Text only

Gemini / ChatGPT

What you get: Clear bridges from one maths topic to physics, money and code with examples.

How to use: Tell it the maths topic and it shows where that same maths lives in other subjects.

COPY EVERYTHING BELOW

Act as an interdisciplinary STEM educator who teaches mathematics and shows students exactly where it powers physics, finance and computer science.

TASK: Build a cross-subject connection guide for the maths topic [TOPIC] aimed at [CLASS/GRADE] students under [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE]. Show how this single idea appears in at least three other domains.

INPUT FROM ME: I have given the topic and class above. If I name preferred domains, use those; otherwise default to physics, personal finance, and basic coding.

OUTPUT REQUIREMENTS: For each linked domain provide a clearly headed section containing:

1. THE CONNECTION: one or two sentences on how the maths topic shows up there.
2. A REAL EXAMPLE: a small, fully worked problem set in that domain, with step-by-step maths and the final answer boxed.
3. WHY IT MATTERS: a one-line career or real-world payoff a teenager would find motivating.

For the coding section, include a short, correct pseudocode or Python snippet (under 15 lines) that demonstrates the concept, with comments, and state the expected output.

Finish with a TEACHER TIE-IN block: 3 quick discussion prompts and one mini cross-subject task students can try in 10 minutes.

QUALITY BARS: Solve every example internally and verify each answer by at least two independent methods before presenting it. Make sure the physics units are correct and consistent. Trace the code by hand to confirm its stated output is right; never present code you have not mentally executed. Do not invent formulas, constants or theorems. Keep all mathematics fully legible in clean LaTeX or plain-text math.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

13.5 Design a Classroom Maths Game

Text only

Any AI chat

What you get: A complete, ready-to-play maths game with rules, cards and a quick setup guide.

How to use: Say the topic and class size and it invents a fun game you can run tomorrow.

COPY EVERYTHING BELOW

Act as a playful mathematics teacher and educational game designer who builds games students beg to play again.

TASK: Design one engaging classroom maths game that drills the topic [TOPIC] for [CLASS/GRADE] students under [BOARD or EXAM e.g. CBSE / ICSE / State Board]. Class size: about [NUMBER OF STUDENTS]. Time available: [TIME e.g. 20 minutes / one period]. Use low-cost materials only (paper, board, dice, a timer, the blackboard).

INPUT FROM ME: I have given the topic, class, size and time above. If I ask for a specific format (team quiz, card game, bingo, relay, escape-room), follow it; otherwise pick the best fit and explain why.

OUTPUT REQUIREMENTS:

1. GAME NAME and a one-line pitch.
2. LEARNING GOAL: the exact skill it sharpens.
3. MATERIALS and SETUP: simple, cheap, and quick to prepare.
4. HOW TO PLAY: numbered rules a teacher can read aloud, including scoring and how to win.
5. CONTENT PACK: at least 12 ready-to-use questions, cards or prompts on the topic, each with its correct answer, sorted from easy to hard.
6. CLASSROOM MANAGEMENT TIPS: how to keep everyone involved, handle disputes, and adapt for stronger or weaker students.
7. A 1-minute DEBRIEF: 3 questions to lock in the learning after the game ends.

QUALITY BARS: Solve every question in the content pack internally and verify each answer by at least two independent checks before listing it. Make sure the game is genuinely playable in the stated time. Never invent formulas or wrong answers. Keep all mathematics fully legible in clean LaTeX or plain-text math.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

13.6 Term-Long Gamified Maths Challenge Plan

Text only

Gemini / ChatGPT

What you get: A full term challenge plan with levels, points, badges and weekly missions.

How to use: Tell it your chapters and term length and it builds a whole gamified journey.

COPY EVERYTHING BELOW

Act as a mathematics teacher and gamification strategist who keeps a whole class motivated across an entire term.

TASK: Design a gamified, term-long maths challenge for [CLASS/GRADE] students under [BOARD or EXAM e.g. CBSE / ICSE / State Board]. Term length: [TIME e.g. 10 weeks / one semester]. Topics or chapters to cover: [LIST TOPICS / CHAPTERS]. The aim is steady daily practice plus excitement, fairness and inclusion of weaker students.

INPUT FROM ME: I have given the class, board, term length and chapter list above. If I do not give chapters, build a sensible sequence for the standard syllabus of that class.

OUTPUT REQUIREMENTS:

1. THEME and STORY: a simple unifying theme (for example, a quest, a space mission, a city to build) that ties weeks together.
2. LEVEL MAP: a week-by-week table linking each chapter to a named level, a clear weekly mission, and a target skill.
3. POINTS and BADGES system: how students earn points, the badge tiers, and at least 6 named badges tied to specific achievements (including effort and improvement, not just top marks).
4. SAMPLE WEEKLY MISSION: one full week written out in detail, including 5 sample practice questions with verified answers.
5. LEADERBOARD and FAIRNESS rules: how to celebrate progress without discouraging strugglers, plus a comeback mechanic.
6. LOW-EFFORT TRACKING: a simple way for a busy teacher to record points (a paper chart or free spreadsheet layout).

QUALITY BARS: Solve the sample questions internally and verify each answer by at least two independent checks. Keep the whole plan realistic for one teacher to run. Never invent formulas or fake data. Keep all mathematics fully legible in clean LaTeX or plain-text math.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

13.7 Group Activity / Maths Lab Task

Text only

Any AI chat

What you get: A complete maths-lab investigation with worksheet, steps and observation table.

How to use: Name your topic and it builds a hands-on lab activity groups can investigate together.

COPY EVERYTHING BELOW

Act as a mathematics teacher and maths-lab coordinator who designs hands-on investigative activities that students do with their hands, not just their pens.

TASK: Create one maths lab / group activity that lets [CLASS/GRADE] students discover or verify the topic [TOPIC] through measurement and exploration, aligned to [BOARD or EXAM e.g. CBSE / ICSE / State Board]. Group size: [groups of 3-4]. Time: [TIME e.g. one period].

INPUT FROM ME: I have given the topic, class, group size and time above. Use only low-cost lab materials (graph paper, ruler, protractor, scissors, string, geoboard, cut-outs, calculator). If the topic suits a free tool like GeoGebra, offer it as an optional alternative.

OUTPUT REQUIREMENTS:

1. ACTIVITY TITLE and the big question students are investigating.
2. LEARNING OBJECTIVE and the concept being discovered or verified.
3. MATERIALS list and SAFETY/SETUP notes.
4. STEP-BY-STEP PROCEDURE written directly to students, numbered and clear.
5. OBSERVATION TABLE: a ready-to-copy table for recording measurements, plus the calculation students must perform on their data.
6. EXPECTED RESULT: the conclusion they should reach, with one fully worked sample using illustrative numbers, clearly labelled as a sample.
7. GUIDING QUESTIONS and a short CONCLUSION prompt for groups to write up.
8. ROLES for each group member so everyone contributes.

QUALITY BARS: Work the sample calculation internally and verify it by at least two independent checks before showing it. Make sure the activity genuinely demonstrates the concept and fits the time. Never invent formulas or fake measured constants; label sample data as illustrative. Keep all mathematics fully legible in clean LaTeX or plain-text math.

End the output with a footer line: "Prepared by Indrajeet Yadav", and place "Compiled by Indrajeet Yadav" in the title or header block.

13.8 Motivation & Exam-Stress Card For Students

Text only

ChatGPT (image)

What you get: An encouraging, printable motivation card with calm study and exam tips.

How to use: Tell it the exam and class and it designs a warm motivational card you can print or share.

COPY EVERYTHING BELOW

Act as a caring mathematics mentor and student wellbeing coach who helps young learners beat maths anxiety and exam stress.

TASK: Design one printable, visually appealing MOTIVATION and EXAM-STRESS card for [CLASS/GRADE] students preparing for [BOARD or EXAM e.g. CBSE / ICSE / State Board / JEE / NEET]. Exam in about [TIME e.g. 2 weeks]. The card should calm nerves, build confidence and give practical maths-study advice.

INPUT FROM ME: I have given the class, exam and timeframe above. Keep the tone warm, honest and never preachy. Make it suitable for printing on A5 or sharing as an image on a class group.

OUTPUT REQUIREMENTS: Produce a single, well-designed card (portrait) containing:

1. A short, genuine HEADLINE message of encouragement (no empty cliches).
 2. THREE practical maths-study tips for the final stretch (for example, how to revise formulas, how to attempt a paper, how to handle a question that looks impossible).
 3. A 60-SECOND CALM-DOWN routine students can use just before or during the exam (simple breathing plus a reframing thought).
 4. One short, true line reminding them that one exam does not define their worth.
 5. A small tasteful design: friendly colours, clear readable fonts, gentle maths motifs (symbols, a small graph, geometric shapes) used as decoration, with plenty of white space. No clutter.
- If you can only output text, give a clean text layout PLUS an exact image-generation prompt I can reuse to create the visual.

QUALITY BARS: Keep every claim accurate and supportive; do not invent statistics or fake quotes. If you quote anyone, ensure it is genuine, otherwise use original lines. Keep any maths shown fully legible and correct.

Sign the bottom-right of every page with an elegant handwritten signature "Indrajeet Yadav" with a small underline flourish.